

Electrical Project Design

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Acknowledgment

The project team wishes to express his gratitude and sincere thanks to
Dr. Mohamed AbdElwahab For his supervision, encouragement and his
creative guidance during the work at the project

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Index

Chapter 1	➔ Indoor Lighting	Page 6
Chapter 2	➔ Outdoor Lighting.....	Page 30
Chapter 3	➔ Accessories	Page 57
Chapter 4	➔ Cables	page 70
Chapter 5	➔ Load Estimation and Calculation	page 83
Chapter 6	➔ Voltage Drop and Short Circuit	page 98
Chapter 7	➔ Feeding System	page 107
Chapter 8	➔ Power Factor Correction	page 117
Chapter 9	➔ Emergency	page 129
Chapter 10	➔ Earthing	page 138
Chapter 11	➔ protection	page 155
Chapter 12	➔ Fire Alarm	page 166
Chapter 13	➔ CCTV and IP telephone	page 184
Chapter 14	➔ Sound System	page 202
Chapter 15	➔ Smart Home	page 212
References		page 224

Project Description.

This project is electrical distribution of the low voltage, Medium voltage circuits and light current circuits of a Touristic resort with samples of a smart home design. The Touristic resort consists of 25 building on area equal 60,000 including swimming pools and roads of the village. And at this book we illustrate the design principles of the electrical distribution of the resort.

Chapter 1 ➔ **indoor lighting**

It explains the lighting design for indoor application manually and using dialux program. The chapter has information about each luminaire used and no of luminaire at each floor.

Chapter 2 ➔ **outdoor lighting**

It has the principles of street lighting design and swimming pools lighting design.

Chapter 3 ➔ **Accessories**

At this chapter we will discuss the different types of sockets, switches and other accessories used in the project

Chapter 4 ➔ **Cables**

This chapter explains the classification of cables, the de-rating factors and some examples from our project.

Chapter 5 ➔ **Load Estimations and Calculations.**

This chapter explains how to compute the primary rating of transformers and the design of the branch circuits and Main Circuits.

Chapter 6 ➔ **Voltage Drop and Short Circuits calculations**

This chapter explains the method of calculations of voltage drop and short circuits provided with examples from our project.

Chapter 7 ➔ **Feeding System**

This chapter explains the low voltage and medium voltage feeding systems including transformers, ring main units.

Chapter 8 ➔ **power factor correction**

This chapter explains the different methods of power factor correction, the advantage of each one and the method of calculation.

Chapter 9 ➔ **Emergency**

This chapter explains the ups and Diesel generators and ATS. The advantages of the system

Chapter 10 ➔ **Earthing**

This chapter explains the advantage of earthing, the manual method calculating the number of electrodes used and the types of earthing systems.

Chapter 11 ➔ **Protection**

This chapter describes the different types of Circuit Breakers and fuses used in our project and the protection devices on the transformer.

Chapter 12 ➔ **Fire Alarm**

This chapter explains the different systems used in fire alarm design, the component of each systems, the principles of design of the system.

Chapter 13 ➔ **CCTV and Ip Telephone**

This chapter illustrates the CCTV system and the IP telephone system and the principles of design of the two systems.

Chapter 14 ➔ **sound system**

This chapter illustrates the component of the sound systems the principles of design

Chapter 15 ➔ **smart home design**

This chapter explains how to control the lighting or anything in the home, the chapter also includes the advantage of the system. The equipment and the software used

CHAPTER 1

INDOOR LIGHTING



Indoor lighting

1.1-Lighting:

Light is the prime factor in the human life as all activities of human depend upon the light. Where there is no natural light, a source of artificial light was needed. Light may be produced by passing electric currents through filaments as in the incandescent lamps, through arcs between carbon or metal rods, or through suitable gases as in neon and other gas tubes. In some forms of lamps the light is due to fluorescence excited by radiation arising from the passage of electricity through mercury vapor

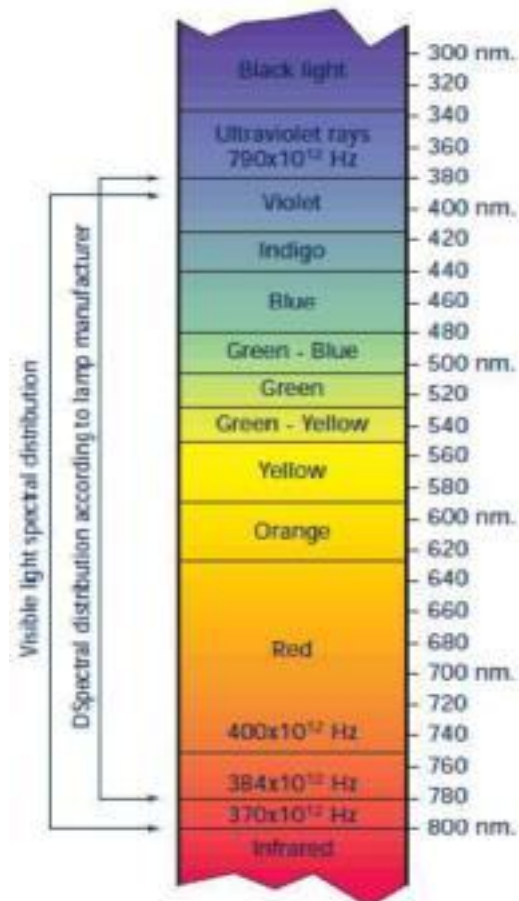
1.2-Nature of Light:

Various forms of incandescent bodies are the sources of light and the light emitted by such bodies depend upon its temperature. Energy is radiated into the medium by a body which is hotter than the medium surrounding it, in the form of electromagnetic waves of various wavelengths. The velocity of propagation of radiant energy is approximately 3×10^8 m/sec. The properties and Behavior of the radiant energy depends on the wavelength

1.3-Types of lighting schemes:

The interior lighting schemes may be classified as:

- 1- Direct lighting
- 2- Semi-direct lighting.
- 3- Semi indirect lighting.
- 4- Indirect lighting.
- 5- General lighting



1- Direct Lighting

In this type of lighting the light from the source falls directly on the object or the surface to be illuminated.

2- Semi-direct Lighting

Only 60-90 % of the flux reaches the working plan directly while the rest is reflected to illuminate the ceiling and walls.

3- Indirect Lighting:

In this form of lighting, light does not reach the surface directly from the source but indirectly as the ceiling works as light source and reflects the light: 90% of light upwards, 10% downwards.

4- Semi-Indirect Light

Like the previous method but 75% of light upwards and 25% downwards.

5- General lighting:

Lamps made of diffusing glass are used which give nearly equal illumination in all directions

Types of lamps:

1-Incandescent lamps:

- a) standard (A)
- b) Halogen

2-discharge lamp:

A-High intensity discharge lamp:

- a) Mercury vapor
- b) Metal Halide
- c) High pressure sodium

B-Low pressure

sodium C-Fluorescent:

- a) compact fluorescent
- b) Straight tube
- c) circular fluorescent

3- Arc lamps:

- a) Carbon arc lamps.
- b) Flame arc lamps.
- c) Magnetic lamps.





CDM-TDC(METAL HALIDE LAMP).



Fluorescent Lamps, TL5.



Incandescent lamps.



HALOGEN Lamp.(color rendering reach 95%).



PL-L (COMPACT FLOURECENT NON INTEGRATED)



PL-C/13W (Compact Fluorescent)



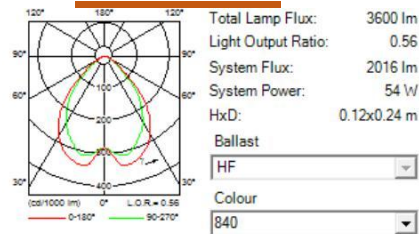
PL-C/18W(Compact Fluorescent)

3) Types of reflectors:

		
Reflector : Opal <i>for healthcare and hospital applications</i>	Reflector : Prismatic <i>for educational facilities & corridors</i>	Reflector : Parabolic (high polished) <i>for both office and school applications</i>

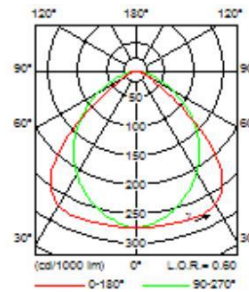
Specification of Luminaire

TYPE:1



NAME	SPOT (DOWNLIGHT).
SURFACE TYPE	RECESSED MOUNTED
BALLAST	Electronic ballast.
LAMP TYPE	Compact saving lamps(pl-c/2p)
NO. OF LAMP	2x26W
USAGE	ball room and buffet zone
CATALOGUE	• PHILIPS CATALOGUE. • CODE:: DOWNLIGHTS LUMINAIRE CATEGORY • HOUSING: FBS120.
Degree of protection	IP20
Total lamp flux:	3600 lm
Number of lamps from this type In our project	ground : first :26 second :112

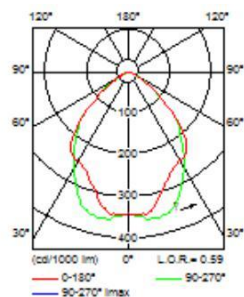
TYPE:2



Total Lamp Flux: 5000 lm
 Light Output Ratio: 0.60
 System Flux: 3000 lm
 System Power: 63 W
 LxBxH: 0.60x0.60x0.05 m

LAMP NAME	TL5-14W
SURFACE TYPE	RECESSED MOUNTE D
BALLAST	Electronic ballast.
LAMP TYPE	Fluorescent lamps
NO. OF LAMP	4x14W
USAGE	Office space
CATALOGUE	-PHILIPS CATALOGUE. CODE: LUMINAIRE CATEGORY: Recessed Mounted HOUSING: TBS165
Degree of protection	IP20.
Total lamp flux:	5000 lm
Number of lamps from this type In our project	ground :293*4 first :175*4 second :165*4

TYPE:3

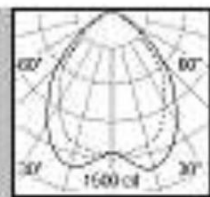


Total Lamp Flux: 1200 lm
 Light Output Ratio: 0.59
 System Flux: 708 lm
 System Power: 18 W
 HxD: 0.13x0.19 m
 Ballast

 Colour

LAMP NAME	SPOT (DOWNLIGHT).
SURFACE TYPE	RECESSED MOUNTED
BALLAST	Electronic ballast.
LAMP TYPE	Compact saving lamps(pl- c/4p)
NO. OF LAMP	1x18W
USAGE	BED ROOM
CATALOGUE	-PHILIPS CATALOGUE. -CODE: LUMINAIRE CATEGORY: DOWNLIGHTS HOUSING: FBS261
Degree of protection	IP20.
Total lamp flux:	1200 lm
Number of lamps from this type In our project	ground : 2634 first :2634 second :2634

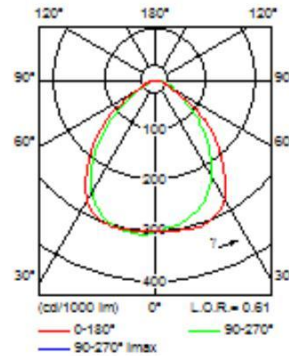
TYPE:4



46654.000
Quintessence Double
washlight

LAMP NAME	SPOT (DOWNLIGHT).
SURFACE TYPE	RECESSED MOUNTED
BALLAST	Electronic ballast.
LAMP TYPE	CompactFluorescent lamps
NO. OF LAMP	2x32W
USAGE	Reception and public service areas
CATALOGUE	ERCO CATALOGUE. HOUSING: TC-TELI 32W
Degree of protection	IP20.
Total lamp flux:	4800 lm
Number of lamps from this type In our project	ground :83*2 first :64*2 second :64*2

TYPE:5

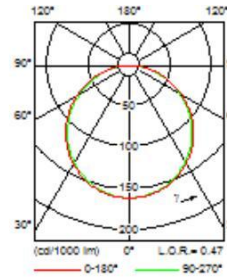


Total Lamp Flux: 3600 lm
 Light Output Ratio: 0.61
 System Flux: 2196 lm
 System Power: 66 W
 HxD: 0.26x0.14 m
 Ballast

 Colour

LAMP NAME	SPOT (DOWNLIGHT).
SURFACE TYPE	SURFACE MOUNTED
BALLAST	Conventional
LAMP TYPE	Compact saving lamps(pl-c/2p)
NO. OF LAMP	2x26W
USAGE	Restaurants, cafeterias and hotels Fashion shops and boutiques
CATALOGUE	-PHILIPS CATALOGUE. -CODE: LUMINAIRE CATEGORY:DOWNLIGHT HOUSING: FBH059
Degree of protection	IP20.
Total lamp flux:	3600 lm
Number of lamps from this type In our project	ground :20*2 first :54*2 second :

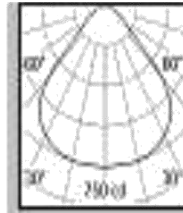
TYPE:6



Total Lamp Flux: 5400 lm
 Light Output Ratio: 0.47
 System Flux: 2538 lm
 System Power: -1 W
 LxBxH: 0.60x0.60x0.08 m
 Ballast
 HFE
 Colour
 840

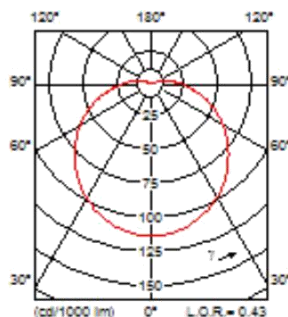
LAMP NAME	MASTER TL-D
SURFACE TYPE	Recessed Mounted
BALLAST	Electronic ballast.
LAMP TYPE	Fluorescent lamps
NO. OF LAMP	4x18W
USAGE	WE USE THIS TYPE IN MAIN KITCHTEN
CATALOGUE	-PHILIPS CATALOGUE. -CODE: LUMINAIRE CATEGORY: Recessed Mounted HOUSING: TBS318
Degree of protection	IP20.
Total lamp flux:	5400 lm
Number of lamps from this type In our project	ground : first :170*4 second :

TYPE:7



85592.000
Panarc Surface-mounted
downlight

LAMP NAME	(DOWNLIGHT).
SURFACE TYPE	SURFACE MOUNTED
BALLAST	Electronic ballast.
LAMP TYPE	CompactFluorescent lamps
NO. OF LAMP	1x36W
USAGE	Corridors
CATALOGUE	-ERCO CATALOGUE. HOUSING: TC-F 36W
Degree of protection	IP44
Total lamp flux:	2800 lm
Number of lamps from this type In our project	ground : 310 first :310 second :310



Total Lamp Flux: 3600 lm
 Light Output Ratio: 0.43
 System Flux: 1548 lm
 System Power: 54 W
 HxD: 0.14x0.42 m

Ballast

HF

Colour

840

TYPE:8

LAMP NAME	Gondola
SURFACE TYPE.	SURFACE MOUNTED
BALLAST	Electronic ballast.
LAMP TYPE	Compact Fluorescent lamps
NO. OF LAMP	2 x26 W
USAGE	toilet
CATALOGUE	-PHILIPS CATALOGUE. -CODE: LUMINAIRE CATEGORY:Decorative Luminaires HOUSING: FWG251
Degree of protection	IP65
Total lamp flux:	3600 lm
Number of lamps from this type In our project	ground : 250*2 first :250*2 second :250*2

TYPE:9



LAMP NAME	Standard Can (B-shape)
LAMP TYPE	Incandescent
Lamp wattage	60 w
USAGE	Ballrooms and Restaurants
Luminous Flux Lamp (Lm)	690 Lm
CATALOGUE	-PHILIPS CATALOGUE

TYPE:10



LAMP NAME	MASTER PL-C 2 Pin
LAMP TYPE	Compact Fluorescent Non-Integrated
Lamp wattage	26 W
USAGE	Ballrooms
Luminous Flux Lamp (Lm)	1800 Lm
CATALOGUE	-PHILIPS CATALOGUE
Number of lamps from this type In our project	ground : first : second :17

TYPE:11



LAMP NAME	Tornado
LAMP TYPE	Compact Fluorescent Integrated lamps
Lamp wattage	20w
USAGE	bathrooms and balconies
Luminous Flux Lamp (Lm)	1250 lm
CATALOGUE	-PHILIPS CATALOGUE
Number of lamps from this type In our project	ground : 734 first :734 second :734

Lighting Calculations of any Compartment (Steps of design):

A) The manual Method:

Step no.(1):

Determine the room dimensions a & b & H

Where: a = length of room (m).

b = width of room (m).

H = height of room (m).

Calculate the Area (A) of any Compartment will be:

$$\text{Area} = \text{length (a)} * \text{width (b)} \text{ (m}^2\text{)}$$

Step no. (2):

- Specify the amount of luminance (E) Lux required, according to the space purpose
- Specify the lamp type according to purpose of the place from(Philips,Erco) catalogues.
- Determine the Utilization factor (U.F) of the room this number reveals how much of the flux emitted by the lamp reach the working plane; this is obtained from the tables ceiling and wall reflectances are of the factors affecting the (U.F).
- Choose the appropriate Maintenance factor (M.F).

Selected based on where the luminaries (fixture) are to be used;

- 0.8 For the indoor regular luminaries cleaning applications.
- 0.75 For the indoor irregular luminaries cleaning applications.

Step no. (3):

- Calculate the No. of the lamps (N) required to give the calculated amount of lumen:

$$N = (\text{Lux(E)} * \text{Area (A)}) / (\text{M.F} * \text{U.F} * \Phi)$$

E: average horizontal intensity of illumination in lux (lx)
A: Area to be illuminated in square meter
Φ: luminous flux per lamp (lm)
U.F: utilization factor
M.F: Maintenance factor



- Calculate Actual Power (watt) in the compartment: • Actual Power = Lamp Wattage * No. of Lamps (watt) • Choose appropriate No. of lamps / Luminaire.
- Calculate the No. of the Luminaries (n):

$$\text{No. of Luminaries (n)} = \frac{\text{Total No. of Lamps (N)}}{\text{No. of Lamps / Luminaire}}$$

B) Using software program (Dialux):

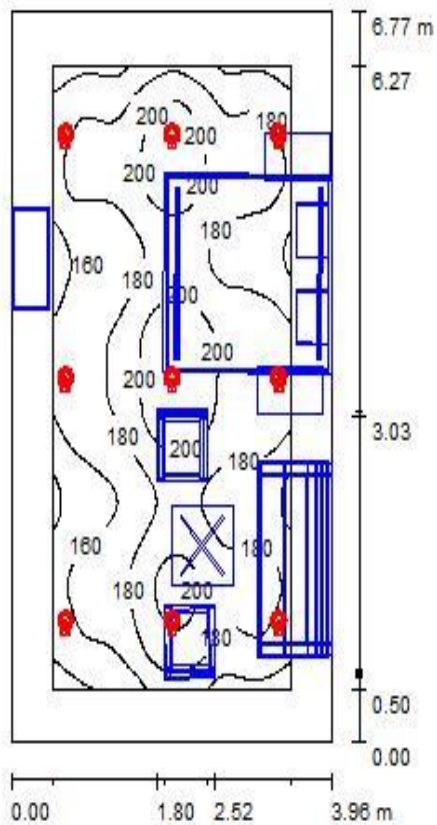
In this method we use a software program (Dialux) to make the required calculations to get the number of lamps and their spacing layout for specified room by using data given by the user such as

• Room dimensions.

- Types of luminaires to be used.
- The value of illumination required.
- Reflection factors of walls, ceiling and floor.
- Maintenance factors for the lamp
- Working plane height

Room name	Illumination (lx) LUX
Bed room	150
toilet	300
Kitchen	500
Office	500
corridor	60
Restaurants	120

For example: bed room

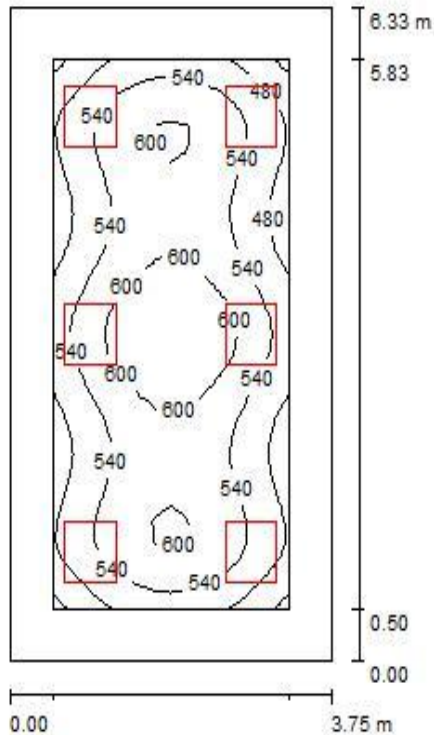


Height of Room: 2.600 m, Mounting Height: 2.727 m, Light loss factor: 0.80

Values in Lux, Scale 1:87

Surface	ρ [%]	E_{av} [lx]	E_{min} [lx]	E_{max} [lx]	$u0$
Workplane	/	181	127	221	0.701
Floor	20	89	1.53	168	0.017
Ceiling	70	36	25	47	0.675
Walls (4)	50	60	6.58	167	/

Office

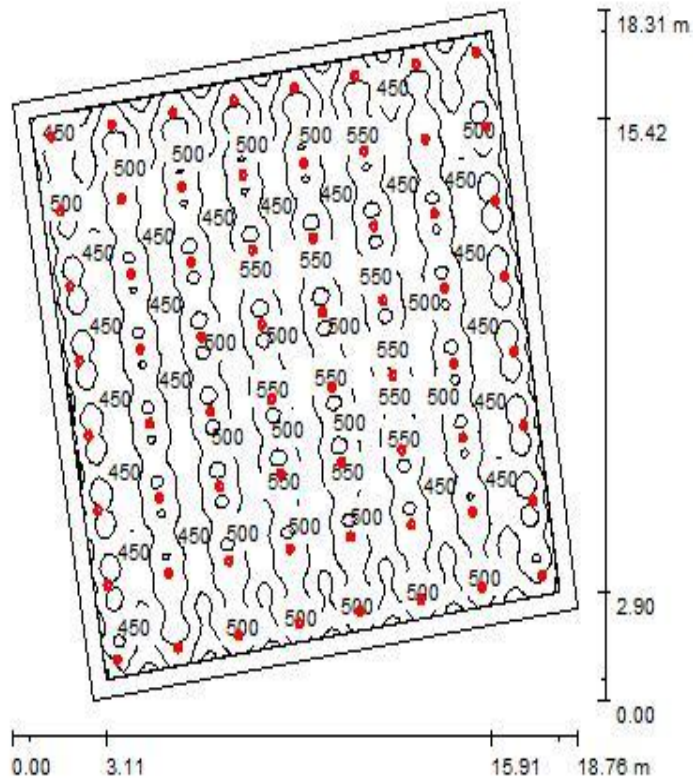


Height of Room: 2.600 m, Mounting Height: 2.651 m, Light loss factor: 0.80

Values in Lux, Scale 1:82

Surface	ρ [%]	E_{av} [lx]	E_{min} [lx]	E_{max} [lx]	u_0
Workplane	/	548	410	662	0.749
Floor	20	414	263	523	0.636
Ceiling	70	97	76	104	0.789
Walls (4)	50	222	88	382	/

Reception

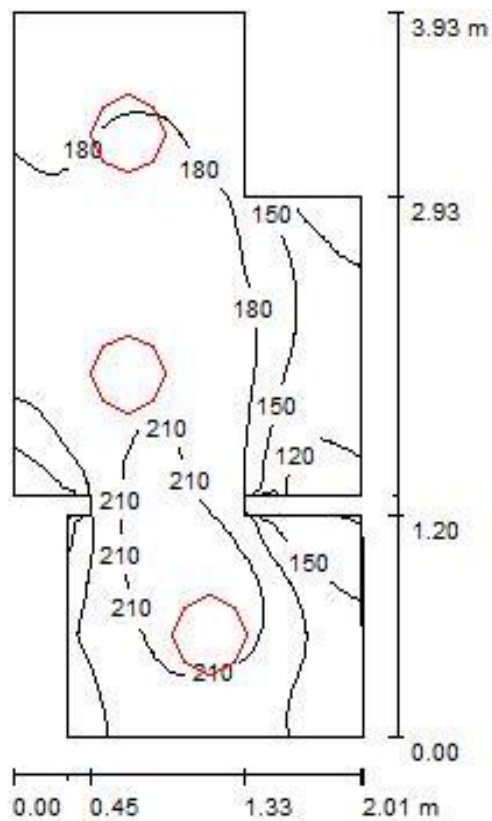


Height of Room: 2.600 m, Mounting Height: 2.600 m, Light loss factor: 0.80

Values in Lux, Scale 1:236

Surface	ρ [%]	E_{av} [lx]	E_{min} [lx]	E_{max} [lx]	u0
Workplane	/	491	342	572	0.695
Floor	20	454	214	523	0.471
Ceiling	70	89	69	102	0.775
Walls (4)	50	175	73	281	/

toilet

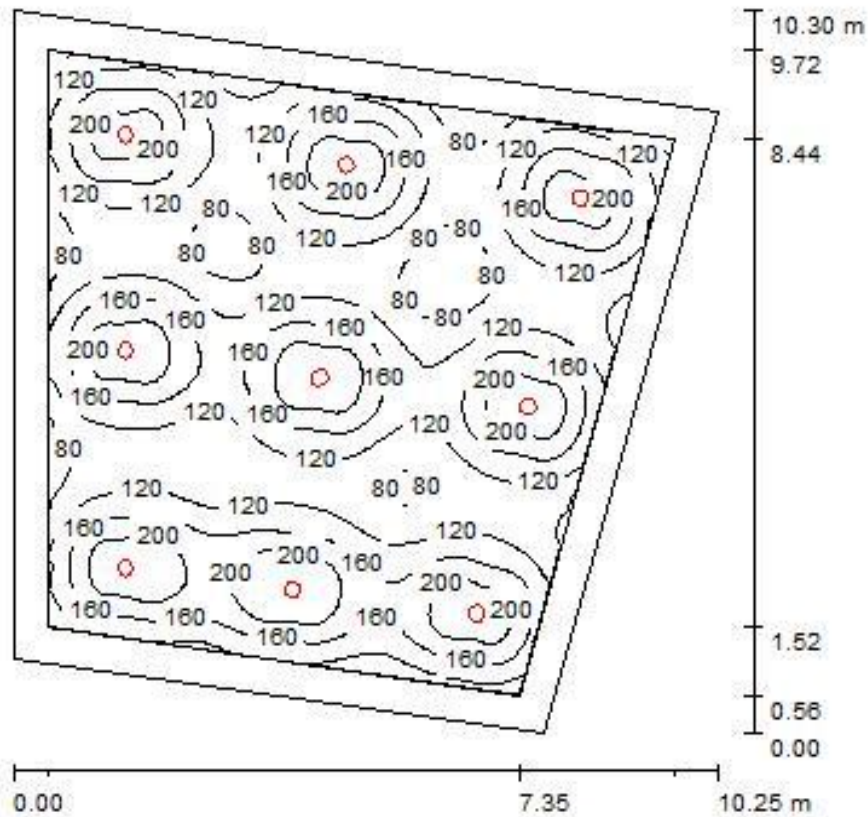


Height of Room: 3.000 m, Light loss factor: 0.80

Values in Lux, Scale 1:51

Surface	ρ [%]	E_{av} [lx]	E_{min} [lx]	E_{max} [lx]	$u0$
Workplane	/	180	99	228	0.550
Floor	20	127	82	152	0.644
Ceiling	70	108	52	946	0.480
Walls (14)	50	139	50	486	/

Restaurants

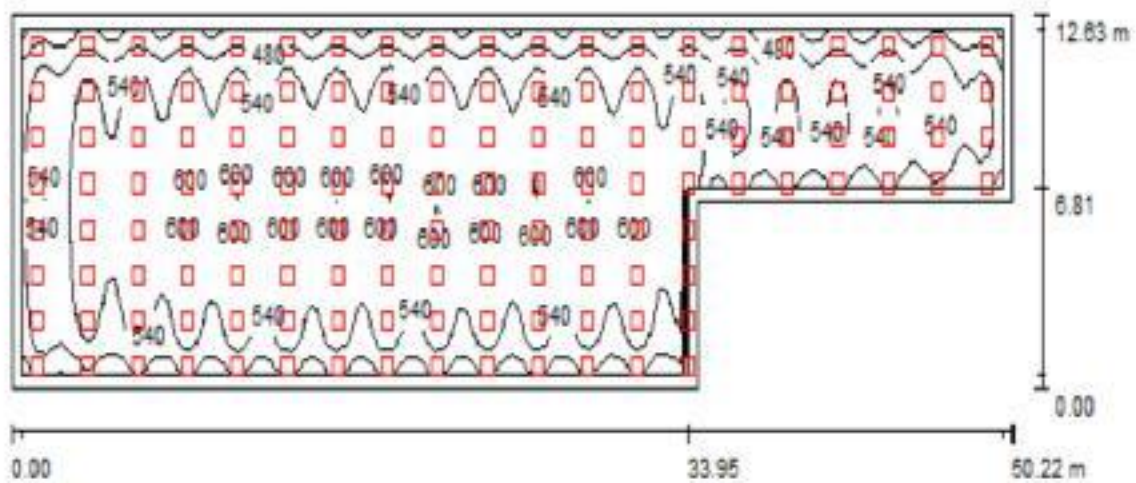


Height of Room: 2.600 m, Mounting Height: 2.722 m,
Light loss factor: 0.80

Values in Lux, Scale 1:133

Surface	ρ [%]	E_{av} [lx]	E_{min} [lx]	E_{max} [lx]	$u0$
Workplane	/	140	57	236	0.406
Floor	20	117	42	167	0.357
Ceiling	70	21	14	25	0.662
Walls (4)	50	37	13	74	/

Main kitchen



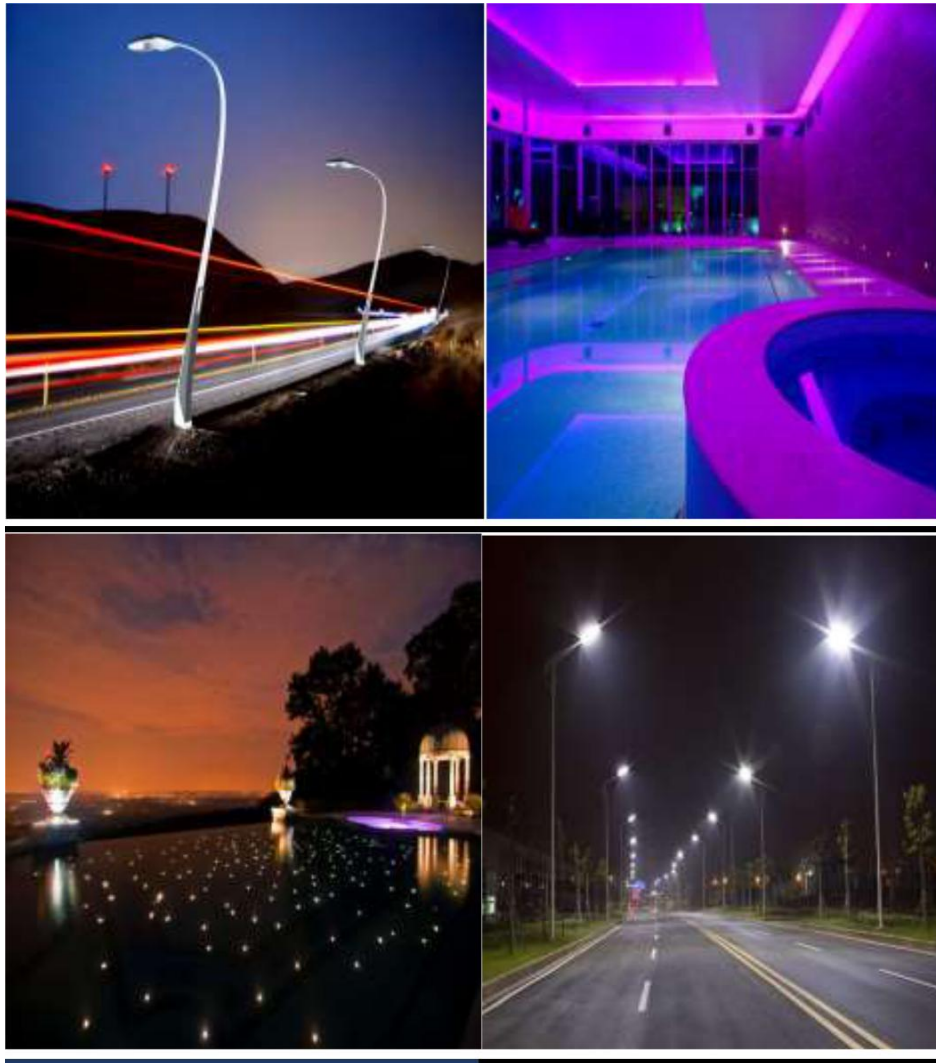
Height of Room: 2.600 m, Mounting Height: 2.676 m, Light loss factor: 0.80

Values in Lux, Scale 1:360

Surface	ρ [%]	E_{av} [lx]	E_{min} [lx]	E_{max} [lx]	$u0$
Workplane	/	535	342	617	0.639
Floor	20	490	253	574	0.517
Ceiling	70	111	90	198	0.816
Walls (7)	50	292	129	823	/

CHAPTER 2

OUTDOOR LIGHTING



Street lighting

The difference between Noor and Light

ارون رمقلا و اءايض سمشلا انلعجو " :بلاعت لاق"

Light is the rays which emitted from the light source like (sun and lamp)
While the Noor is reflected rays from body or thing and it is not the light source like (moon and mirror).

Introduction

This chapter specifies the general requirements for lighting columns, mounting height, arrangement, spacing, lighting sources and distribution box of street lighting. It also covers the type of cable and feeder lengths recommended for the street lighting distribution.

Definitions

➤ Arrangement

The pattern according to which luminaires are sited in plan for instance staggered, opposite, single side and twin central.

➤ Glare

The Condition of vision in which there is discomfort or reduction of ability to see, or both, caused by an unsuitable distribution or range of luminance.

➤ Luminance (B) = I/A

The physical quantity corresponding to the brightness of a surface (e.g. a lamp, luminaire, sky or reflecting material) in a specified direction. It is the luminous intensity of the surface divided by that area, measured in units of cd/ m².

➤ Projection

The distance measured horizontally from the centre of a bracket-mounted luminaire, to the centre of the column or pole.

➤ Overhang

The distance measured horizontally between the photometric centre of a luminaire and the adjacent edge of a carriageway.

➤ Set back

The shortest distance from the forward face of a column to the edge of a carriageway.

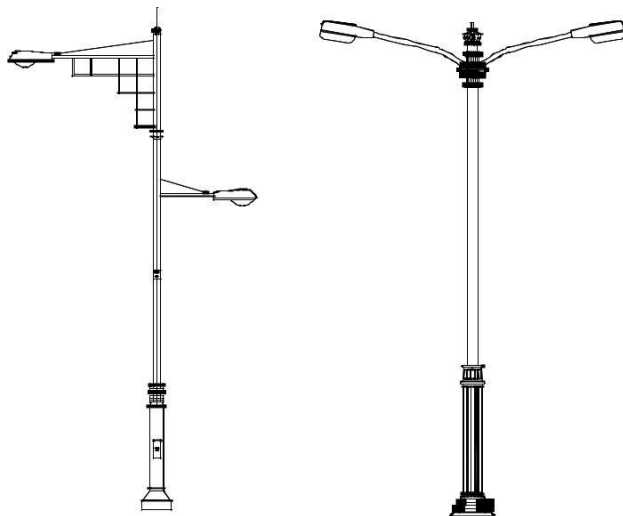
➤ Tilt angle

The angle by which the axis of the fixing spigot entry is tilted above the horizontal when the luminaire is installed.

Lighting Columns

➤ Typical forms of columns

Figure (1) shows the exterior appearances of typical street lighting with one and two arms.



(a) One arm

(b) Two arms

[Figure 1] Typical forms of street lighting

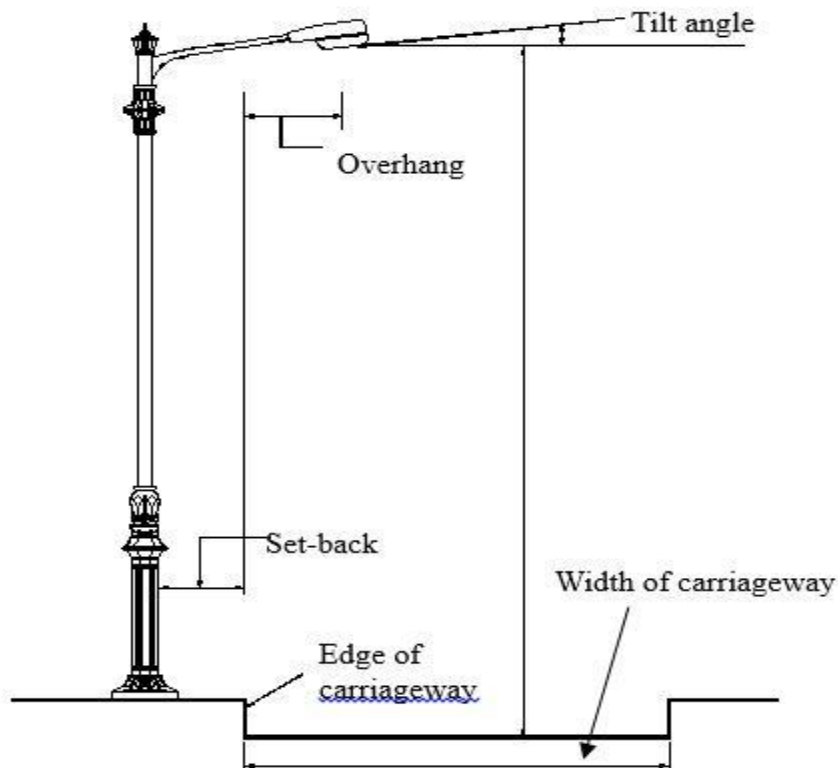
➤ Locations of columns

A minimum clearance of 800 mm. The minimum clearances according to the design speed of the road are given below.

Design speed (km/h)	Clearance (m)
50	0.8
80	Absolute minimum 1.0 Recommended minimum 1.5
100	Absolute minimum 1.5
120	Absolute minimum 1.5

Lighting Columns

➤ General



[Geometry in elevation of street lighting]

When a mounting height is generally higher, a glare is decreasing and overall quality of surrounding circumstances is improving, but a capital cost is higher and overall efficiency is lower. The same mounting height shall be kept not only in the same width of roadways, but also at road junctions. As far as possible. Therefore, before a mounting height is determined, luminance distribution, overall lighting effect and economic efficacy shall be considered and compared.

6 M mounting height or less than

6 M or less than shall be installed at some locations where the majority of side streets, public gardens and parking lots to make people feel safe

8 M mounting height.

8 M shall be installed at some locations where the urban traffic route where, owing to the multiplicity of road junctions, no reduction in the number of lighting points is to be gained by the use of a higher mounting height, or relatively narrow roads such as local distributor and access roads to residential areas in which a mounting height between 10 M or 12 M and 5 M or 6 M is desirable.

10 M mounting height

10 M shall be installed at some locations where the majority of urban traffic routes, or lighting for traffic routes in which a mounting height of 12 M would be an undesirable appearance, or lighting for wide heavily used routes where a large number of intersection, bends, etc, could lead to a short spacing making the use of 12 M mounting height uneconomical.

12 M mounting height

12 M shall be installed at the location of wide or heavily used routes where advantage can be taken of a longer spacing of luminaires.

18M mounting height or more than

High mast lighting poles shall be installed at large-scale area such as airports, dockyards, large industrial areas, sports areas and road intersections.

Lighting Arrangement

Methods of pole arrangement according to the no. of lanes in the road.

<i>Numbers of lane</i>	<i>2 ~ 3 lanes</i>	<i>Over 4 lanes</i>
<i>Arrangement</i>	<i>Staggered or single side</i>	<i>Opposite or twin central</i>

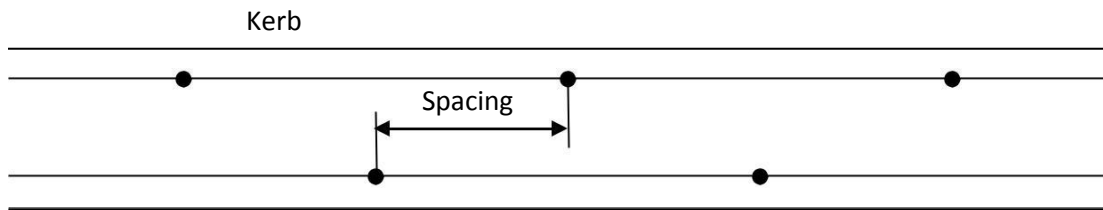
- *Single side arrangement ($W \leq h$).*
- *Opposite arrangement ($W = 3/2h$).*
- *Twin central arrangement on dual roadways ($W > h$).*

Where $\rightarrow$ W = width of road. & h = height of column

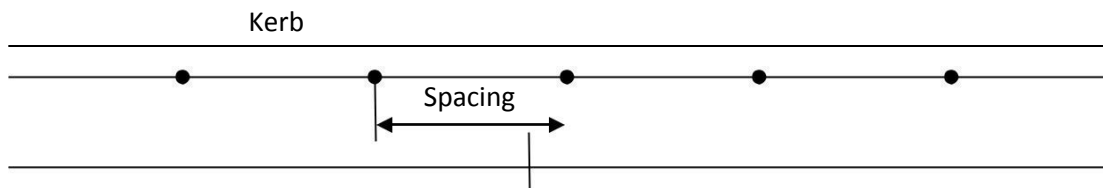


For straight streets.

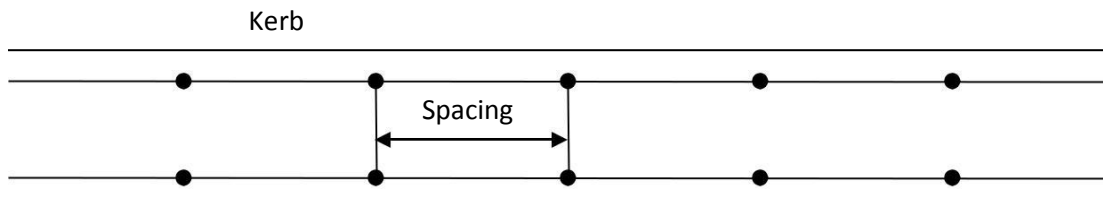
• **Staggered arrangement.**



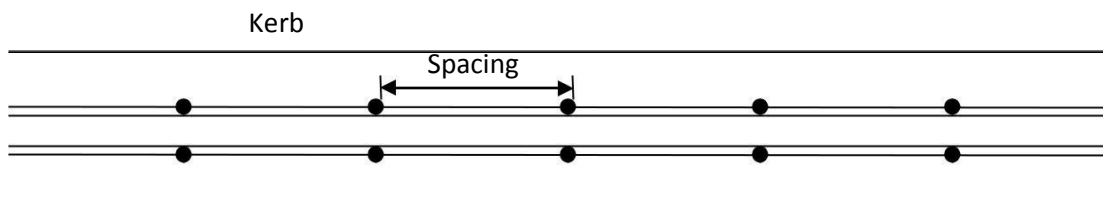
• **Single sided arrangement.**



• **Opposite arrangement.**



• **Twins central**

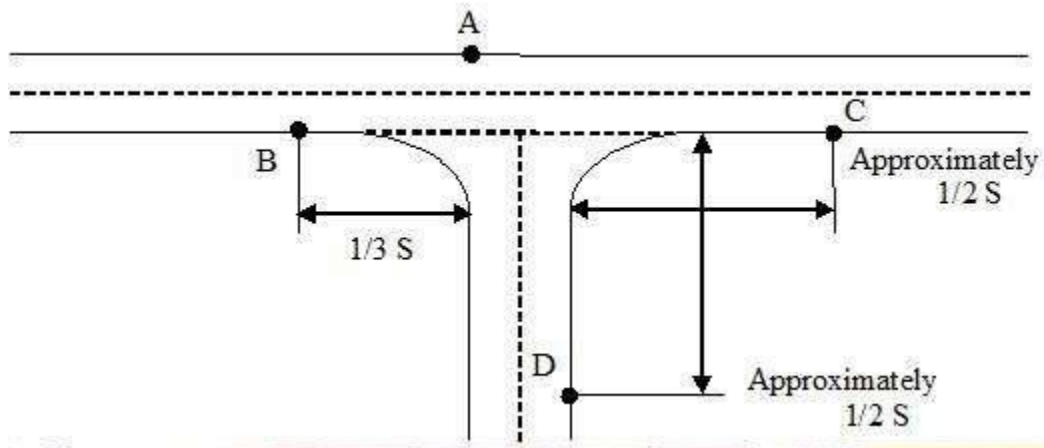


Single-level road junctions

The lighting at a junction shall be related to the need of drivers approaching the junction to see vehicles approaching from other directions. The illuminance, on the carriageway, throughout a junction shall be not less than the illuminance provided on the main roads leading to the junction

• T junction

To achieve the required luminance on the T junction, typical lighting positions shall be as shown in figure.

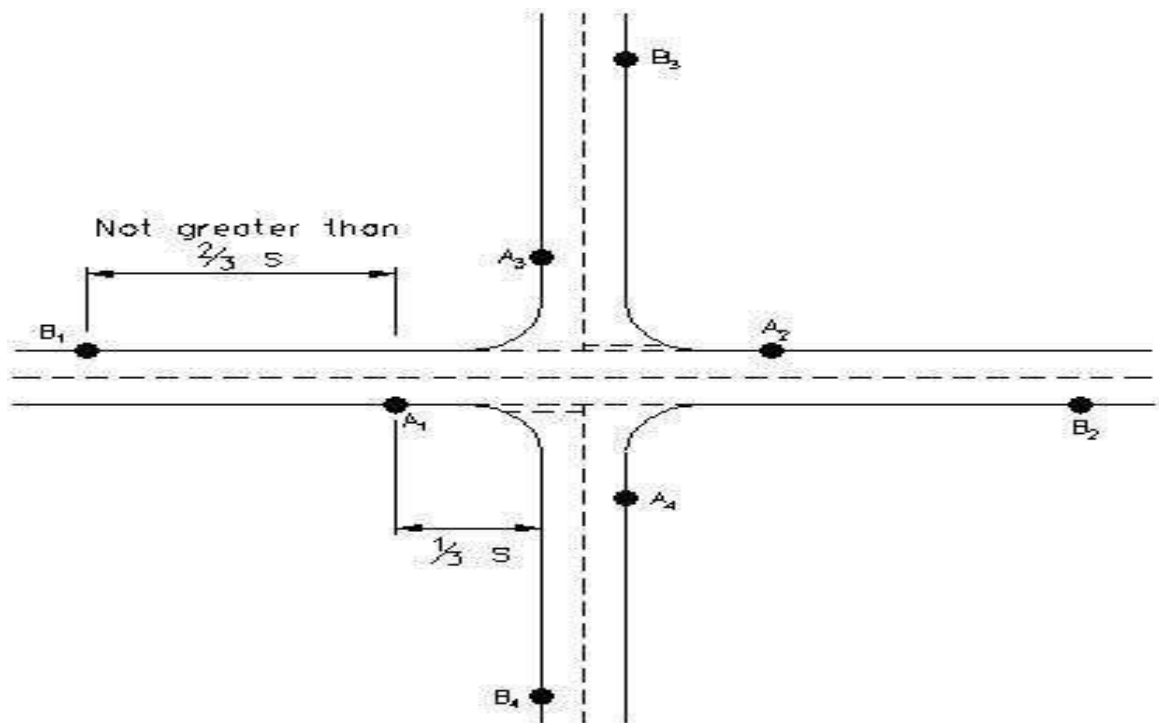


Luminaire A on the major road shall be sited in line with centre line of the approach lanes of the minor road. Luminaire B on the major road shall be sited at a distance from the kerb of the minor road approximately equal to one-third of the design spacing for the major road and preferably not greater than 12m along the major road. Luminaire C on the major road shall be sited at a distance from the kerb of the minor road of about one-half of the same design spacing along the major road. Luminaire D on the minor road shall be sited at a distance from the kerb of the major road of about one-half of the design spacing for the major road.

- Cross roads

Typical light positions for a cross roads are shown below. On each road, there shall be a luminaire A on the nearside for traffic leaving the junction at a distance from the kerb of the cross roads approximately equal to one-third of the design spacing for the major road. These four luminaires shall be of the same type and mounted at the same height.

The next luminaire B, on each road, on the offside for traffic leaving the junction, shall be at not more than two-thirds of the design spacing for the major road, from luminaire A.



- Bridges

When street lighting is installed at bridges The appearance of opposite arrangements is preferred particularly as this may make possible a lower mounting height. A central arrangement appears as a regularly spaced array of columns from any viewpoint and

needs fewer columns than an opposite arrangement. All luminaires shall be at the same mounting height at bridges.

Types of lamps used in street lightning

We have to choose suitable kinds of lamps for different streets. The lamp must be convenient for vehicles and pedestrians.

1- High pressure sodium



- *high efficiency (has an initial efficiency in excess of 100 lumen per watt). good color rendition.*
- *Median lamp life is 6000 hours (may be improved with improved construction techniques). •*
- *Used in high ways . •*
- *Its light is yellow and its light is suitable for such kinds of lightning even cloudy weather •*

2-Low pressure sodium lamps:



- *give very strong light under small power •*
- *has dark yellow light*
- *have relatively long life. •*
- *This type of lamps used in tunnels and closed public places •*

3- Metal halide lamps:-



- turning off quickly and turning on slowly.
- this type of lamps used in medical operation room and flood lighting,

4- Mercury lamps:-

- There are several types of Mercury lamps such as:-
- High pressure, Low pressure and Compound mercury lamps.
- This type has special

Protection

➤ Earthing and fuses

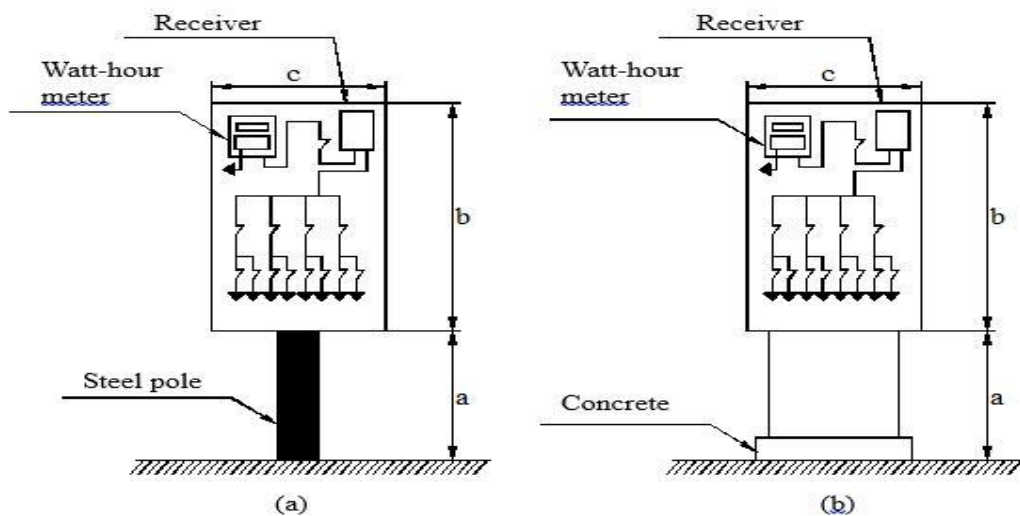
All metal parts of columns shall be earthed , The grounding wire shall be a minimum size of 22mm² PVC insulated wires. The earth shall be terminated on the body of the pole by drilling and affixing with a suitable galvanized nut. The rated current of a fuse shall be at least twice the normal full load current. The ampere rating of the fuse which is installed in each column shall be not less than a 10 Amp Fuse to ensure protection against electric shock and over current.

• Distribution Box

Locations of distribution boxes

Distribution boxes shall be of sheet steel construction, galvanized after fabrication, of weatherproof construction and shall be large enough to afford easy access to the

control equipment. They shall be provided with a concrete foundation, be free standing with necessary ducts for incoming and outgoing cables and Distribution boxes shall not be sited in indentations in landscaping features where water is likely to collect, or in position vulnerable to impact from vehicles and shall normally be positioned a minimum of 800 mm from the edge of carriageways. Ground electrodes shall be installed at every distribution box.



The bottom of the door (a) shall be located between 700 and 1200 mm above finished ground level and the door (bc) shall be a minimum of 900 mm high $\times$ 600 mm wide. The size of distribution boxes can be changed from some situations. The door shall be prevented from being opened by unauthorized persons. A ground electrode shall be installed at every distribution box.

- Circuit breakers

MCCB(molded case circuit breaker) which is no need to replace a broken fuse, shall be used as a primary and secondary circuit, and the rated current of primary MCCB shall be not less than 100 A. When such a primary MCCB is installed, the total

number of lamps in a distribution box shall not exceed 60 lamps for 400 W or 102 lamps for 250 W.

Minimum rated capacity

<i>Secondary CB (ELB)</i>	<i>Primary CB (MCCB)</i>	<i>Category</i>
<i>30 A</i>	<i>100 A</i>	<i>Rated current</i>

Generally the rated current of secondary ELB (Earth Leakage Breaker) shall not be less than 30 A. When such a secondary breaker is installed, the number of lamps within one ELB shall not exceed 10 lamps for 400 W or 17 lamps for 250 W.

Maximum number of lamps

<i>Maximum number of lamps (EA)</i>		<i>Category</i>
<i>400W lamp</i>	<i>250W lamp</i>	
<i>60</i>	<i>102</i>	<i>Main breaker</i>
<i>10</i>	<i>17</i>	<i>Secondary breaker</i>

The rated capacity of magnetic contactors shall be not less than 30A. When the number of street lighting increases within one distribution box, it shall not replace the existing MC with a new MC which has much higher rated current than the old one. The number of branch circuits per distribution box shall not exceedsix.the distribution box shall have expansion space for breakers or magnetic contactors when they need to be installed.

➤ **Size and installation of cables**

The incoming cable for the main breaker of distribution boxes shall be a minimum size of 3.5C ×35 mm² XLPE insulated PVC sheathed copper cable or equivalent to meet the load and the outgoing cable for a group of streetlights shall be a minimum size of 2C ×16 mm² XLPE insulated PVC sheathed copper cable or equivalent. The cable for internal wiring of columns shall be a minimum size of 2C × 2.5 mm² PVC insulated copper cable. Street lighting loads shall be balanced over the

three phases. Therefore, lighting designers shall indicate on the street lighting plan the phase to which each luminaire is to be connected.

Minimum size of cables

<i>Internal wiring</i>	<i>Outgoing cable</i>	<i>Incoming cable</i>	<i>Category</i>
<i>2.5 mm²</i>	<i>16 mm²</i>	<i>35 mm²</i>	<i>Cable size</i>

Maximum cable length from the distribution box to the last street light shall not exceed 600 m to ensure voltage drops.

Design of street lighting

They are a few steps to design street lighting

The method according to the CIE only to know the lighting requirements for example

Step no. 1 → specify the type of street according to

1. Typical speed of main user type.
2. Main user type.
3. Other allowed user type.
4. Excluded user type.

The user type may be:-

- Motorized traffic.
- Slow moving vehicles
- Cyclists
- Pedestrians.

Typical speed of main user type	User types in the same relevant area			Set of situations
	Main user type	Other allowed user type	Excluded user type	
High >60 km/h	M		SCP	A1
		S	CP	A2
		SCP	-	A3
Moderate 30 km/h - 60km/h	MS	CP	-	B1
	M S C	P	-	B2
	C	P	M S	C1
Low 5 km/h - 30 km/h	C	P	M S	C1
	M P	-	S C	D1
		S C	-	D2
	M C	S P	-	D3
	M S C P	-	-	D4
Very low Walking speed	M S C P			D4
	P	-	M S C	E1
		M S C	-	E2

M: Motorized traffic
S: Slow moving vehicles
C: Cyclists
P: Pedestrians

For example from our project.

- ✓ A street of speed more than 60 km/hour.
- ✓ The main user for this street is motorized traffic

✓

The excluded user types are pedestrians ,slow moving vehicles and cyclists

So according to the CIE standard from the above table the type of the street is A1

Step no. 2 →determine the status of lightning

This can be done by the following steps.

- 1- Determine the status of public weather (dry or wet)
- 2- Determine if there is separation in the lanes in the carriageway
- 3- Determine the type of junctions.
- 4- Determine the number of cars (usually when you aren't provided with sufficient information , take the extreme case 25000 Car per day.)

After specifying the conditions in the previous page we

We can easily know the status of of lighting from the table below which is concerned with the type 1 of street

Main weather type	Separation of carriageways	Type of junctions		Traffic flow Amount of vehicles								
		Interchanges	Intersections									
		Interchanges spacing Distance between bridges	Intersection density	< 15000			15000 bis 25000			> 25000		
												
Dry	Yes	> 3 km		ME5	ME4a	ME3a	ME4a	ME3a	ME2	ME4a	ME3a	ME2
		<= 3km		ME4a	ME3a	ME2	ME4a	ME3a	ME2	ME3a	ME2	ME1
		< 3 intersections /km	ME5	ME4a	ME3a	ME5	ME4a	ME3a	ME4a	ME3a	ME2	
			>=3 intersections /km	ME4a	ME4a	ME3a	ME4a	ME3a	ME2	ME3a	ME2	ME1
	No	<3 km		ME4a	ME4a	ME3a	ME4a	ME3a	ME2	ME3a	ME2	ME1
		>=3 km		ME3a	ME2	ME1	ME3a	ME2	ME1	ME2	ME2	ME1
		<3 intersections /km	ME4a	ME4a	ME3a	ME4a	ME3a	ME2	ME3a	ME2	ME1	
			>=3 intersections /km	ME4a	ME3a	ME2	ME4a	ME3a	ME2	ME3a	ME2	ME1
		Wet			Choice as above, but select MEW Classes							

After we specify the status of lighting we go to the next step

Step no. 3 ➔ Third:-Knowledge requirements of lighting condition

1) Luminance (B) = $I/Acd/ m^2$

2) U_0 (min value/average value of lightning on (street))

3) U_1 (min value/average value of lightning on (lane))

4) T_1 (Value of glow lighting of devices)

5) SR (lightning of surroundings)

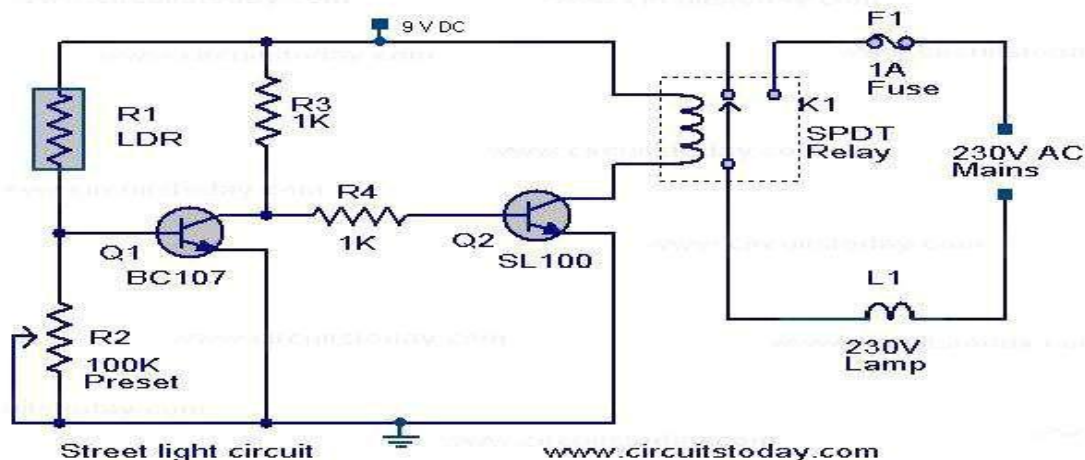
➤ Street lighting control

There are three ways of lighting control methods such as:-

- 1- timer control
- 2- photocell control
- 3- Wireless remote control.

Most of main streets shall be controlled by the wireless remote control system which consists of a central controller and a receiver installed at a distribution box, if necessary a pole receiver. In case of controlling the range of streetlights, for instances by each branch circuit or column, the way of controlling each branch circuit must be more economical, suitable than controlling each column. In rural areas, a timer control or photocell control shall be applied as a control way of street lighting.

The circuit diagram



The circuit diagram present here is that of a street light that automatically switches ON when the night falls and turns OFF when the sun rises. In fact you can this circuit for implementing any type of automatic night light. The circuit uses a Light Dependent Resistor (LDR) to sense the light .When there is light the resistance of LDR will be low. So the voltage drop across R2 will be high. This keeps the transistor Q1 ON. The collector of Q1 is coupled to base of Q2 So Q2 will be OFF and so do the relay. The bulb will remain OFF. When night falls the resistance of LDR increases to make the voltage across the R2 to decrease below 0.6V. This makes transistor Q1 OFF which in turn makes Q2 ON. The relay will be energized and the bulb will glow.

Design steps using dialux

Step no 1 → Determine the street specifications

DIALux Street Light Wizard

Valuation Fields and Lighting Classes

Enter the various or combined valuation fields for the streets. Select a lighting class for each valuation field in order to define the photometric requirements of the street.

☒ Use a single valuation field for all elements.

Lighting Class for this Valuation Field: ME4a Selection

Separate Valuation Fields:	Common Valuation Fields:	Lighting Classes for separate Valuation Fields:	Lighting Classes for common Valuation Fields:
☐ Sidewalk 2	☐	S1 Selection	S1 Selection
☐ Bicycle Lane	☐	S1 Selection	S1 Selection
☐ Roadway 2	☐	ME1 Selection	ME1 Selection
☐ Median	☐	CE2 Selection	ME1 Selection
☐ Roadway 1	☐	ME1 Selection	ME1 Selection
☐ Bicycle Lane	☐	S1 Selection	S1 Selection
☐ Sidewalk 1	☐	S1 Selection	S1 Selection

[< Back](#) [Next >](#) [Cancel](#)

Step no 2 → specify the street lighting status from the tables

DIALux Street Light Wizard

Valuation Field

Select a valuation field for the optimisation.

The following valuation fields are available as basis for the optimisation:

Relevant area for all elements (ME4a)

The possible targets for the optimisation are defined by means of the illumination class of the valuation field.

Parameter	Threshold value	J...	Foundation
☒ Lm	0.75	...	(Luminance)
☒ U0	0.40		(Luminance)
☒ U1	0.60		(Luminance)
☒ TI	15	%	(Luminance)
☒ SR	0.50		(Illuminance)

Select all parameters that have to be taken into account for the optimisation.
Adapt the limit values of the illumination class to the previously selected valuation field, if necessary.

< Back

Next >

Cancel

Step no 3 → **choose the luminaire from Philips catalogue**

DIALux Street Light Wizard

Luminaire Selection

Select a luminaire for the arrangement from your favourites or search the installed database for a suitable luminaire.

Recently used Luminaires

Select >

< Delete

 Catalogues

Selected luminaires

1. Philips SPP202 GB 1xSON-TPP70W SGR OR

Currently selected luminaire

Luminaire: Philips SPP202 GB 1xSON-TPP70W SGR OR

Luminous flux of luminaire: 5412 lm

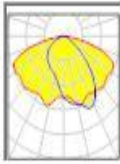
Wattage of luminaire: 80 W

Luminous emittance 1

Lamp: SON-TPP70W/220

Luminous 6600 lm

Wattage: 80 W





< Back

Next >

Cancel

50

Step no 5 ➔ specify the poles specification like pole height , the pole distance and the street arrangement , slope angle

DIALux Street Light Wizard

Variable arrangement parameters

Specify which parameters of the luminaire arrangement are allowed to vary at which intervals. Alternatively edit the values of the fixed parameters.

Parameters that may be varied for the optimisation:

Parameter	Minimum	Maximum	Increment	J...
☒ Pole Distance	10.000	50.000	1.000	m
☐ Height				m
☐ Light overhang				m
☐ Slope angle				°

Number of combinations to investigate: 41

Fixed parameters for the optimisation

Pole Distance: 20.000 m

Height: 8.000 m

Light overhang: 0.500 m

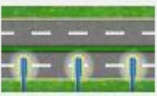
Inclination: 10 °

Distance Pole to Roadway: 0.800 m

Boom Length: 0.000 m

Arrangement Type

Single row, bottom



☒ Fixed distance

☐ Fixed Boom Length

< Back

Next >

Cancel

Step no 6 → the output of the dialux program

X

DIALux Street Light Wizard

Positioning suggestions

Select a suggested luminaire arrangement for implementation.

Classification of the arrangement variants:

Suitable: 13, largely suitable: 0, inadequate d: 28

Distance [m]	Height [m]	Overhan [m]	Slope angle [°]	Type	Lm [cd/m²]	U0	UI	TI [%]	SR
☐ Suitable variants (all target values met)									
22.000	8.000	0.500	10	(1)	0.77	0.67	0.86	8	0.59
21.000	8.000	0.500	10	(1)	0.80	0.67	0.87	8	0.59
20.000	8.000	0.500	10	(1)	0.84	0.67	0.89	8	0.59
19.000	8.000	0.500	10	(1)	0.89	0.67	0.89	8	0.59
18.000	8.000	0.500	10	(1)	0.94	0.68	0.89	7	0.59
17.000	8.000	0.500	10	(1)	0.99	0.68	0.90	7	0.59
				Target ...	0.75	0.40	0.60	15	0.50

Philips SPP202 GB 1xSON-TPP70W SGR OR (SON-TPP70W/220)

If you want to implement a positioning suggestion, select it in the list and subsequently complete the wizard. Else simply cancel the wizard.

☐ Save set values for further optimisations.



CSV Export

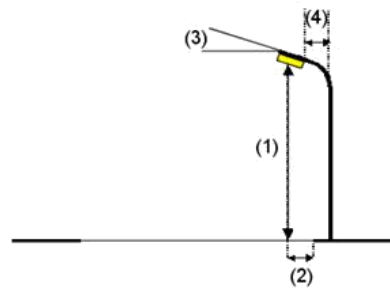
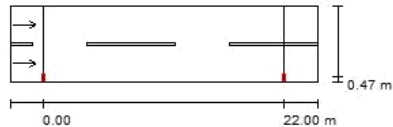
< Back
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Cancel

Street Profile

Roadway 1

Light loss factor: 0.80

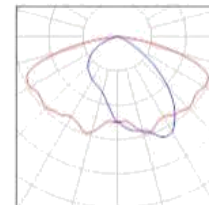
Luminaire Arrangements



Luminaire:	Philips SPP202 GB 1xSON-TPP70W SGR OR	
Luminous flux (Luminaire):	5412 lm	Maximum luminous intensities
Luminous flux (Lamps):	6600 lm	
Luminaire Wattage:	80.0 W	
Arrangement:	Single row, bottom	
Pole Distance:	22.000 m	
Mounting Height (1):	8.177 m	
Height:	8.000 m	
Overhang (2):	0.500 m	
Boom Angle (3):	10.0 °	
Boom Length (4):	1.269 m	

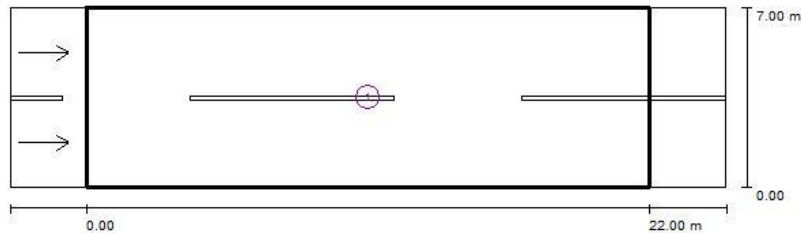
Street 3 / Luminaire parts list

Philips SPP202 GB 1xSON-TPP70W SGR OR
Article No.:
Luminous flux (Luminaire): 5412 lm
Luminous flux (Lamps): 6600 lm
Luminaire Wattage: 80.0 W
Luminaire classification according to CIE: 100
CIE flux code: 40 74 96 100 82
Fitting: 1 x SON-TPP70W/220 (Correction Factor 1.000).



The figure below shows that the poles arrangement of the pole and they met the lighting requirements

Street 3 / Photometric Results



Light loss factor: 0.80

Scale 1:201

Calculation Field List

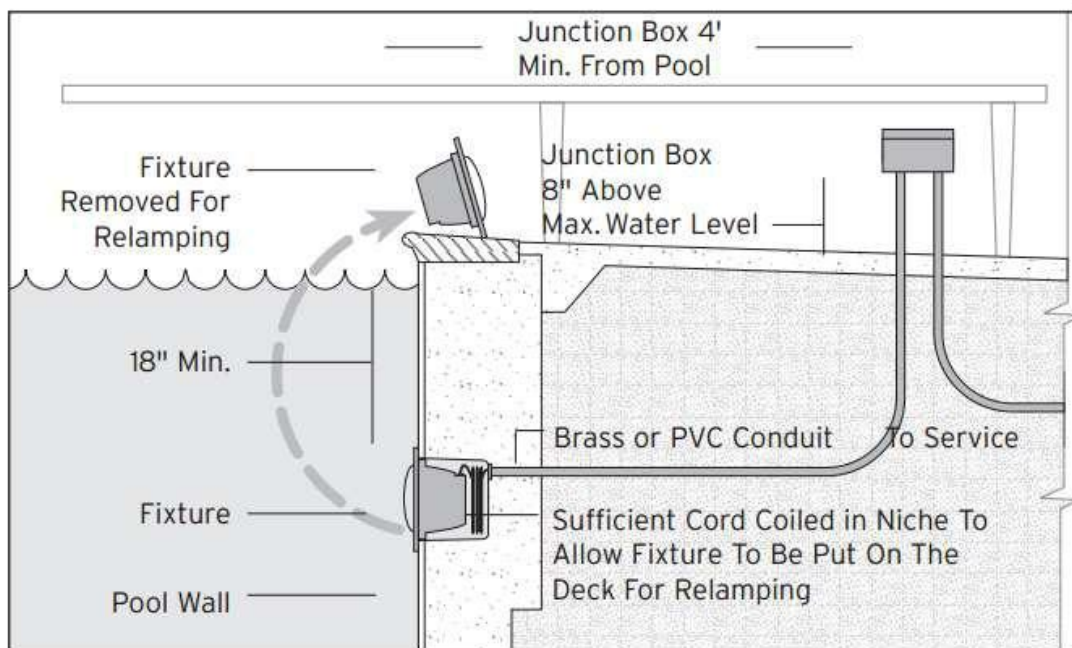
- 1 Valuation Field Roadway 1
Length: 22.000 m, Width: 7.000 m
Grid: 10 x 6 Points
Accompanying Street Elements: Roadway 1.
tarmac: R3, q0: 0.070
Selected Lighting Class: ME4a

(All lighting performance requirements are met.)

	L_{av} [cd/m²]	U0	UI	TI [%]	SR
Calculated values:	0.77	0.67	0.86	8	0.59
Required values according to class:	≥ 0.75	≥ 0.40	≥ 0.60	≤ 15	≥ 0.50
Fulfilled/Not fulfilled:	✓	✓	✓	✓	✓

Lighting Design of a Swimming Pool

The N.E.C. requires that fixtures must be installed so that the top of the fixture is at least 18' below normal water level. The junction box shall be located at least 8" above maximum water level, pool deck, or ground, whichever is greater. The junction box shall be located not less than 4' from the perimeter of the pool, unless separated from the pool by a fence, wall or other permanent barrier.

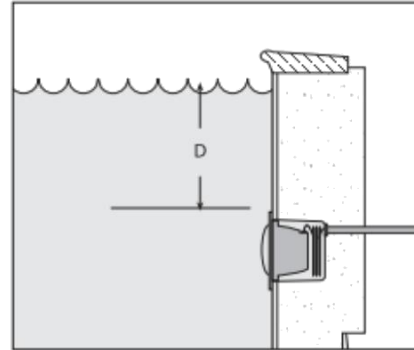
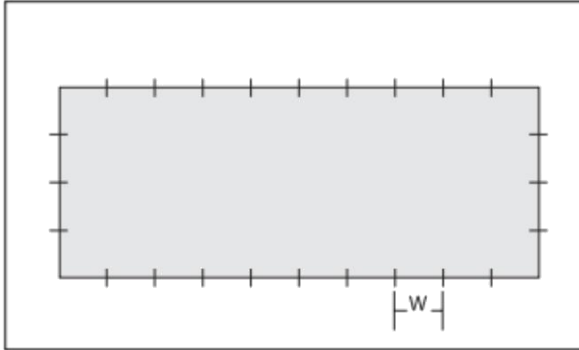


Recommended Lighting Design

The following information will be useful in determining the lighting requirements for swimming pools.

Step no. 1 → *Establish the lamp wattage and number of fixtures per square feet required based on the size of the pool and spacing as shown on the chart below.*

OUTDOOR* POOL	LAMP WATTS PER SQ. FT.	"W" = MAX. FEET		"D" = DEPTH BELOW WATER
		POOL DEPTH OVER 5 FT.	POOL DEPTH UNDER 5 FT.	
RESIDENTIAL	0.5 TO 1	--	--	18"
COMMERCIAL	1 TO 15	16'	18'	18"
COMPETITION	1.5 TO 2**	14'	16'	18"
DIVING POOL	2 TO 3	14'	--	36"



Swimming pool

$$\frac{\text{Pool Area} \times \text{Watts per Sq. Ft}}{\text{Number of Fixtures}} = \text{Watts per Fixture}$$

Step no. 2 ➔ Establish the approximate fixture wattage using the following formula:

Step no. 3 ➔ Select the fixture that comes closest to meeting wattage requirements
For in-between wattages, use next highest wattage or increase the number of fixtures.

In our project , the table below shows the selected luminaire.

POOL 1	Luminaire Module	No of luminaires	Area (M ²)	Distance
Left region	4425 SWM	30	817	5
Middle region	4425 SWM	15	415	5
Right region	4425 SWM	10	281	7

CHAPTER: 3

ACCESSORIES



Accessories

Sockets

In this chapter we will discuss AC sockets

Chapter contents:

1. *Types and uses of sockets*
2. *Types of switches*
3. *Rules of socket distribution*
4. *types of switches*
5. *contactor*

Types of sockets:

1. *Single socket*
2. *Double socket*
3. *Power socket*
4. *Three phase socket*
5. *UPS socket*
6. *Weather proof socket*

Single socket



That can be described according to two standards

- [According to Egyptian Code \(EC\):](#)

V=250 volts, I=10, 13 or 16 Amperes with 250 VA

- [International Electro technical Commission \(IEC\):-](#)

V=250 volts, I=10, 13 or 16 Amperes with 180 VA

1) Double socket



Also can be described according to two standards

- [According to the Egyptian Code \(EC\):](#)

V=250 volts, I=10, 13 or 16 Amperes with 500 VA

- [According to International Electro technical. Commission \(IEC\):](#)

V=250 volts, I=10, 13 or 16 Amperes with 360

VA These two types of sockets are generally used

Power socket



With standard

V=250 volts, I=20 or 30 Amperes with power range from 1000 VA to 2000 VA

Applications: -

Power sockets are used in places where motors can be found as:

- *Kitchens*
- *Bath rooms (heaters, hand dryers)*

Three phase socket



With standard

$v=400$ volts, $I=16, 20, 30\ldots\ldots\ldots 100$ Amperes (depend on load)

Applications:-

This type is used with three phase loads like welding and drilling machines in factories



UPS socket

$V=250$ volts, $I=10, 13$ or 16 Amperes with 250 VA

This type is used for computer loads

Weather proof socket

It is characterized with high IP to withstand water and dust.



Can be described according two standards

- According to the Egyptian Code (EC):

V=250 volts, I=10, 13 or 16 Amperes with 250 VA

- International Electro technical commission (IEC):

V=250 volts, I=10, 13 or 16 Amperes with 180

VA It is used in:

1. Kitchens
2. Bath room
3. Outdoor sockets
4. Corridors

- Sockets type form control

1. Unswitched socket

Contacts permanently connected to wiring therefore the appliance to connected is turned on as soon as the plug is pushed into the socket.

2. switched socket:

A switch is added in the socket outlet and has an advantage of avoiding the arc may be occur by making the socket is not switched on until after the plug has been pushed in.

Socket mounting:

It is divided in three types:

1. Wall mounting

*On 30 cm from floor if there is not furniture.
On 120 cm from floor if there is furniture.*

2. Floor mounting

It must have very high IP to withstand dust and water

3. Furniture mounting

Installation is refer to furniture position

Distribution of sockets:

- 1. According to furniture demand*
- 2. According to codes*

It is divided into two places

- **Inside rooms.**

Distance between two sockets is 4 meters

- **. In corridors**

Distance between two sockets is 6 meters

Distribution board and wiring

According to the code any line carry 2000VA and to calculate number of sockets per line :

According IEC:

Number of sockets per line=2000/200=10 sockets

According EC:

Number of sockets per line = 2000/250 = 8 sockets

So, we take (8:10 sockets in one line)

For double sockets (4:5 sockets in one line)

For UPS sockets must take one socket per line.

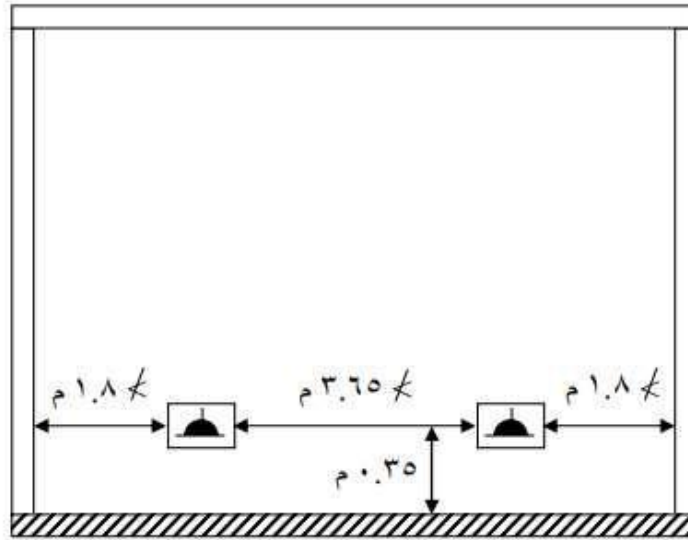
Also three phase sockets must be taken in one line

For power socket: 2000/2000 = 1 socket per line

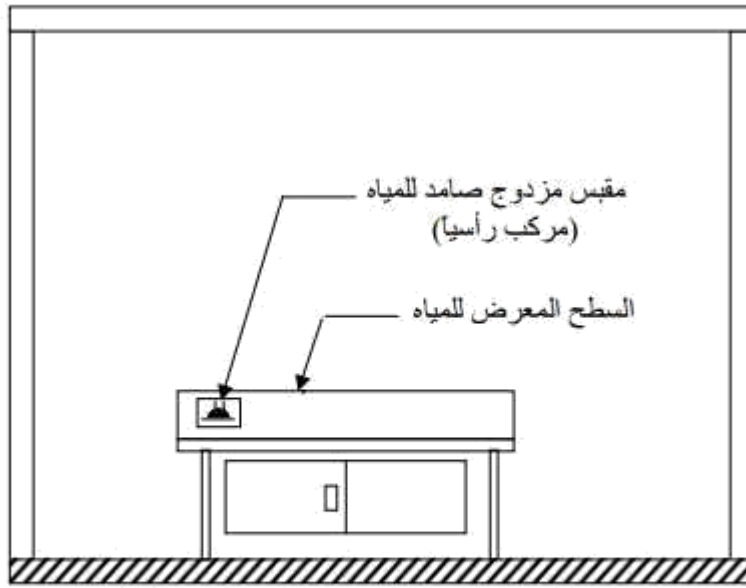
➤ Some notes from the Egyptian Electric code.

- *If the sockets are subjected to the mechanical damage so we must cover it with a metal case and this case should be earthed*
- *When a different level of voltage of sockets every socket should be with a different shape and color.*
- *If a number of sockets is used in a room with area of 50 M² so they must be fed from one phase .*
- *If the rooms with area greater than 50 M² the sockets should be in a different phases and every phase should supply a different separate area and every socket case should be with a specific color.*
- *When sockets are used on both sides of a wall , the horizontal distance between them must be at least 150 mm to avoid sound transmission across it.*

- The sockets should be installed at a height between 30 and 40 cm over the finished floor at office and residential areas except the kitchens and bathrooms the sockets should be installed at a height of 120 cm to 135 cm.
- The distance between the sockets and the vertical Culum shouldn't exceed 180 cm. and the distance between the sockets and the other shouldn't exceed 3.65 M.
- Sockets shouldn't be installed on the surface of the labs. To avoid the dust and moisture from leakage to the sockets.



شكل رقم (٢-٤): أقصى مسافة بين المقابس وبعضها البعض وبين المقابس والحائط المجاور



شكل رقم (٢-٥): وضع المقاييس بترابيزات المعامل

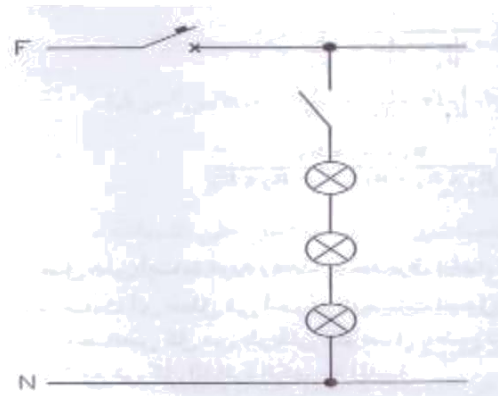
Types of switches:

1. *Single pole switch*
2. *It opens only the phase circuit*
3. *Used for switching the light circuit*

Types of single pole switch:

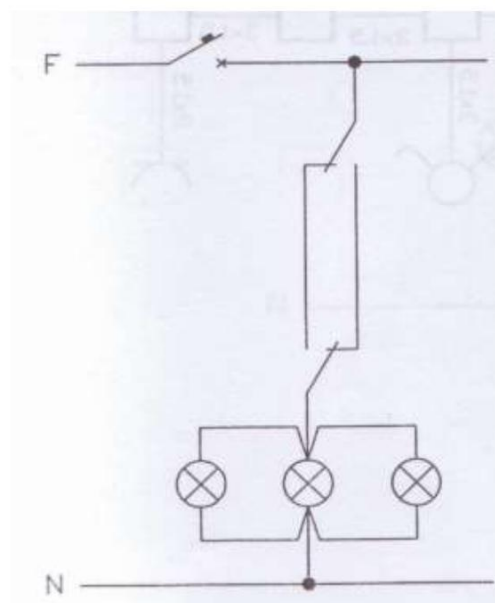
1. *One way one gang switch*

The single pole switch is the general purpose workhorse of switches. Single pole switches turn a light, receptacle or device on and off from a single location. A characteristic of a single pole toggle switch is that it has an on and off marking on the switch



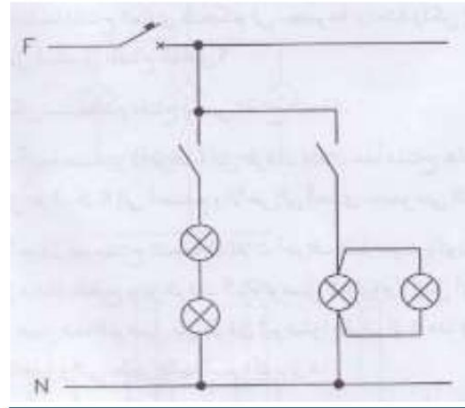
2. Two way one gang switch

To control on only one group of loads (lamps) from two places.



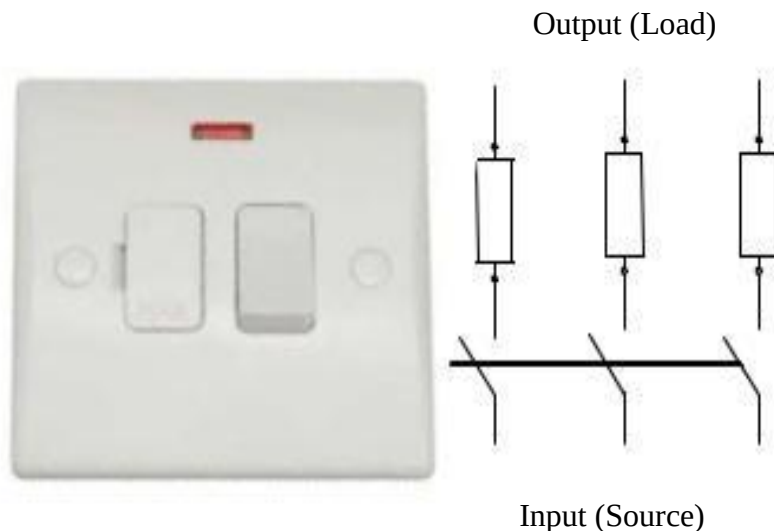
3. one way two gang switch

The double pole switch also has "on" and "off" markings and functions similar to a single pole switch in that it turns something on and off from one location



Switch fuse

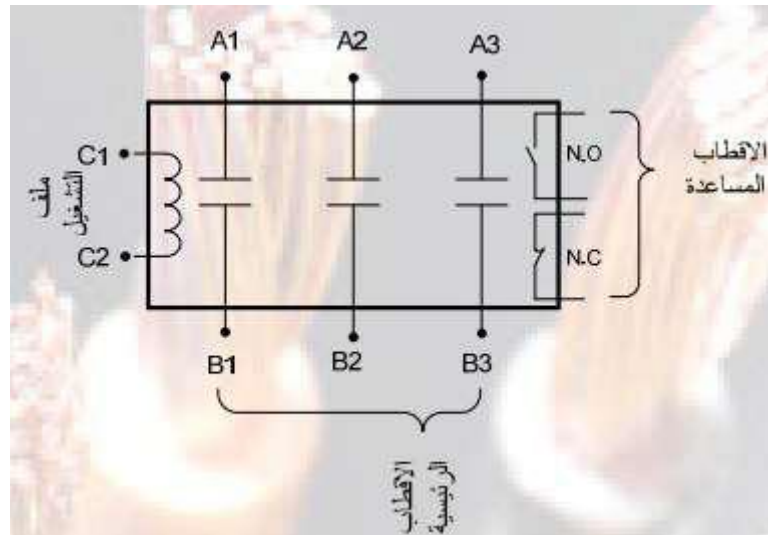
It's a switch is with a fuse is connected in series with the fuse to protect the load. As shown in the figure the fuse isn't responsible for conduction it's only for protection.



It is used with the heavy loads like air conditioning devices, heaters and ovens.

Contactors

The contactor is considered the main element in the control circuits. It consists of main poles able to carry large values of currents.



As shown in the figures it contains Main poles and Auxiliary points

The main poles used to supply currents and voltage for loads

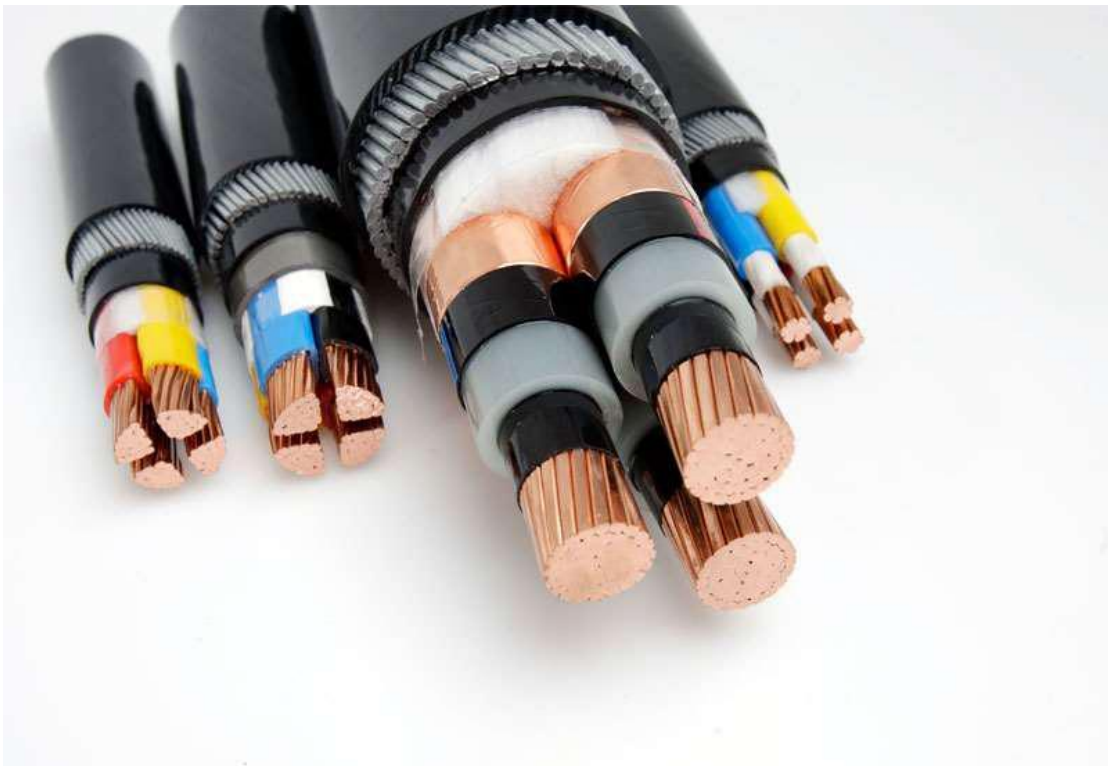
The contactor rating depends on the load

Application in our projects :

- Used for the elevators

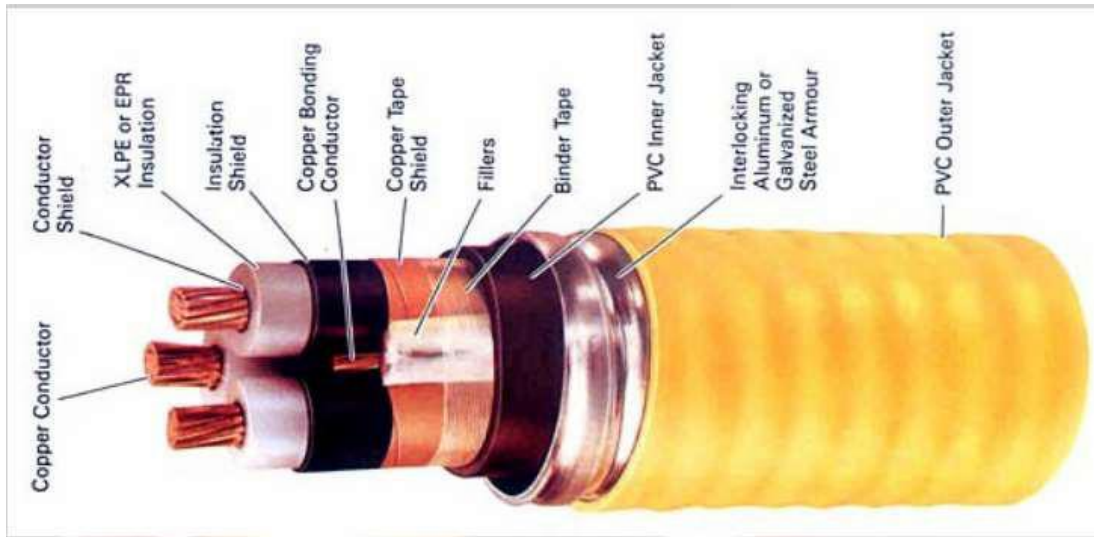
CHAPTER 4

Cables



1. Introduction

The necessary requirements of a cable mean that it should conduct electricity efficiently, cheaply and safely. It shouldn't be too long to limit the cost. It shouldn't be small to limit the large interval voltage drop. Its insulation should be such as to prevent leakage of current and minimizing the losses.



2. General Construction of power cable

- **Note:**

at high voltage the construction of cables is very complicated, until the components of cable in high voltage reach to eight layer due to the need for very high insulation efficiency

1) Conductor:

It is the current carrying element, it's made of Copper or Aluminum it's known that copper can carry 15% current more than Aluminum . on another hand Aluminum is cheaper than copper . so for the economic cost we use Aluminum in large distances .

2) Conductor shield:

It is a thin layer from a semiconductor material used to make the best distribution at the common surface between the conductor and the dielectric.

3) Insulation or Dielectric:

The most common used is XLPE and PVC.

4) Insulation shield:

Its function as the conductor shield , it's used to make regular electric distribution on the outer surface of the insulation.

5) Metallic sheath or tape shield:

It's a copper tape with a thickness of 0.1 mm , used for the purpose of protection of the cable from mechanical stress and prevent moisture leakage to the cable, nowadays the Aluminum as a material of metallic sheath due to its low cost and also as a ground conductor.

6) Filler:

It's used to fill the gaps between the cores of the multi core cables

7) Armour:

It's steel wires or Aluminum wires used to protect cables from mechanical stresses, the standards states that cables of voltage greater than 1 KV must be provided with Armour.

8) Outer jacket:

Used to protect the metallic parts from corrosion, specially the Aluminum parts.

3. Cable classifications:

➤ According to operating voltage:

1. High voltage cables (up to 66kv).
2. Medium voltage cables (up to 3,3kv).
3. Low voltage cables.

Note➔there is no world agreement on specific values for these classifications, and may vary from place to another.

➤ According to type of conductor:

There are many types of conductors used inside the cables like (CU, AL, GOLD, Silver, etc..). and the CU, AL is commonly used because:

1. copper conductors:

1. *Copper has low resistivity and thus, higher conductivity than Aluminum.*
2. *Copper conductors are comparatively soft and strong, and are most suitable for indoor and outdoor wires.*
3. *Copper is easier to install.*
4. *Copper is corrosion resistant.*

2. Aluminum conductors.

- a- *Al conductor is cheaper than copper conductor.*
- b- *Al is lighter and more available than copper conductor.*
- c- *Al conductors commonly.*

➤ According to type of insulation:

❖ The main properties of the insulating material:

- 1- *High insulation resistance.*
- 2- *High dielectric strength.*
- 3- *Good mechanical properties.*
- 4- *Capable of being operated at high temperature.*
- 5- *Low thermal resistance.*
- 6- *Low power factor.*

• Types of insulations:

- 1- *PVC (poly vinyl chloride).*
- 2- *XLPE (Cross Linked Polyethylene).*
- 3- *EPR (Ethylene Propylene Rubber).*
- 4- *Butyl rubber.*
- 5- *silicon rubber.*
- 6- *Mineral insulation.*
- 7- *LSF(low smoke and fume)*
- 8- *Paper.*

• PVC insulation

It is both lighter and tougher and inherently more flexible than paper. PVC is resistant to most chemicals although care must be taken with installations in petrochemical environments. It is a thermoplastic material which softens at high temperatures and therefore cannot withstand the thermal effects of short circuit currents as well as paper insulation.

➤ Advantages of PVC cables:

1. *Compacted conductors to reduce AC losses which are due to skin and proximity effect.*
2. *Complete protection against most forms of electrolytic and chemical corrosion.*
3. *Not affected by vibration.*
4. *Reduce internal stress and low dielectric loss.*
5. *Higher current and short circuit current rating.*
6. *Good ageing characteristics.*

• XLPE insulation

XLPE is a thermo setting material hence XLPE has no true melting point and remains elastic at high temperatures therefore permitting greater current carrying capacity, overload and short circuit performance in comparison with PVC- and paper-insulated cables.

• EPR insulation

It has the same thermal characteristics as XLPE but has a higher degree of elasticity which is maintained over a wide temperature range EPR is water resistant and cables are able to operate without a metallic sheath in waterlogged areas, but it isn't resistant to oil and gasoline.

➤ According to number of cores:

a) Multi-core cables

consists of (3 phases + neutral) they are all inside the outer cable insulation



b) Single-core cables

Generally prefer to (multi- core) For easy to deal with in terms of transportation and extended in the pipes, but with large sections (up to 240 mm)There is difficulty in extending cable inside pipes due to excess weight so use (single-core).



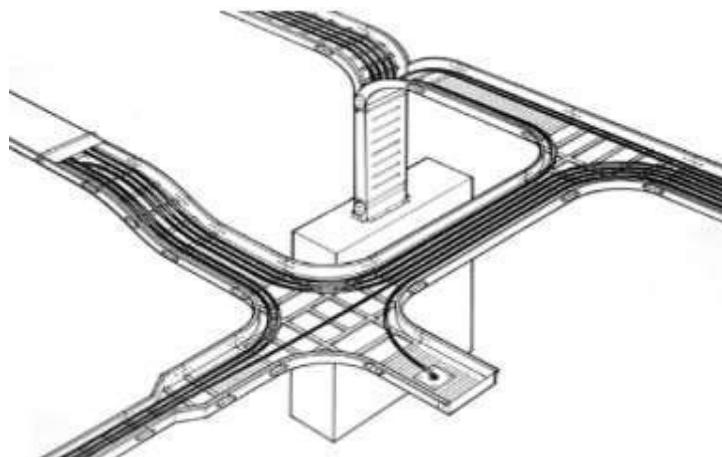
The methods of cables installations

There are a lot of methods to install cables

- 1- *Using different types of conduits.*
- 2- *Using cable trays*
- 3- *Direct burial under ground.*

➤ Cable trays

it's used in the industrial applications as the cables are put on the cable trays are exposed to air and the heat transfer is better than the method used in conduits. And it's easy to follow faults in the cable trays.



➤ Direct burial in the ground

It's preferred to install cables in ground in case of main networks in cities. It's good in heat transfer.

But when we bury cables we should take in our mind the below.

- *The burial depth is not less than 80 CM.*
- *A Layer of sand with a thickness 10 cm then put the cable over it directly*
- *Then put a layer of sand at a height of 20 cm from the burial depth of the cable.*
- *Bricks are put over the step above as assign*
- *Put the rest of the digging and over it a yellow tapes.*

Some notes about cables burial

1- In case of more than one cable in the ground we should put a separator between the cables to reduce the derating factors.

2- When the cables cross any street , the cable must be put inside a PVC conduits.

3- The diameter of the conduit must be 4 cm more than the cable outer diameter .

4- We should apply the derating factors to the cables.

5- Any cable mustn't put single cable inside a metallic tube because it may induced currents inside the conduit causing excessive heat inside the conduit then the cable will burn.

6- We should put fire barrier between the cables of emergency and the cables of main supply , we may make two separate paths to avoid it's high price .

7- We must put the cables of the control , light current at a distance is not less than 30 cm from the power cables.

❖ We can follow the Egyptian code as below

(أ) كابل جهد منخفض بجانب كابل اتصال	المسافة 300 مم على الأقل
(ب) كابل جهد منخفض بجانب كابل تحكم	بدون مسافة فاصلة
(ت) كابل تحكم بجانب كابل تحكم	بدون مسافة فاصلة
(ث) كابل جهد منخفض بجانب كابل جهد منخفض	قطر الكابل وبحد أدنى 10 مم وتستعمل فواصل بين الكابلات توضع كل مسافة تتراوح ما بين 1 و 1.5 متر
(ج) كابل جهد متوسط بجانب كابل جهد متوسط	150 مم ويفصل بينهما إذا كانت مدفونة في الأرض بقوالب طوب توضع رأسياً على جانبها

➤ Current carrying capacity

Any cable can carry current according to the method of the installation , for example the cable that is put inside conduits carry less current than the value of current if it is buried in the ground .So it is not accurate that a cable of cross sectional area 16 mm^2 can carry current 80 A , this statement will be accurate of we say the method of installation is this will be in air , pipes or in ground.

➤ Cables formula

For multi core cables we can name it is a below

$$(3 \times 185 + 1 \times 95)$$

This means that the cable is 4 cores with reduced neutral of a cross section area is 185 mm^2 and the neutral cross sectional area 95 mm^2

Cable of 4x (1*10) ➔ This means that they are a 4 single core cables with a cross sectional area 10 mm^2

➤ The neutral line specifications.

The neutral cross sectional area is always one half of the phase cross sectional area .this is because and this is because in case of balance there is no current passes through the neutral.

But it's not usual that the neutral cross sectional area is half the phase C.S.A and this due to the following reasons.

- 1- *In case of the small cables cross sectional areas , it hasn't large economic to reduce the cables cross section area specially the cables with cross sectional area less than 35 mm²*
- 2- *In case of industrial application that used power electronics it's preferred to make the neutral C.S.A equal to phase C.S.A*
- 3- *Also full cables are used in case of the fluorescent lamps used because the 3rd harmonic value is very high.*

➤ The derating factors

These de-rating factors are applied to the main cables of main feeders only it have no meaning if it's applied to the lighting circuits

In this case we should revise the cables Ampere capacity according to the environment where it will be installed

- *The derating factors that we used it in our project are ground temp. De-factors , depth derating factor and grouping factors.*

The table below shows the ground temp derating factor

Ground Temperature Derating Factor									
Ground Temperature °C	15	20	25	30	35	40	45	50	55
PVC cables rated 70 °C	1.25	1.19	1.13	1.07	1.00	0.93	0.85	0.76	0.65
XLPE cables rated 90 °C	1.16	1.13	1.09	1.04	1.00	0.95	0.90	0.85	0.80

- *In our project we take the ground temp 50° so we divide the current on 0.85 for XLPE cable*

The burial depth de-rating factors

Table 4

Burial Depth Derating Factor

Depth of Laying mt.	Cables Cross Section		
	Up to 70 mm ²	95 up to 240 mm ²	300 mm ² & above
0.50	1.00	1.00	1.00
0.60	0.99	0.98	0.97
0.80	0.97	0.96	0.94
1.00	0.95	0.93	0.92
1.25	0.94	0.92	0.89
1.50	0.93	0.90	0.87
1.75	0.92	0.89	0.86
2.00	0.91	0.88	0.85

The low voltage cables are buried at a depth not less than 60 cm So for cables of cross sectional area up to 70 mm² we take 0.99 factor. And for cables from 95 to 240 mm² the derating factors we take is 0.98

The grouping factors is calculated as below

Trefoil or Flat Formation Derating Factors for Multi-core Core Cables Laid Direct in Ground

Number of Circuits								
	Trefoil Formation				Flat Formation			
	Touching		Spacing = 0.15 M		Spacing = 0.30 M		Spacing = 0.30 M	
nr	Trefoil	Flat	Trefoil	Flat	Trefoil	Flat	Trefoil	Flat
2	0.81	0.81	0.87	0.87	0.91	0.91	0.91	0.91
3	0.69	0.70	0.76	0.78	0.82	0.84	0.82	0.84
4	0.62	0.63	0.72	0.74	0.77	0.81	0.77	0.81
5	0.58	0.60	0.66	0.70	0.73	0.78	0.73	0.78
6	0.54	0.56	0.63	0.67	0.70	0.76	0.70	0.76

* L = Spacing

➤ The method we chose for installation in our project is trefoil it's better than the flat because in trefoil the magnetic fields are vanished

For example of our project

A cable of a current is 200 A with a length of 65 m xlpe /pvcinsulation , is put next to an of cables is 3cables in a depth of 60 cm in ground , assume that the ground temperature is 50° . Find the cable cross sectional area under these conditions if the method of installing is trefoil method.

ANSWER

The ground temp derating factors is from the table is 0.85

The burial depth derating factors is 0.98

The grouping factors is 0.62 assume they are touching.

So the cable should carry current after applying the derating factors

New cable current = $\frac{200}{0.85 \times 0.98 \times 0.62} = 388 \text{ A.}$

From the cable catalogue the cable cross sectional area

Is → 3*185 + 1*95 XLPE / PVC.

This means that the cable is 4 core cable with reduced neutral.

The table in the next page shows the cables we used for 5 building in our project

Transformer B for building 6						
DB Name	fed from	length (M)	C.B Current (A)	Derating factors Spacing /temp/ depth	CABLE USED SHOULD CARRY (A)	C.S.Area (mm2)
Tr. Terminal voltage ----- full load current						
SMD801	TrL Main DB	60	200	0.62*0.85*0.99	365.64A	3*185+1*95
SMD8-B1	SMD801	2	80	0.78*0.82	125A	3*50+1*25+1*25
SMD8-F101	SMD801	5	80	0.70*0.82	139.37A	3*50+1*25+1*25
SMD8-F201	SMD801	12	80	0.70*0.82	139.37A	3*50+1*25+1*25
DB-f201	SMD8-F201	32	32	1.2	38.4	2*10+1*10
Emergency B for building 6						
DB Name	fed from	length (M)	C.B Current (A)	Derating factors	New cable (A)	C.S.Area (mm2)
SMD801	Diesel Main DB	60	40	0.62*0.85*0.99	77 A	3*16+1*16
SMD8-B1	SMD801	2	32	0.78*0.82	50	3*10+1*10+1*10
SMD8-F101	SMD801	5	32	0.70*0.82	55	3*10+1*10+1*10
SMD8-F201	SMD801	12	32	0.70*0.82	55	3*10+1*10+1*10
EDB-f201	SMD8-F201	16	25	1.2	30	2*6+1*6
Transformer B for building 7						
DB Name	fed from	length (M)	C.B Current (A)	Derating factors	New cable (A)	C.S.Area (mm2)
Tr. Terminal voltage ----- full load current						
SMD802	TrL Main DB	27	200	0.62*0.85*0.99	365.64A	3*185+1*95
SMD8-B2	SMD802	2	80	0.78*0.82	125A	3*50+1*25+1*25
SMD8-F102	SMD802	5	80	0.70*0.82	139.37A	3*50+1*25+1*25
SMD8-F202	SMD802	12	80	0.70*0.82	139.37A	3*50+1*25+1*25
DB-f201	SMD8-F202	32	32	1.2	38.4	2*10+1*10

CHAPTER 5

Load estimation and calculation



Introduction

There is a step before getting starting in design of any project. It's estimating the total load of any project without entering to any details of the project.

The electrical loads are classified as below :-

- 1) Load of lighting (general - gaps - corridors).*
- 2) Load of Small appliance (sockets -offices devices - televisions).*
- 3) Load of Air conditioning devices and heating devices.*
- 4) Load of Firefighting pumps and water heating devices.*
- 5) Load of Light current devices (fire alarm - CCTV - Sound system)*
- 6) Load of Elevators and electrical stairs.*

Load estimation stages

1) Load estimation in the primary stages.

It's calculated by using the standard electric load for square meter. And by this method we can get the licenses and estimate the transformers power rating.

For moderate residential areas the load demand is about 2 : 4 KVA per 100 M².

For rich residential areas the load demand is about 6 : 10 KVA per 100 M².

For commercial areas the load demand is about 8 : 12 KVA per 100 M².

2) Load estimation in the final stages.

At the end of the design every load rating is specified accurately. So we can go to the loads catalogue and note them in excel sheet and design the main distribution boards.

➤ Lighting load estimation

Lighting represent about 20 to 50 % from the total electric loads. And the standard lighting load in watt/ M^2 changes from place to other according to its nature and application. The below table shows that.

نوع الحيز أو المرفق	الحمل النوعي لكل متر مربع (واط)
البنوك	25
أماكن العبادة	20
النوادي الملاعب	50-20
المستشفيات	35-20
الفنادق ومباني الشقق المفروشة	15
المدارس	20-16
المكتبات	20-15
المتاجر	25
السالن	10

➤ Sockets load estimation

There is a lot methods to estimate the general socket load , in our project we consider that every socket has a load of 180VA. In the NEC the sum of loads all general sockets at any flat is equal to 3000VA at least plus the heavy loads like dryer , washing machine and ovens.

المكان	الحمل التقريبي W/m^2
المكتبة / غـ ر ف	50 - 30
الاجتماعات/المنازل	60 - 40
المحلات	20-10
الفصول	
المطابخ	2 : 6 Circuits (each of 20A)

➤ Air conditioning load estimation

The air conditioning estimation is the function of Mechanical engineer ,but the electrical engineer must know a simple principles of estimation of it.The Egyptian electric code offers a table below

المساحة المخدومة (م ²)	نظام التشغيل	قدرة الوحدة الكهربائية		القدرة الحرارية	
		بارد/ساخن (ك.و.)	بارد فقط (ك.و.)	طن (تبريد)	(و.ح.ب/ساعة)
12 – 10	1/50/220	1.5	1.540	1	12000
18 – 15	1/50/220	1.6	1.930	1.5	18000
25 – 20	1/50/220	2.670	2.670	2	24000
30 – 25	1/50/220	4.20	3.745	3	36000
40 – 35	3/50/380	4.50	3.5	3.5	42000
50 – 40	3/50/380	5.00	4.5	4	48000
60 – 50	3/50/380	7.00	6.25	5	60000

➤ Elevators load estimation

We can estimate the primary load of the elevator in steps:-

1. Specify if the used machine supplied with gears or not
2. No of elevators inside the building.
3. The speed of the elevator.
4. The weight of elevator in Kg.
5. The demand factor for a number of elevators.

By knowing these steps we can easily estimate the elevator horse power.

Step no 1 ➔ usually high speed motors are supported with gears.

Step no. 2 ➔ easily calculated .

Step no. 3 ➔ it depends on the number of floors of the building and the table below from the Egyptian electrical code shows that.

عدد الطوابق	السرعة المقترحة متر / دقيقة
5	60 - 90
10	120 - 105
15	150 - 120
20	210 - 180
25	210 - 240
30	240 - 300

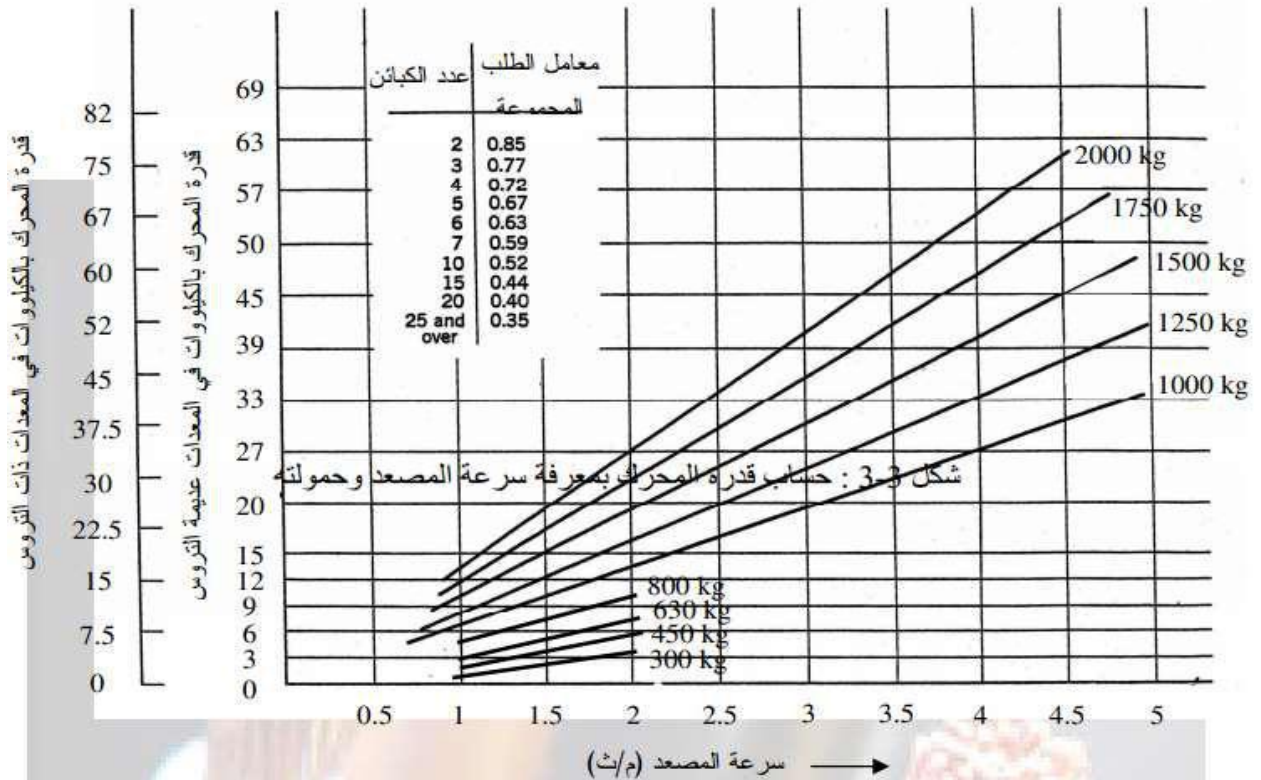
And this table compare between the machines supported with gears or not .

الماكينة	ارتفاع المبنى (م)	السرعة (م/ث)	التحكم	العمر الافتراضي	الاحتياج للصيانة	التكلفة	درجة النعومة
ذات تروس	45 – 15	1 – 0.25	تغيير الجهد	40 - 30	متوسط	منخفضة إلى متوسطة	منخفضة إلى متوسطة
بدون تروس	50 <	2 <	تغيير الجهد	غير محدد	منخفض	عالية	عالية

Step no. 4 → calculate the weight of the elevators this depends on the weight of the cabin, load inside the cabin, assume a cabin of weight 750 Kg and the load inside depends on the number of persons if a person of 80 Kg and 5 person inside the cabin so total load will be

$$750 \text{ Kg} + 5 * 80 \text{ Kg} = 1150 \text{ Kg}.$$

After these step we can compute the rating of the motor of the elevators.



Step no. 5 → specify the demand factor according the table below.

عدد الكابائن	معامل الطلب	
	المجمعة	
2	0.85	—
3	0.77	
4	0.72	—
5	0.67	
6	0.63	
7	0.59	—
10	0.52	
15	0.44	
20	0.40	—
25 and over	0.35	—

Example from our project:-

4 elevators are used, each one has a cabin with a load 750 Kg. in a building of 3 floors. Each cabin can carry 5 persons.

To calculate the motor rating by applying the previous steps.

- 1) Speed will be 1 M/S
- 2) Weight of elevator = $750 + 5 \times 80 = 1150$ Kg.
- 3) Assume that the elevator are without gears so the motor rating is 7 KW.
- 4) But the number is 4 so we apply a demand factor = 0.72
- 5) The motor power → $7 \times 0.72 = 5.04$ Kw.

➤ **Water pump load estimation**

For buildings with 3 floors they can be fed from the main water network depending on its pressure (3.5 to 4.8 bar).it can reach to the 3rd floor without using water pumps . for medium height building (10 floors , 30 m height) usually used a pump of 7.5KW .

And we can estimate the pump power easily from this equation.

$$P = \frac{\omega \cdot Q \times H}{75 \times \eta} \times 0.746 \text{ KW}$$

Where : Q → the water flow rate M³/S.

W → the water density in Kg/M³

H → the static height of building.

η → the pump efficiency (80%).

P → pump power in KW.

➤ Firefighting pump load estimation.

It is the function of the mechanical engineer, but in the same case the electrical engineer must know the minimum information to estimate the firefighting pump power. This depends on the height of the building , the flow rate of liquid and the pressure at this rate.

The following table is according the LPC standard shows required flow rate and its pressure according to the degree of fire and the height of the building.

جدول (1-2/2) السعة المطلوبة (التدفق والضغط) لمضخات المرشات حسب LPC

الخصائص المطلوبة ليست أقل من		المقدر الاسمي بالإضافة إلى أي مخرج صفيحة ضغط		اختلاف الارتفاع من المضخة (المباني المنخفضة) أو فرق المسافة بين أدنى وأعلى مرش في المباني المرتفعة		درجة الخطورة	
التدفق المنخفض	التدفق العالي	التدفق	الضغط	التدفق	الضغط	ليست أكثر من	أكثر من
عند المقياس	عند المقياس	عند المقياس	عند المقياس	عند المقياس	عند المقياس	عند المقياس	عند المقياس
(ل/د)	(بار)	(ل/د)	(بار)	(ل/د)	(بار)	(م)	(م)
---	---	225	3.7	300	1.5	15	0
---	---	225	5.2	340	1.8	30	15
---	---	225	6.7	375	2.3	45	30
375	2.5	540	2.2	900	1.2	15	0
375	4.0	540	3.7	1150	1.9	30	15
375	5.5	540	5.2	1360	2.7	45	30
725	2.9	1000	2.5	1750	1.4	15	0
725	4.4	1000	4.0	2050	2.0	30	15
725	5.9	1000	5.5	2350	2.6	45	30
1100	3.2	1350	2.9	2250	1.4	15	0
1100	4.7	1350	4.4	2700	2.0	30	15
1100	6.2	1350	5.9	3100	2.5	45	30
1800	3.5	2100	3.0	2650	1.9	15	0
1800	5.0	2100	4.5	3050	2.4	30	15

And we can estimate the pump power from this equation

$$W = \frac{QP}{600E}$$

Where : Q ➔ is the liquid flow rate in litre/minute

E ➔ the pump efficiency (60 : 70 %)

P ➔ the required pressure at the flow rate (bar)

W ➔ the power of the pump in KW.

Branch circuit design

Introduction

According to the NEC code the branch circuits are defined as the final circuits in the distribution networks and end with a load points. These circuits are gathered at a small sub main distribution boards. then we design a main distribution board to feed the sub main distribution boards. These main distribution boards are interconnected with the feeding network of the city or the project. In our project, we have 5 transformers and 5 diesel generators, the transformers are the main supply for our project and the diesel generator are used to supply the emergency circuits in case of outage and faults of the main supply.

So we have two types of circuits .

- 1) Branch circuits (it started from a small sub main distribution board and ended with a load like lamp , socket , power socket and etc)*
- 2) Main feeders (it started from main distribution board and ended with a sub main distribution boards)*

There a large difference in the design between these circuits because the Branch circuits feed a load with a constant value, but the other circuits feed a group of loads of different values and they may operate at the same time or every load operates separately so its load is not constant.

Principles of Branch circuits design

The purpose of the design of these circuits is to determine :-

- 1- The suitable circuit breaker value I_{CB} in Ampere.*
- 2- The cross sectional area of the used cable according to C.B. value in mm^2 .
And the design rules depends on the type of load*
 - 1) Static loads lighting, heating loads*
 - 2) Dynamic loads like motors.*



The NEC code states some rules.

- *We mustn't feed a different types of loads in the same circuit (sockets and lighting shouldn't be fed from the same circuit)*
- *The power loads must be fed from a separate circuits (every heater and Air conditioning device must be in separate circuits)*
- *The circuit of lighting is with a load from 4 : 6 A. The circuit breaker used for it should be 10 A at least .or according to the lighting load but at least 10 A.*
- *The sockets are in most residential applications are with a load of 8 A. the circuit breaker used must be at least 16A.*
- *The adjacent luminaires are gathered in one circuit except if they are with a different types such as main supply luminaire and emergency luminaire .*
- *The total load of any branch circuit mustn't be more than 80% of circuit breaker value .*
- *Any load is considered as a connected load if it operates continuously more than 3 hours.*
- *The wire of the branch circuit must be able to carry current more than C.B rating.*



Additional notes from Egyptian Electric Code

- *If the sockets are subjected to the mechanical damage so we must cover it with a metal case and this case should be earthed*
- *When a different level of voltage of sockets every socket should be with a different shape and color.*
- *If a number of sockets is used in a room with area of 50 M² so they must be fed from one phase .*

- If the rooms with area greater than 50 M^2 the sockets should be in a different phases and every phase should supply a different separate area and every socket case should be with a specific color.
 - When sockets are used on both sides of a wall , the horizontal distance between them must be at least 150 mm to avoid sound transmission across it.
 - The sockets should be installed at a height between 30 and 40 cm over the finished floor at office and residential areas except the kitchens and bathrooms the sockets should be installed at a height of 120 cm to 135 cm.
 - The distance between the sockets and the vertical Column shouldn't exceed 180 cm. and the distance between the sockets and the other shouldn't exceed 3.65 M.
 - Sockets shouldn't be installed on the surface of the labs. To avoid the dust and moisture from leakage to the sockets.
- After we read the previous rules and instructions we make a layout of lighting , sockets and accessories on the archeologist's graphs.

Design of circuits of the static loads.

The method in steps :-

- 1- Calculate the load current I_L
- 2- Choose a C.B. current rating standard and must be at least equal to $1.25 * I_L$

$$I_{C.B.} = 1.25 * I_L$$

- 3- Use the de-rating factors of the cable to ensure that cable can carry the C.B. current

And can bear the thermal stresses of the C.B current rating value.

- 4- Choose a cable from the catalogue can carry current more than C.B. current.

$$I_{CABLE} > I_{C.B.}$$

Design of circuits of the Dynamic loads.

- The big difference in design between static loads and dynamic loads is the starting current.
- To calculate the rated current of motor can be easily calculated from this eq.

$$I_m = P / (V * p.f.)$$

To calculate the value of starting current from motor nameplate using KVA code like this

SIEMENS									
PE•21 PLUS™					PREMIUM EFFICIENCY				
ORD. VO.	1LA02864SE41				E NO.				
TYPE	RGZESD				FRAME	286T			
H. P.	30.00				SERVICE FACTOR	1.15		3 PH	
AMPS	34.9				VOLTS	460			
R.P.M.	1765				HERTZ	60			
DUTY	CONT 40°C AMB.				OPT. CODE				
CLASS INSUL.	F	NEMA DESIGN	B	KVA CODE	G	NEMA, NOM. EFF.	93.6		
SP. END BRG.	50BC03JPP3				SP. END BRG.	50BC03JPP3			
MILL AND CHEMICAL DUTY QUALITY INDUCTION MOTOR									
Siemens Energy & Automation, Inc. Little Rock, AR						MADE IN U.S.A.			

By knowing this code we can determine the starting current form this table .

kVA Code	(kVA/HP) at starting
A	0-3.14
B	3.15-3.43
C	3.44-3.99
D	4 - 4.49
E	4.5 - 4.99
F	5 - (5.59)
G	5.6 – 6.29
H	6.3 - 7.09
J	7.1 - 7.99
K	8 - (8.99)



Design steps for small motors

- 1- Calculate the motor rated current as we explained in the prev. page.
- 2- Calculating the starting current for the motor
- 3- Choose a C.B. of $I_{C.B} > I_{st}$.
- 4- Choose a cable with a current $I_{CABLE} > I_{C.B.}$ and take in your mind the de-rating factors



Design steps for large motors

- 1- Calculate the motor rated current.
- 2- Calculate the motor starting current
- 3- Choose a C.B. of $I_{C.B} > I_{st}$.
- 4- Apply the derating factors for the cables if it's required
- 5- Choose a cable of $I_{CABLE} > I_{LOAD}$, but take in your mind the derating factors.
- 6- Adjust the overload at a value of 1.05 of the motor rated current.



Design steps for a group of motors.

Steps :-

- 1- Calculate the total starting current for a group of motor from this equation and choose a C.B. of the nearest standard current rating.

$$I_{st} (Group) = I_{st-max} + DF \left[\sum I_{rated} - I_{rated-max} \right]$$

- 2- Calculate the rated current for a group of motor from this equation.

$$I_{rated} (Group) = 1.25 \times I_{rated-max} + DF \left[\sum I_{rated} - I_{rated-max} \right]$$

- 3- Apply the de-rating factors to the second step and choose a cable
- 4- Choose the cable cross sectional area can carry current after applying the de-rating factors.
- 5- Make sure that the cable current can carry more than the C.B. rating , otherwise choose another cable with a higher cross sectional area.

LOAD CALCULATION

At the end of the project design every load is accurately determined, specified and calculated. According to the load information at every circuit , it easily to calculate the connected load of every distribution panel and we can apply two factors are the diversity and demand factor.

The demand factor → *is concerned by the loads of the same type at the same panel for example:-*

In our project we have a several building every building has a number of Pavilions. We will take a pavilion load from our project

Load of lighting is → 6 luminaire of 18 watt, 2 of 26 watt and 2 of 30 watt.

*So the total connected loads of lighting = $6*18+2*26+2*30 = 220$ watt assuming unity power factor. So the total connected load of lighting is 220 VA. But not all the luminaires operate the same time. Assume that 75% (from Egyptian electrical code) of lighting is actually operated.*

So the demand factor is defined as = *the maximum demand of a certain load type / the total connected load of the same load type.*

The Egyptian electric code specify the demand factor for the load type according to its application and nature of the area of the load (hotel,residential, commercial and etc). and the table below shows this.

نوع الحمل	عشرات تتكون من عدة وحدات سكنية	وحدة سكنية أو وحدات سكنية خاصة	فنادق صغيرة أو مباني عامة للتوم والمعيشة	مكاتب ومشاير ومبان عامة بخلاف الورش والمصانع
الإضاءة	50% من الحمل الكلي	66% من الحمل الكلي	75% من الحمل الكلي	90% من الحمل الكلي
المأخذ الكهربائية (البرايز)	100% من التيار التصميمي لأكثر مأخذ بالدائرة. 40+ % من مجموع التيارات التصميمية لباقي مأخذ الدائرة	100% من التيار التصميمي لأكثر مأخذ بالدائرة. 40+ % من مجموع التيارات التصميمية لباقي مأخذ الدائرة	100% من التيار التصميمي لأكثر مأخذ بالدائرة. 40+ % من التيار التصميمي لباقي مأخذ الدائرة	100% من التيار التصميمي لأكثر مأخذ بالدائرة. 75+ % من مجموع التيارات التصميمية لباقي مأخذ الدائرة
الأجهزة الكهربائية الثابتة بخلاف المحركات والمضخات وأجهزة الطهي	100% من الحمل الكامل لأكثر جهاز. 50+ % من الحمل الكامل للجهاز الأول الذي يلي أكبر جهاز. 33+ % من الحمل الكامل للجهاز الثاني الذي يلي أكبر جهاز. 20+ % من الحمل الكامل لباقي الأجهزة	100% من إجمالي الحمل الكامل لمجموع الأجهزة حتى سعة 10 أمبير. 50+ % من الحمل الكامل للأجهزة التي حملها يزيد عن 10 أمبير	100% من الحمل الكامل لأكثر جهاز. 80+ % من الحمل الكامل للجهاز الأول الذي يلي أكبر جهاز. 60+ % من الحمل الكامل لباقي الأجهزة	100% من الحمل الكامل لأكثر جهاز. 75+ % من الحمل الكامل لباقي الأجهزة

أجهزة الطهي الثابتة	100% من الحمل الكامل لأكبر جهاز 50+% من الحمل الكامل للجهاز الأول الذي يلي أكبر جهاز 33% من الحمل الكامل للجهاز الثاني الذي يلي أكبر جهاز 20+% من الحمل الكامل لباقى الأجهزة	100% من الحمل الكامل للأجهزة حتى 10 أمبير 30+% من الحمل المقنن الزائد على 10 أمبير 5+ أمبير إذا كان يوجد بالجهاز مخرج إضافي.	100% من الحمل الكامل لأكبر جهاز 80+% من الحمل الكامل للجهاز الأول الذي يلي أكبر جهاز 60+% من الحمل الكامل لباقى الأجهزة	100% من الحمل الكامل لأكبر جهاز 80+% من الحمل الكامل للجهاز الأول الذي يلي أكبر جهاز 60+% من الحمل الكامل لباقى الأجهزة
المحركات الكهربائية "خلاف محركات المضاد التي لها اعتبارات خاصة"	100% من الحمل الكامل لأكبر محرك 50+% من الحمل لباقي المحركات	100% من الحمل الكامل لأكبر محرك 50+% من الحمل لباقى المحركات	100% من الحمل الكامل لأكبر محرك 50+% من الحمل لباقي المحركات	100% من الحمل الكامل لأكبر محرك 80+% من الحمل الكامل للمحرك الذي يلي أكبر محرك 60+% من الحمل الكامل لباقى المحركات
المسخنات الكهربائية متقطعة التشغيل	100% من الحمل لأكبر سخان 100+% من الحمل الكامل للسخان الذي يلي أكبر سخان 25+% من الحمل الكامل لباقي السخانات	100% من الحمل لأكبر سخان 100+% من الحمل الكامل للسخان الذي يلي أكبر سخان 25+% من الحمل الكامل لباقي السخانات	100% من الحمل لأكبر سخان 100+% من الحمل الكامل للسخان الذي يلي أكبر سخان 25+% من الحمل الكامل لباقي السخانات	تقدر بمعرفة المختصين تبعاً لظروف التشغيل الفعالية المحتملة
المسخنات الكهربائية مستمرة التشغيل	100% من الحمل الكامل في جميع الحالات.	100% من الحمل الكامل في جميع الحالات.	100% من الحمل الكامل في جميع الحالات.	100% من الحمل الكامل في جميع الحالات.

The diversity factor

It is the probability of operating different types of load in the same time. For example the probability to operate a socket and a lamp in the same time. And it is calculated from $\Rightarrow$ the total maximum demand of the panel divided by the total connected loads.

Example from our project.

Room loads $\Rightarrow$ sockets + lighting load + air conditioning load.

Sockets $\Rightarrow$ room of 6 sockets every socket load is 180VA so socket loads = $6 \times 180 = 1080 \text{ VA}$.

Lighting $\Rightarrow$ 6 luminaires of 18watt, 2 of 26watt and 2 of 30watt. So lighting load = 220 watt.

Air conditioning load $\Rightarrow$ from catalogue 2.25 HP split unit. Assume p.f.

= 0.8 $\Rightarrow$ load = 2098 VA

Assume for lighting P.F. = 1. So total connected loads $\Rightarrow 1080 + 220 + 2098 = 3165 \text{ VA}$.

From Egyptian electric code.

Demand factor for

*Lighting for hotels = 0.75 ,Sockets = Max socket VA + 0.4*n*socket VA.*

For split unit →

Demand factor = 1. Because (it's on or off).

So max demand by applying demand factor →

*Max demand = $(0.75*220) + (180+0.4*5*180) + (2098*1) = 2911 \text{ VA}.$*

The diversity factor = max demand / total connected loads.

→ $2911/3165 = 0.91.$

But in our project the diversity factor is equal to unity. Because of it is the designer view.

So for this room a load of 2911 VA we can use this value to calculate the main breaker rating of the room.

CHAPTER 6

Voltage drop and Short circuit calculation



Voltage drop calculation

After applying de-rating to the cables according to the nature and the way of the installation. We must choose calculate voltage drop in the cables and it shouldn't exceed 5% or 8% in case of emergency and for Some standard (3 : 6 %).

Why we make voltage drop calculation?

- *The reduction in voltage by a ratio of 1% make incandecant lamps light reduce 3%.*
- *Some devices that used in homes that operates with motors. Any reduction in voltage make the devices motors draw more current if this current increased by 10% the breaker will not sense this current if it continue for a long time it may burn the motorized devices.*

Methods of voltage drop calculation.

There is 4 methods to calculate the voltage drop in cables.

➤ **First method**

It is the most accurate method by using this equation.

$$\Delta V_0 = I_{ph} (R \cos \phi + jX \sin \phi)$$

Factors affecting the cable.

R : Cable resistance.

X : cable reactance.

p.f: power factor.

I_{ph} : The phase current.

➤ **Second method**

$$\Delta V_0 = I_{ph} \times Z(\Omega)$$

Or it can be more simplified as below

$$\Delta V_0 = I_{ph} \times R(\Omega) \quad \text{or} \quad \Delta V_0 = I_{ph} \times X(\Omega)$$

We notice here that either the R or X of the cable is negligible, and this is logical in small cross sectional area cables.

➤ Third method

By using the tables of catalogue of cable companies, from these tables we use the voltage drop mV/M/A . And this method use the cable length and the current passing through the cable.

Voltage Drop for Single Core L.V Cables

C.S.A mm^2	Copper Conductor Voltage Drop (mv / AMP / Meter)			
	PVC Insulation & PVC Sheathed		XLPE Insulation & PVC Sheathed	
	Flat	Trefoil	Flat	Trefoil
				
4	7.83	7.770	8.337	8.277
6	5.287	5.226	5.628	5.568
10	3.184	3.124	3.401	3.341
16	2.068	2.008	2.203	2.142
25	1.357	1.297	1.440	1.380
35	1.034	0.971	1.085	1.024
50	0.793	0.732	0.836	0.776
70	0.595	0.534	0.624	0.564
95	0.469	0.408	0.490	0.430
120	0.410	0.349	0.417	0.357
150	0.354	0.294	0.366	0.305
185	0.312	0.252	0.322	0.262
240	0.272	0.211	0.278	0.218
300	0.247	0.187	0.253	0.192
400	0.224	0.164	0.220	0.159
500	0.208	0.148	0.211	0.150
630	0.194	0.134	0.191	0.131

For example:-

Cable with cross sectional area 70 mm^2 and with insulation PVC/PVC And its installation will be trefoil. The voltage drop will be $0.534 \text{ mV/AMP/Meter}$. If the cable length is 500 m and current passing across it is 100 A . So voltage drop inside through the

cable is $100 \times 500 \times 0.534 = 26.7 \text{ Volt}$.

➤ Fourth method

If the impedance of the cable is known in per unit. So the voltage drop from this equation

$$\Delta V_o = Z_{p.u} \times V_{line}$$

Some notes about voltage drop in cables

- *for single phase the V.D is multiplied by 2.*
- *For three phase the V.D is multiplied by 1.732.*
- *The voltage drop in three phase as a ratio of line voltage.*
- *Most of Companies tables are given multiplied by 2 for single phase and 1.73 for three phase.*

A real example from our project

A transformer of 500KVA has a percentage impedance of 4 %. The voltage drop across the transformer terminal is 0.04×410 ,

410 → is the secondary voltage of the transformer.

So voltage drop → $0.04 \times 410 = 16.4$ Volt.

The cable length from the transformer to the main distribution board is negligible.

And the cable from the main distribution of the transformer to the main distribution board of building no. 1 is 60 M protected with current of C.B. 200A .

The cable mv/AMP/M.Is equal to 0.241.

So the voltage drop at the Main distribution board of Building no. 1 = $6.4 +$

$60 \times 200 \times 0.241 \times 10^{-3} = 9.292$ V.

The voltage drop at the 3rd floor of the building no. 1 =

$9.292 + \text{length} \times \text{mv/amp/m} \times \text{Cable Amp} = 19.292 + 12 \times 0.72 \times 80 \times 10^{-3} = 9.98$ volt

Transformer B for building 5									
DB Name	fed from	length (M)	C.B Current (A)	Derating factors Spacing /temp/ depth	Cable used (SPECIAL CARRY (A)	C.S.Area (mm2)	notes	Voltage drop (mV/M/A)	Check
Tr. Terminal voltage	medium voltage	---	full load current					Percentage impedance 4.5%	
SMD881	TR. Main DB	60	200	0.62*0.85*0.99	365.64A	3*185+1*95	CU XLPE/PVC STEEL WIRE ARMOURED	0.241	391.55 V
SMD8-B1	SMD881	2	80	0.78*0.82	125A	3*50+1*25+1*25	CU PVC/PVC	0.72	accepted
SMD8-F1B1	SMD881	5	80	0.70*0.82	139.37A	3*50+1*25+1*25	CU PVC/PVC	0.72	accepted
SMD8-F2B1	SMD881	12	80	0.70*0.82	139.37A	3*50+1*25+1*25	CU PVC/PVC	0.72	accepted
EDB-F2B1	SMD8-F2B1	32	32	1.2	38.4	2*10+1*10	CU / PVC	3.124	accepted
Emergency B for building 6									
DB Name	fed from	length (M)	C.B Current (A)	Derating factors	New cable (A)	C.S.Area (mm2)	notes	Voltage Drop (mV/M/A)	Check
SMD881	Diesel Main DB	60	40	0.62*0.85*0.99	77 A	3*16+1*16	CU XLPE/PVC STEEL WIRE ARMOURED	1.982	ACCEPTED
SMD8-B1	SMD881	2	32	0.78*0.82	50	3*10+1*10+1*10	CU PVC/PVC	3.101	2.22
SMD8-F1B1	SMD881	5	32	0.70*0.82	55	3*10+1*10+1*10	CU PVC/PVC	3.101	2.29
SMD8-F2B1	SMD881	12	32	0.70*0.82	55	3*10+1*10+1*10	CU PVC/PVC	3.101	2.47
EDB-F2B1	SMD8-F2B1	16	25	1.2	30	2*6+1*6	CU / PVC	5.226	2.99
Transformer B for building 7									
DB Name	fed from	length (M)	C.B Current (A)	Derating factors	New cable (A)	C.S.Area (mm2)	notes	Voltage Drop (mV/M/A)	Check
Tr. Terminal voltage	medium voltage	---	full load current					Percentage impedance 4.5%	
SMD882	TR. Main DB	27	200	0.62*0.85*0.99	365.64A	3*185+1*95	CU XLPE/PVC STEEL WIRE ARMOURED	0.241	391.55 V
SMD8-B2	SMD882	2	80	0.78*0.82	125A	3*50+1*25+1*25	CU PVC/PVC	0.72	accepted
SMD8-F1B2	SMD882	5	80	0.70*0.82	139.37A	3*50+1*25+1*25	CU PVC/PVC	0.72	accepted
SMD8-F2B2	SMD882	12	80	0.70*0.82	139.37A	3*50+1*25+1*25	CU PVC/PVC	0.72	accepted
EDB-F2B2	SMD8-F2B2	32	32	1.2	38.4	2*10+1*10	CU/PVC	3.124	accepted
Emergency B for building 7									
DB Name	fed from	length (M)	C.B Current (A)	Derating factors	New cable (A)	C.S.Area (mm2)	notes	Voltage Drop (mV/M/A)	Check
SMD882	Diesel Main DB	27	40	0.62*0.85*0.99	77 A	3*16+1*16	CU XLPE/PVC STEEL WIRE ARMOURED	1.982	ACCEPTED
SMD8-B2	SMD882	2	32	0.78*0.82	50	3*10+1*10+1*10	CU PVC/PVC	3.101	1.56
SMD8-F1B2	SMD882	5	32	0.70*0.82	55	3*10+1*10+1*10	CU PVC/PVC	3.101	1.64
SMD8-F2B2	SMD882	12	32	0.70*0.82	55	3*10+1*10+1*10	CU PVC/PVC	3.101	1.81
EDB-F2B2	SMD8-F2B2	32	25	1.2	30	2*6+1*6	CU/PVC	5.226	2.86
Transformer B for building 8									
DB Name	fed from	length (M)	C.B Current (A)	Derating factors	New cable (A)	C.S.Area (mm2)	notes	Voltage Drop (mV/M/A)	Check
SMD8A2	TR. Main DB	27	200	0.81*0.85*0.99	293.02A	3*185+1*95	CU XLPE/PVC STEEL WIRE ARMOURED	0.241	391.55 V
SMD8-A2	SMD8A2	2	80	0.78*0.82	125A	3*50+1*25+1*25	CU PVC/PVC	0.72	1.31
									1.33

Short circuit

Introduction

The first step to ensure that system protective devices have the proper interrupting rating and provide component protection is to determine the available short-circuit currents. The application of the Point-To-Point method can be used to determine the available short-circuit currents with a reasonable degree of accuracy at various points for either 3 phase or 1 phase electrical distribution systems

Calculation procedure

Step no. 1 → Determine the transformer full load current from transformer nameplate or formula.

For three phase transformer use the below formula.

$$3\phi \text{ transf. } I_{FLA} = \frac{KVA \times 1000}{E_{L-L} \times 1.732}$$

For single phase transformer use the below formula.

$$1\phi \text{ transf. } I_{FLA} = \frac{KVA \times 1000}{E_{L-L}}$$

Step no. 2 → Determine the transformer multiplier.

$$\text{Multiplier} = \frac{100}{\text{Transf. \% Z}}$$

Where Z → is the transformer percentage impedance.

Step no. 3 → Determine the transformer short circuit current.

$$I_{SCA} = \text{Transf.}_{FLA} \times \text{multiplier}$$

Step no. 4 → Calculate the f factor.

For three phase faults

$$3\phi \text{ faults} \quad f = \frac{1.732 \times L \times I_{L-L-L}}{C \times n \times E_{L-L}}$$

For line to line faults

$$f = \frac{2 \times L \times I_{L-L}}{C \times n \times E_{L-L}}$$

For line to neutral faults

$$f = \frac{2 \times L \times I_{L-N}^*}{C \times n \times E_{L-N}}$$

Where $\rightarrow L \rightarrow$ length (feet) of conduit to the fault.

$C \rightarrow$ conductor constant. See Tables below

$n \rightarrow$ number of conductors per phase

(Adjusts C value for parallel runs)

Copper												
AWG or kcmil	Three Single Conductors Conduit						Three-Conductor Cable Conduit					
	Steel	5kV	15kV	Nonmagnetic			Steel	5kV	15kV	Nonmagnetic		
	600V	5kV	15kV	600V	5kV	15kV	600V	5kV	15kV	600V	5kV	15kV
14	389	-	-	389	-	-	389	-	-	389	-	-
12	617	-	-	617	-	-	617	-	-	617	-	-
10	981	-	-	982	-	-	982	-	-	982	-	-
8	1557	1551	-	1559	1555	-	1559	1557	-	1560	1558	-
6	2425	2406	2389	2430	2418	2407	2431	2425	2415	2433	2428	2421
4	3806	3751	3696	3826	3789	3753	3830	3812	3779	3838	3823	3798
3	4774	4674	4577	4811	4745	4679	4820	4785	4726	4833	4803	4762
2	5907	5736	5574	6044	5926	5809	5989	5930	5828	6087	6023	5958
1	7293	7029	6759	7493	7307	7109	7454	7365	7189	7579	7507	7364
1/0	8925	8544	7973	9317	9034	8590	9210	9086	8708	9473	9373	9053
2/0	10755	10062	9390	11424	10878	10319	11245	11045	10500	11703	11529	11053
3/0	12844	11804	11022	13923	13048	12360	13656	13333	12613	14410	14119	13462
4/0	15082	13606	12543	16673	15351	14347	16392	15890	14813	17483	17020	16013
250	16483	14925	13644	18594	17121	15866	18311	17851	16466	19779	19352	18001
300	18177	16293	14769	20868	18975	17409	20617	20052	18319	22525	21938	20163
350	19704	17385	15678	22737	20526	18672	22646	21914	19821	24904	24126	21982
400	20566	18235	16366	24297	21786	19731	24253	23372	21042	26916	26044	23518
500	22185	19172	17492	26706	23277	21330	26980	25449	23126	30096	28712	25916
600	22965	20567	17962	28033	25204	22097	28752	27975	24897	32154	31258	27766
750	24137	21387	18889	29735	26453	23408	31051	30024	26933	34605	33315	29735
1,000	25278	22539	19923	31491	28083	24887	33864	32689	29320	37197	35749	31959

$I \rightarrow$ available short-circuit current in amperes at beginning of circuit.

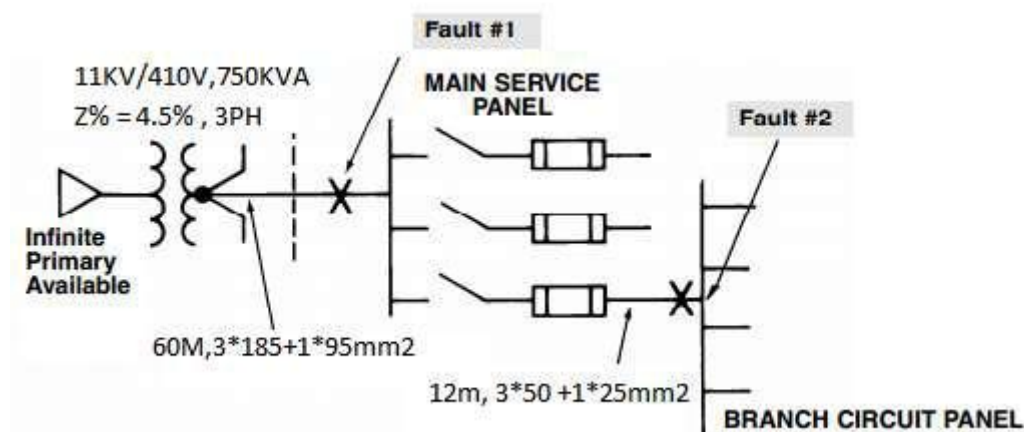
Step no. 5→Calculate the M multiplier factor.

$$M = \frac{1}{1 + f}$$

Step no. 5→Calculate the short circuit current multiplier factor.

$I_{sc} \text{ (at the fault)} = I_{sc} \text{ (at the beginning of the circuit)}$
* Multiplier factor

For example from our project



Given:- as shown in the figure above

Required :

Calculate the short circuit currents at fault1 and at fault2. And at the secondary of the transformer.

Answer :-

Step 1 → calculate transformer full load current = $\frac{1000}{\sqrt{3} \times 11} = 1126A$

Step 2 → calculate transformer Multiplier = $\frac{1}{0.045} = 22.22$

Step 3 → I_{sc} at the secondary of transformer = $22.22 \times 1126 = 25.023 KA$

Step 4 → calculating f factor → $\frac{1}{\sqrt{3} \times 11} = 0.054$

Step 5 → calculate m multiplier = $\frac{1}{1.8} = 0.54$

Step 6 → the short circuit current fault1 = $0.54 \times 25.023A = 13512 A$.

Step 7 → calculating f factor $\frac{1}{\sqrt{3} \times 11} = 0.244$

Step 8 → calculate m multiplier = $\frac{1}{1.25} = 0.8$

Step 9 → the short circuit current fault2 = $0.8 \times 13512 = 10908 A$.

CHAPTER 7

Feeding system



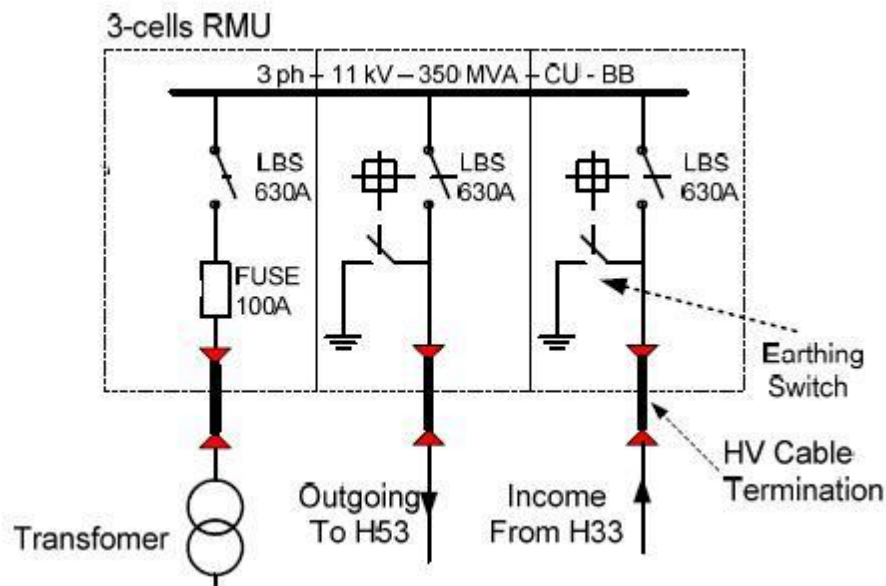
Medium voltage distribution network

Main component mv distribution network:

1) Ring main unit (RMU)

It consist of 3 load break switches (LBS) and sometimes of 4 load break switches

- a) Input load break switch (630 A)
- b) Output to another load break switch (630 A)
- c) Entry to transformer load break switch (400 A)
- d) Spare load break switch (630 A)



- Classification of ring main units:

A) According to operating voltage :

All ring main units used in medium voltage side (11kv or 22kv)

b) according to spark cut field :

- *sf6 ring main unit*

- *air ring main unit*

- *Ring main unit should be equipped with earthing switches mechanically interlocked*
- *Ring main unit should be located at a separate room with adjacent size according to number of RMU inside this room*
- *Ring main unit room should be equipped with ventilation source , double doors open outwards , adjacent cable trench*
- *RMU should be with adjacent degree of protection*

- Difference between sf6 & air ring main unit

	Sf6 RMU	Air RMU
cost	expensive	cheap
weight	light	heavy
size	Compact size	Larger size

2) High rupture capacity fuse (HRCF)



- This fuse is a protection device for transformer
- Before transformer we must put suitable (HRCF) with adjacent rating

100%	Transformer rating (kVA)																CEF
Un (kV)	25	50	75	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	
3	16	25	25	40	40	50	50	80	100	125	160	160					7.2kV
3,3	16	25	25	40	40	50	50	83	80	100	125	160					
4,15	10	18	25	25	40	40	50	50	63	80	100	125	160				
5	10	18	25	25	25	40	40	50	50	63	80	100	160	160			
5,5	8	18	16	25	25	25	40	50	50	63	80	100	125	160			
6	8	16	16	25	25	25	40	40	50	50	80	100	125	160	160		
6,6	8	16	16	25	25	25	40	40	50	50	63	80	100	125	160		12kV
10	8	10	10	16	16	25	25	25	40	40	50	50	80	80	125	125	
11	8	8	10	16	16	25	25	25	25	40	50	50	63	80	100	125	
12	8	8	10	16	16	16	25	25	25	40	40	50	63	80	100	125	17.5kV
13,8	8	8	10	10	16	16	25	25	25	25	40	50	50	63	80	100	
15	8	8	10	10	16	16	16	25	25	25	40	40	50	63	80	100	
17,5	8	8	8	10	10	16	16	16	25	25	25	40	50	50	63	80	24kV
20	8	8	8	10	10	16	16	16	25	25	25	40	40	50	63	63	
22	8	8	8	8	10	10	16	16	16	25	25	25	40	50	50	63	
24	8	8	8	8	10	10	16	16	16	25	25	25	40	40	50	63	

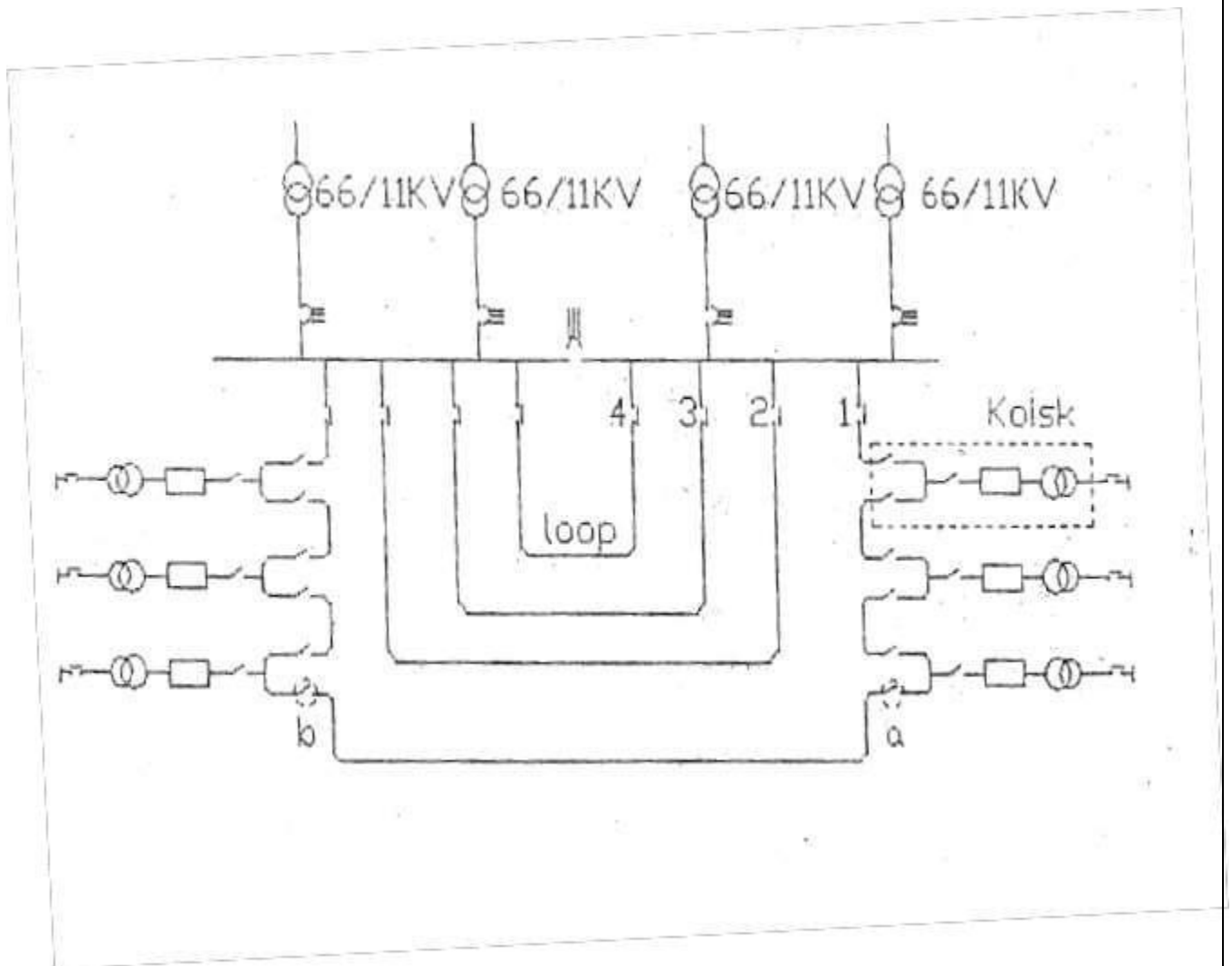
Fuse selection table

The table is based on using fuses type ABB CEF

3) Medium voltage distributor

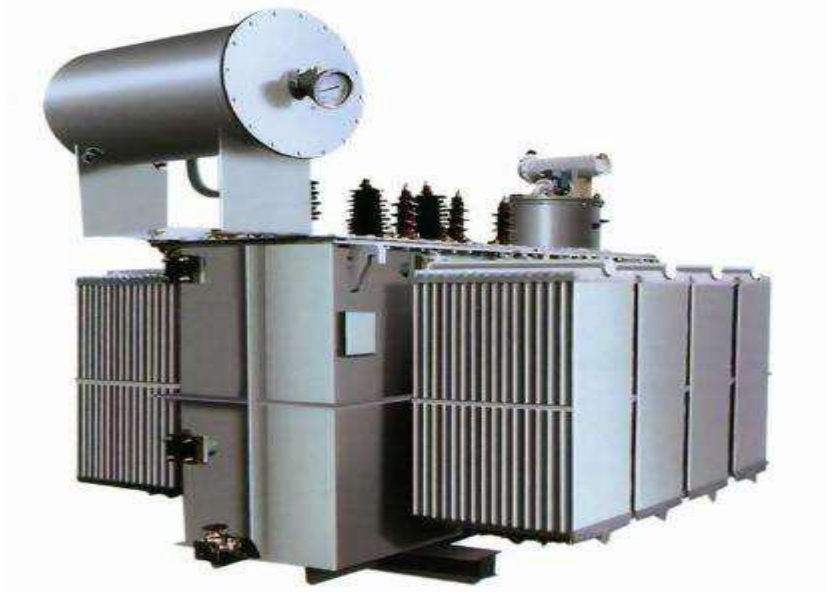
Component of MV distributor :

- I. 4 incoming circuit breaker with 4 cells
- II. 8 outgoing circuit breaker with 8 cells
- III. 1 bus coupler with two cell one for bus coupler and other for bus riser



- *There are two bars each one has 2 incoming equally loaded for high reliability and has 4 outgoing and there is B.C to connect two bus bars to each other*
- *Each outgoing from distributor feed some of elec. Rooms contain (RMU , transformer , low voltage MDB) and all rooms should be connected together via RMU and last room in outgoing line should be connected to another elec. Room from another bus bar.*
- *Each loop must be opened to prevent short circuit fault*
- *When a fault happen we change the location of opened LBS to make maintenance and prevent electricity break*
- *Each loop can feed 4-6 MVA*
- *In our project we can get feeding from RMU that connected to MV distributor*

DISTRIBUTION TRANSFORMER



1. Introduction:

- *A transformer is a static device constructed with two or more windings used to transfer alternating-current electric power by electromagnetic induction from one circuit to another at the same frequency but with different values of voltage and current.*
- *The capacity of a transformer is determined by the amount of current it can carry continuously at a rated voltage without exceeding the design temperature.*

- *The transformers are rated in kilovolt-amperes (KVA), the capacity is limited by the load current which is proportional to the KVA. Distribution transformers are used for distribution networks in urban cities and used in order to reduce the voltage from that of the primary distribution systems (usually 11 KV or 13.8 KV, nowadays 22 kV is also used) to the service voltage level of the secondary distribution network (less than 1 KV).*

2. Importance of distribution transformers:

The transformer allowed electricity to be transmitted over long distances. This made it possible to supply electricity to homes and Locations far from generating stations.

3. Main Parts of Distribution Transformer:

- 1- Core.*
- 2- Winding:*
 - a- High voltage winding.*
 - b- Low voltage winding.*
- 3- Tank.*
- 4- Terminals.*
- 5- Tap changer.*
 - *On load tap changer*
 - *Off load tap changer*
- 6- Bushings:*

4- There are two main types of transformers:

4.1 Dry Type Transformers:

Cooling in dry type transformer takes place by means of air circulation through turns under normal air pressure. The core as well as the turns is exposed directly to air, also the transformer is designed with large paths for ventilation by air.

Note:

- *The transformer must not be unloaded for large duration, since the transformer will be exposed to humidity. Under such conditions the transformer must be loaded to remove the humidity formed inside the insulating terminals. In highly polluted areas, the turns and core are completely enclosed inside a sealed tank to protect transformer.*

4.2. Liquid - filled Transformer:

Liquid-filled transformers offer the advantages of: smaller size, lower cost, and greater overload capabilities compared with dry type transformers that have the same rating.

4.2.i. Oil Immersed Transformers:

There are two types of fluid used with transformer:

- 1) Mineral oil fireproof liquids.*
- 2) Mineral oil extracted from petroleum.*

- *The second is considered the most widely used liquid with transformers. Oil has a dual function; the first is insulated between turns with each other as well between the turns and the core, while the second function is cooling by means of transferring the generated heat energy in the turns and the core to the cooler surrounding.*

5. Transformer energizing inrush current:

- *At the moment of energizing a transformer high values of transient current (which induces a significant dc component) occur and must be taken in account when considering protection schemes.*

The magnitude of the current peak depends on:

- 1. The value of voltage at the instant of energizing.*
- 2. Magnitude and polarity of magnetic flux, if any existing in the core of the transformer.*

3. Characteristics of the load on transformer.

- In distribution type transformers, the first current peak can attain a value equal to 10 to 15 times the full load current, but for small transformers (Less than 50 KVA), may reach 20 to 25 nominal full load current. This transient current decreases rapidly with specified time constant.*

6. Cooling:

ONAN	Oil Natural Air Natural Type
ONAF	Oil Natural Air Forced Type
OFAF	Oil Forced Air Forced Type
OFWF	Oil Forced Water Forced
AN	Air natural(dry type)
AF	Air forced (dry type)

CHAPTER 8

Power factor correction



Introduction:

• The active power P (in kW) is the real power transmitted to loads such as motors, lamps, heaters, computers ... The electrical active power is transformed into mechanical power, heat or light.

• The reactive power Q (in kVAR) is used only to supply the magnetic circuits of machines, motors and transformers.

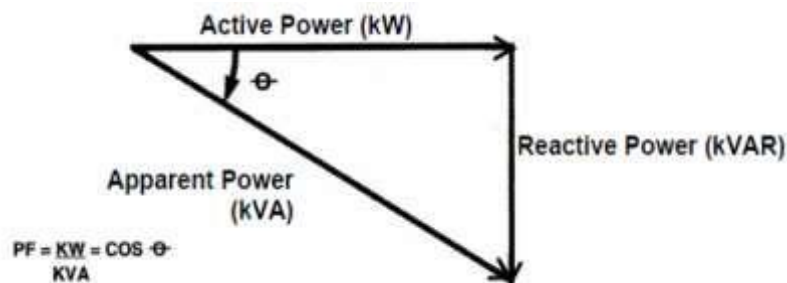
• The apparent power S (in kVA) is the vector combination of active and reactive power.

• Power factor (pf)

Is the ratio of the active (true) power absorbed by the load to the apparent power.

$$P.F = \frac{\text{Real power (P)}}{\text{Apparent power (S)}}$$

• Power triangle



$$S = P + jQ$$

$$|S| = \sqrt{P^2 + Q^2} = VI$$

$$P = |S| \cos \theta = VI \cos \theta$$

$$Q = |S| \sin \theta = VI \sin \theta$$

$$PF = P/S = \cos \theta$$

• The power factor is dimensionless and range between 0 and 1.

• when the nature of the load is purely resistive the kVAR or the reactive component will be nil and thus the angle ϕ will be equal to 0 degrees and the powerfactor will be equal to unity.

• For Inductive loads: → there will be a reactive Power and so power factor is lagging (the current lags behind the voltage so the power factor is said to be lagging).

• For Capacitive loads: → there will be a reactive Power and so power factor is leading (the current leads the voltage so the power factor is said to be leading).

why do we need power factor correction:

The circulation of reactive power in the electrical network has major technical and economic consequences. For the same active power P , a higher reactive power means a higher apparent power and thus, a higher current must be supplied.

Due to this higher supplied current, circulation of reactive energy on distribution networks results in:

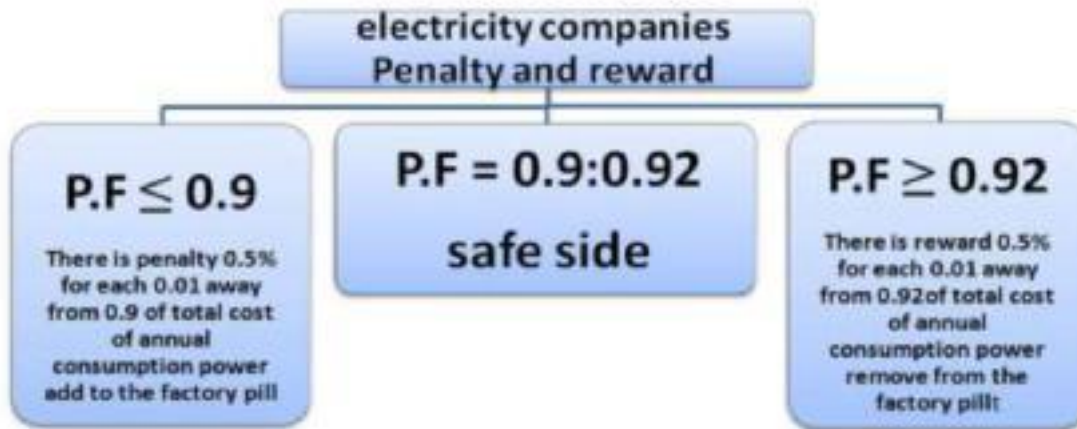
- Overload of transformers,
- Higher temperature rise of the supply cables,
- Additional losses,
- Large voltage drops,
- Higher energy consumption and cost,
- Less distributed active power.

For these reasons we need to make

"Power Factor Correction "

Note:

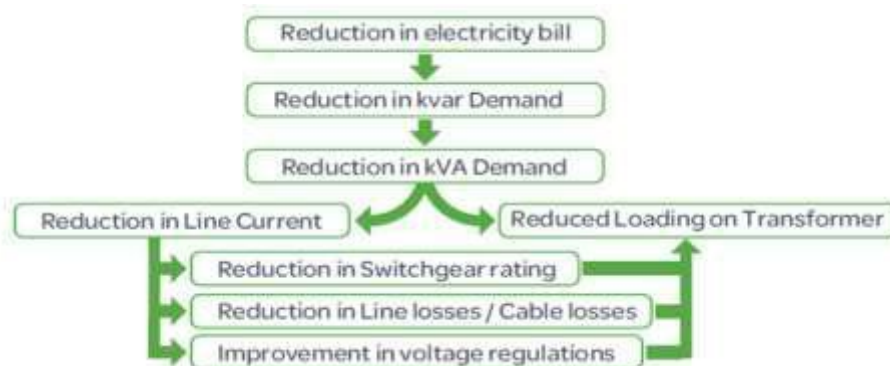
- In Egypt, the power factor penalty depends on how measured power factor is below 0.9.



For example:

- If factory has annual power pill 1,000,000 LE and has average power factor 0.7, so this factory will pay penalty to the electricity company which equal:
 $(0.9 - 0.7) \times 0.5 \times 1000000 = 100,000$ LE add to the annual power pill.
- If factory has annual power pill 1,000,000 LE and has average power factor 0.95, so this factory will have discount from the electricity company which equal:
 $(0.95 - 0.92) \times 0.5 \times 1000000 = 15,000$ LE remove from the annual pill.

Importance of power factor improvement:



1) *Elimination of power factor penalty :*

- *Improves your system power factor, reduces total KVA saving your money on your electric power bill.*

2) *Additional Capacity in Electrical System :*

- *Increase system capacity by reducing loading (KVA) on transformers, saving you from making new capital investment to serve new electrical loads.*

3) *Reduction of Distribution System Currents :*

- *A load or a plant consuming power P and reactive power Q under voltage V draws a current I given by:*

$$I = \sqrt{(\frac{P}{V})^2 + (\frac{Q}{V})^2}$$

4) *Reduction of I^2R Loss :*

- *When power factor improved reduce the drawn current by the load and since power loss is proportional to current squared ($P_{LOSS} = I^2R$), so improving the power factor reduce loss in the used conductors.*
- *Reduces system losses, saving energy costs and allowing your equipment to run cooler and more efficiently.*

5) *Reduction in Cross Sections Of Feeders And Cables :*

- *Improving load power factor means reduction in conductor cross section as a result of reduction in the drawing current.*

Power factor	Cable cross-section multiplying factor
1	1
0.80	1.25
0.60	1.67
0.40	2.50

6) *Reduce voltage drop.*

TYPES OF REACTIVE POWER COMPENSATION SYSTEM

1) static VAR COMPENSATION By using bank of capacitor

1.1) Fixed capacitors :

Consists of capacitor bank connected in parallel with the network, the total Reactive power of capacitor bank is provided by the assembly of capacitors of identical or different unit powers.

The bank is energized by a contactor that simultaneously supplies the capacitors (A single step).so this arrangement is employed to form a constant level of Compensation.

- Drawbacks of fixed power factor correction is the Danger of Excess Capacitance can cause leading Power Factor at light load operation conditions.

1.2) Automatic power factor correction system (Automatic capacitor bank):

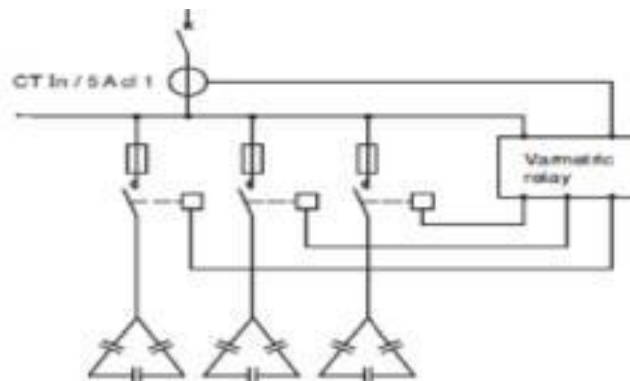
Consists of several capacitor banks of identical or different powers (several steps), energized separately according to the value of the power factor to be corrected.

Types of switching in APFC System:

- 1) Contactor Switching
- 2) Thyristor Switching

Power Factor sensing is done by the APFC Relay (Reactive Power Controller), which sends the control signal to the switching device (Contactor/Thyristor).

- The switching device takes the command (order) from the relay and switches the required Capacitor stages.



1) Synchronous condenser :

Synchronous condenser is a Synchronous motor at certain condition, this condition is:

- 1) Synchronous motor running at no-load
- 2) Synchronous motor operates at over-excited. (Lead current).

Disadvantage:

- 1)Have losses.
- 2) The maintenance cost is high.
- 3) It produces noise.

Location of Capacitors

There are several places to be chosen to install the correction capacitors, these choices are as the following:

1) **Individual compensation:**

Capacitors are installed at individual load directly.

Advantages:

Significant reactive currents no longer exist in the installation, so it reduces the size of all cables as well as the cable losses.

2) **Group compensation:**

Capacitors are installed at distribution points for group of loads.

Disadvantages:

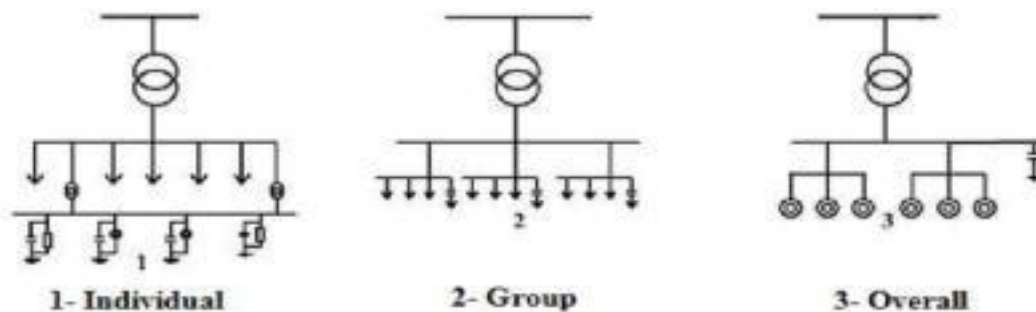
Reduces reactive current flows in cables before capacitors units (upstream of local distribution board), but the reactive current still flows in all conductors after capacitors units (downstream of local distribution board), so the sizing of these cables, and power losses in them, are not improved by the Group compensation.

3) Overall compensation:

Capacitors are installed at the main bus bar.

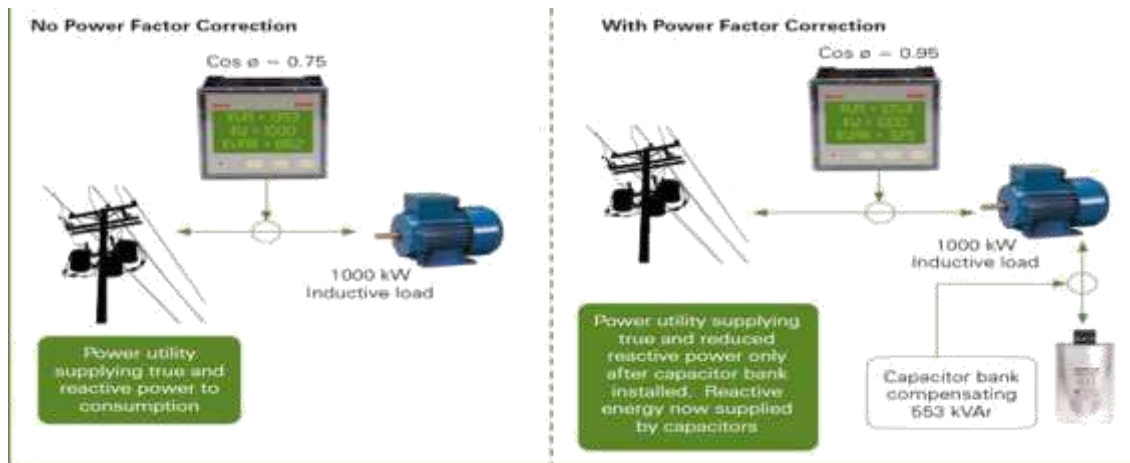
Disadvantages:

Reduces reactive current flows in cables before capacitors units (upstream of main LV distribution board), but the reactive current still flows in all conductors after capacitors units (downstream of main LV distribution board), so the sizing of these cables, and power losses in them, are not improved by the overall compensation.



How to make power factor correction?

• *Power factor correction brings the power factor of an AC power circuit closer to 1 by adding capacitors in parallel with load to supply load by part of required reactive power instead of take all these reactive powers from supply ,so the amount of the taking reactive power from the supply is reduced and since the amount of active power is constant (power of connected loads), So the amount of apparent power that drawn by the load from the source is reduced.*



Calculation of the Capacitors Ratings

Static capacitors units that used for power factor improvements are normally rated in terms of KVAR.

To calculate the KVAR required for particular load power factor improvement, three information's are necessary:

- (a) The load power or KW of the system or load.*
- (b) The existing power factor.*
- (c) The desired power factor.*

• First method

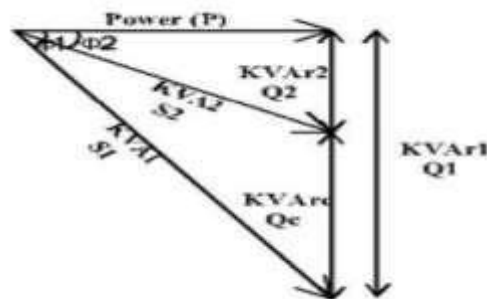


Fig. (a)

Fig. (a) Shows a typical required reactive power Q_c with power factor correction from $\cos \phi_1$ to $\cos \phi_2$ for load power P in KW

$$Q_c = Q_1 - Q_2$$

$$Q_c = P \tan \phi_1 - P \tan \phi_2$$

$$Q_c = P (\tan \phi_1 - \tan \phi_2)$$

$$C = Q_c / (2\pi f V^2)$$

Where

Q_c : The capacitor rating required for power factor correction (KVAR)(reactive power supplied from capacitor).

P : The load active power

C : The capacitor capacitance

f : The system frequency.

V : The capacitor terminal voltage

$\cos \phi_1$: The initial power factor.

$\cos \phi_2$: The new high power factor(desired power factor).

$(\tan \phi_1 - \tan \phi_2)$: The correction factor.

• **Second method**

By using table

Get the intersection between the desired power factor and the old power factor

$$Q_c = \text{intersection} \times P$$

Where

P : active power in (KW)

Before compensation		kvar rating of capacitor bank to install per kW of load, to improve cos φ (the power factor) or tan φ, to a given value													
tan φ	cos φ	tan φ	0.75	0.59	0.48	0.46	0.43	0.40	0.36	0.33	0.29	0.26	0.20	0.14	0.0
		cos φ	0.90	0.86	0.80	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1
2.20	0.40		1.557	1.601	1.905	1.832	1.861	1.895	1.924	1.959	1.998	2.037	2.085	2.146	2.288
2.22	0.41		1.474	1.625	1.742	1.769	1.798	1.831	1.840	1.866	1.935	1.979	2.021	2.082	2.225
2.16	0.42		1.413	1.561	1.681	1.709	1.738	1.771	1.800	1.838	1.874	1.913	1.961	2.022	2.164
2.10	0.43		1.356	1.499	1.624	1.651	1.680	1.713	1.742	1.778	1.816	1.855	1.903	1.964	2.107
2.04	0.44		1.299	1.441	1.558	1.585	1.614	1.647	1.677	1.712	1.751	1.790	1.937	1.899	2.041
1.98	0.45		1.239	1.384	1.501	1.532	1.561	1.592	1.629	1.659	1.695	1.737	1.784	1.846	1.988
1.93	0.46		1.179	1.330	1.448	1.473	1.502	1.539	1.567	1.600	1.636	1.677	1.725	1.786	1.929
1.88	0.47		1.130	1.278	1.397	1.425	1.454	1.485	1.510	1.532	1.588	1.629	1.677	1.758	1.891
1.83	0.48		1.076	1.229	1.343	1.370	1.400	1.430	1.464	1.497	1.594	1.575	1.623	1.684	1.826
1.78	0.49		1.030	1.179	1.297	1.326	1.355	1.388	1.420	1.453	1.480	1.530	1.579	1.639	1.782
1.73	0.50		0.982	1.232	1.218	1.276	1.303	1.337	1.369	1.403	1.441	1.481	1.520	1.590	1.732
1.68	0.51		0.936	1.087	1.202	1.230	1.257	1.291	1.323	1.357	1.395	1.435	1.483	1.544	1.686
1.64	0.52		0.894	1.043	1.160	1.188	1.215	1.249	1.281	1.315	1.353	1.393	1.441	1.502	1.644
1.60	0.53		0.850	1.000	1.116	1.144	1.171	1.205	1.237	1.271	1.308	1.349	1.397	1.458	1.600
1.56	0.54		0.809	0.959	1.075	1.103	1.130	1.164	1.196	1.230	1.268	1.308	1.356	1.417	1.559
1.52	0.55		0.768	0.918	1.035	1.063	1.090	1.124	1.156	1.190	1.228	1.268	1.316	1.377	1.519
1.48	0.56		0.730	0.879	0.996	1.024	1.051	1.085	1.117	1.151	1.189	1.229	1.277	1.338	1.480
1.44	0.57		0.692	0.841	0.958	0.986	1.013	1.047	1.079	1.113	1.151	1.191	1.239	1.300	1.442
1.40	0.58		0.665	0.805	0.921	0.949	0.976	1.010	1.042	1.076	1.114	1.154	1.202	1.263	1.405
1.37	0.59		0.618	0.758	0.864	0.892	0.919	0.953	1.005	1.039	1.077	1.117	1.165	1.226	1.368
1.33	0.60		0.584	0.733	0.840	0.878	0.905	0.938	0.971	1.005	1.043	1.083	1.131	1.192	1.334
1.30	0.61		0.549	0.699	0.815	0.843	0.870	0.904	0.936	0.970	1.008	1.046	1.096	1.157	1.299
1.27	0.62		0.515	0.665	0.781	0.809	0.836	0.870	0.902	0.936	0.974	1.014	1.062	1.123	1.265
1.23	0.63		0.483	0.633	0.749	0.777	0.804	0.838	0.870	0.904	0.942	0.982	1.030	1.091	1.233
1.20	0.64		0.450	0.601	0.716	0.744	0.771	0.805	0.837	0.871	0.909	0.949	0.997	1.058	1.200
1.17	0.65		0.419	0.569	0.685	0.713	0.740	0.774	0.806	0.840	0.878	0.916	0.966	1.027	1.169
1.14	0.66		0.388	0.538	0.654	0.682	0.709	0.743	0.775	0.809	0.847	0.887	0.935	0.996	1.138
1.11	0.67		0.358	0.508	0.624	0.652	0.679	0.713	0.745	0.779	0.817	0.857	0.905	0.966	1.108
1.08	0.68		0.329	0.479	0.595	0.623	0.650	0.684	0.716	0.750	0.788	0.828	0.876	0.937	1.079
1.05	0.69		0.299	0.449	0.565	0.593	0.620	0.654	0.686	0.720	0.758	0.798	0.846	0.907	1.049
1.02	0.70		0.270	0.420	0.536	0.564	0.591	0.625	0.657	0.691	0.729	0.769	0.811	0.870	1.020
0.99	0.71		0.242	0.392	0.508	0.536	0.563	0.597	0.629	0.663	0.701	0.741	0.783	0.850	0.992
0.96	0.72		0.213	0.364	0.479	0.507	0.534	0.568	0.600	0.634	0.672	0.712	0.754	0.821	0.963
0.94	0.73		0.186	0.336	0.452	0.480	0.507	0.541	0.573	0.607	0.645	0.685	0.727	0.794	0.936
0.91	0.74		0.159	0.309	0.425	0.453	0.480	0.514	0.546	0.580	0.618	0.658	0.700	0.767	0.909
0.88	0.75		0.132	0.282	0.398	0.426	0.453	0.487	0.519	0.553	0.591	0.631	0.673	0.740	0.882
0.86	0.76		0.105	0.255	0.371	0.399	0.426	0.460	0.492	0.526	0.564	0.604	0.652	0.713	0.855
0.83	0.77		0.079	0.229	0.345	0.373	0.400	0.434	0.466	0.500	0.538	0.578	0.620	0.687	0.829
0.80	0.78		0.053	0.202	0.319	0.347	0.374	0.408	0.440	0.474	0.512	0.552	0.594	0.661	0.803
0.78	0.79		0.026	0.176	0.292	0.320	0.347	0.381	0.413	0.447	0.485	0.525	0.567	0.634	0.776
0.75	0.80			0.150	0.266	0.294	0.321	0.355	0.387	0.421	0.459	0.499	0.541	0.608	0.750
0.72	0.81			0.124	0.240	0.268	0.295	0.329	0.361	0.395	0.433	0.473	0.515	0.582	0.724
0.70	0.82			0.099	0.214	0.242	0.269	0.303	0.335	0.369	0.407	0.447	0.489	0.556	0.698
0.67	0.83			0.072	0.188	0.216	0.243	0.277	0.309	0.343	0.381	0.421	0.463	0.530	0.672
0.65	0.84			0.046	0.162	0.190	0.217	0.251	0.283	0.317	0.355	0.395	0.437	0.504	0.645
0.62	0.85			0.020	0.136	0.164	0.191	0.225	0.257	0.291	0.329	0.369	0.417	0.478	0.620
0.59	0.86				0.109	0.140	0.167	0.198	0.230	0.264	0.301	0.343	0.390	0.450	0.593
0.57	0.87				0.083	0.114	0.141	0.172	0.204	0.238	0.275	0.317	0.364	0.424	0.567
0.54	0.88				0.054	0.085	0.112	0.143	0.175	0.209	0.246	0.288	0.335	0.395	0.538
0.51	0.89				0.029	0.059	0.086	0.117	0.149	0.183	0.220	0.262	0.309	0.369	0.512
0.48	0.90					0.031	0.058	0.089	0.121	0.155	0.192	0.234	0.281	0.341	0.484

Ex: Factory has total power 666KVA and PF=0.75 and we required make PF correction to be 0.93 ,calculate KVAR of capacitor . Total power of Factory $p=0.75*666=500KW$

• FIREST Method

$\therefore \cos \theta_{old}=0.75; \theta_{old} = \cos^{-1} (0.75) = .$
 $\therefore \cos \theta_{new}=0.93; \theta_{new} = \cos^{-1} (0.93) = .$

$$\begin{aligned}
 &= P [\tan \theta_{old} - \tan \theta_{new}] \\
 &= 500 [\tan (41.409) - \tan (21.56)] \\
 &= 244 \text{ KVAR}
 \end{aligned}$$

• Second Method

By using the table

The intersection between the row of $\cos\phi=0.75$

(before correction) & the row $\cos\phi=0.75$

(after correction) indicate the value of 0.487 KVAR of compensation per KW of load.

$$= 500 \times 0.487 = 244 \text{ KVAR}$$

• In our project

➤ *for second transformer:*

$$\text{total power} = 489078 \text{ w}$$

$$PF(\text{old}) = .8$$

$$PF(\text{new}) = .93$$

by using the table method

= .

$$= 173.6 \text{ kvar}$$

The used compensators are multi-stage (6-stage) as

1 stages of 50 kvar

5 stages of 25 kvar

Rating of CB that protects the capacitor bank is:

$$\therefore = \sqrt{3} \times V \times I \times \sin\phi$$

$$\therefore = \frac{1.732 \times 400 \times 266}{1.732} = 266 \text{ A}$$

$$\therefore = 1.25 \times 266 = 332.5 \text{ A}$$

$$\therefore \text{ the CB rating} = 400 \text{ A}$$

CHAPTER 9

Emergency system



Automatic transfer switch (ATS):

- Automatic Transfer Switches are an integral part of the power generation process, allowing smooth and immediate transfer of electrical current between multiple sources and the load .

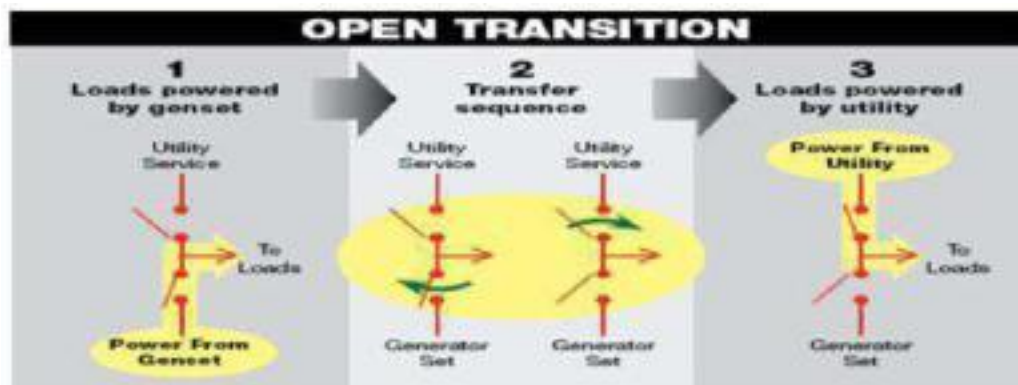
Theory of operation:

- The transfer switches senses the interruption of the utility power by means of sensors and relays and transfers electrical loads from a normal source to an emergency source (generator set). When a normal power is restored, the switch automatically retransfers to the normal source by the same way.

Types of ATS:

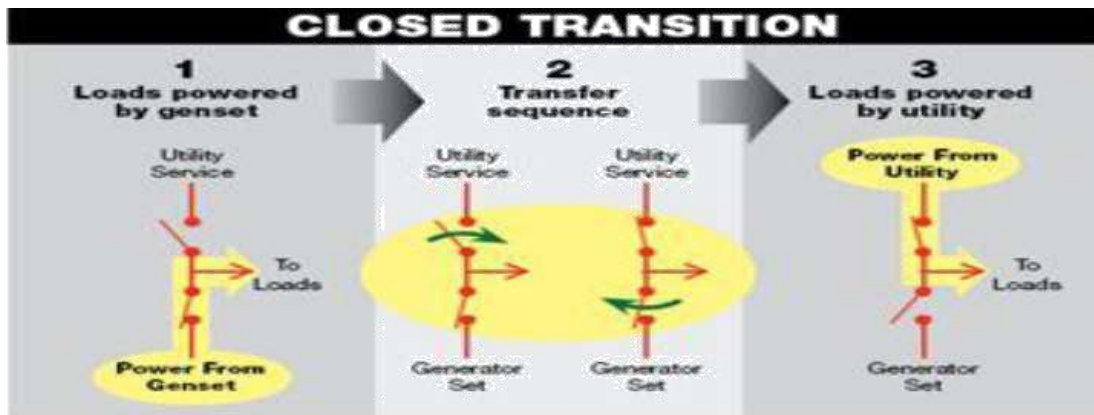
1) Open transition transfer device:

Open the connected source before closing the new source, causing a total power interruption for a short period of time.

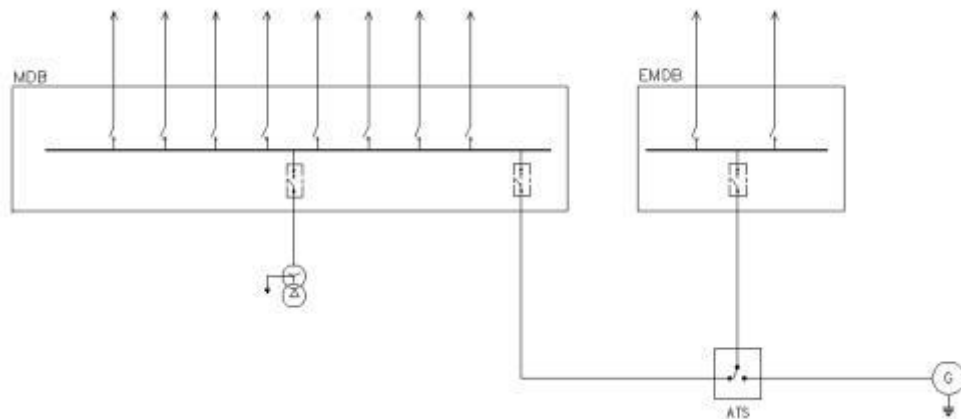


2) Fast closed-transition transfer devices:

Operates like an open-transition transfer switch when a source has failed, But will parallel the two sources for 100 milliseconds or less and then disconnect when both sources are available, so a total interruption of power is avoided.



The operational states of ATS:



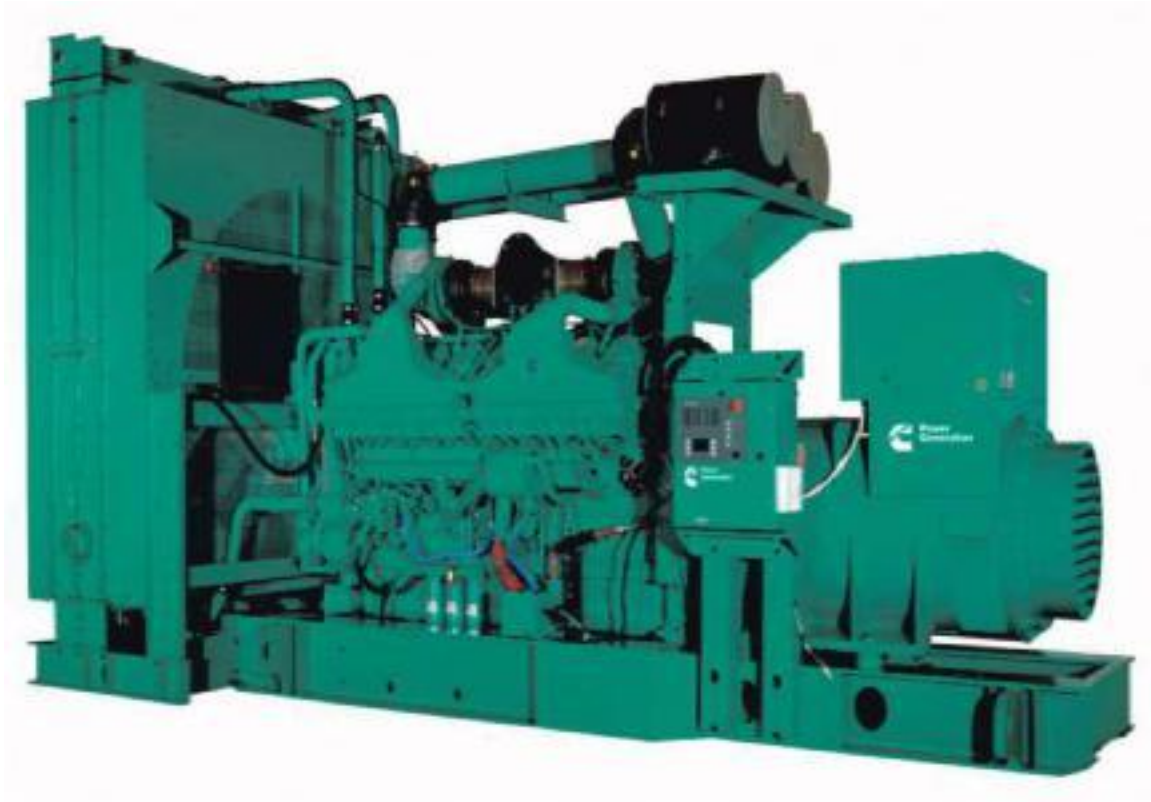
ATS operates when the original Ac-input voltage supply fails or when Loss of voltage on bus bar occurs.

Importance of Using ATS:

To keep continuity of supplying the loads.

To keep outages time to minimum value as the load automatically switched to an alternative stand by source, if the original preferred source is lost for any reason. It also ensures that different power sources are fully synchronized before their power is combined or loads are transferred, which is imperative for safe operation.

Diesel generator:



Basic Components of Standby Generator Set:

- 1) Diesel generator (engine & generator).
- 2) Over current protective devices.

Factors of diesel generator selection:

- 1) The required kilowatts or kilo-volt-ampere.
- 2) The required reserve capacity of the generator for future expansion in the load.
- 3) The system voltage level.
- 4) The required operation mechanism
 - Manual.
 - automatic
- 5) The generator size and cost.

Installation Conditions of standby generator:

- 1) The room should be great enough to contain the machine with its accessories with the existing of spaces, which allow safe, right maintenance and the ability of removing any part outside the room if any external maintenance is needed.
- 2) Good ventilation for this room is a must due to the large radiated heat from the generating unit.
- 3) The location of fuel tank and the feeding type are affected by the room height, usually large tanks are located in the ground level, and pump is needed to feed the machine with fuel.
- 4) We should care about exhaust path, it must be as short as possible. Also we must increase the cross section area of exhaust pipe.
- 5) The connection between the machine and exhaust pipe must be flexible to prevent the vibration transfer to exhaust pipe, which may damage it.
- 6) In case of using two-generating units, we must design the exhaust system of each generator separate than the other.
- 7) The temperature of exhaust pipe should not reaches 500 degree C beside the machine otherwise the heat reradiating from exhaust pipe to machine room.
- 8) The cables exiting from the generator terminal box must be flexible and withstand mechanical and thermal stresses.

The used diesel generators in this project:

For the second transformer:

$$\begin{aligned} \text{(KVA) required} &= 20301 + 22083 + 22083 + 22083 + 22083 \\ &= 108633 \text{ VA} \end{aligned}$$

$$\begin{aligned} \text{(KVA) required by taking reserve load into account} &= 108633 \times 1.2 \\ &= 130359.6 \text{ VA} \end{aligned}$$

The required diesel generator is [150 KVA]

Uninterruptible Power Supplies

Function of UPS

The UPS systems are especially required in places where power outages and fluctuations occur frequently. A UPS provides a backup power circuitry to supply vital systems when a power outage occurs. In situations where short time power fluctuations or disturbed voltage occur, a UPS provides constant power to keep the important loads running. During extended power failures, a UPS provides backup power to keep the systems running long enough so that they can be gracefully powered down.

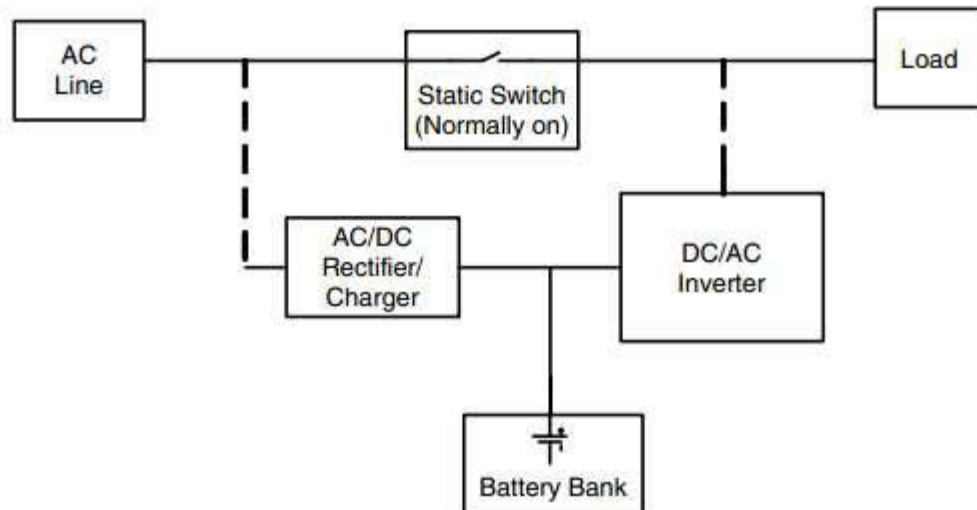
Classification of UPS



Standby UPS

Configuration

As shown in the figure the configuration of a typical standby UPS system. It consists of an AC/DC converter, a battery bank, a DC/AC inverter, and a static switch.



Operation

The static switch is on during the normal mode of operation. Therefore, load is supplied from the AC line directly without any power conditioning. At the same time, the AC/DC rectifier charges the battery set. This converter is rated at a much lower power rating than the power demand of the load. When a power outage occurs or the primary power is out of a given preset tolerance, the static switch is opened and the DC/AC inverter provides power to the load from the battery set for the duration of the preset backup time or till the AC line is back again. This inverter The transition time from the AC line to DC/AC inverter is usually about one quarter of the line cycle, which is enough for most of the applications such as personal computers.

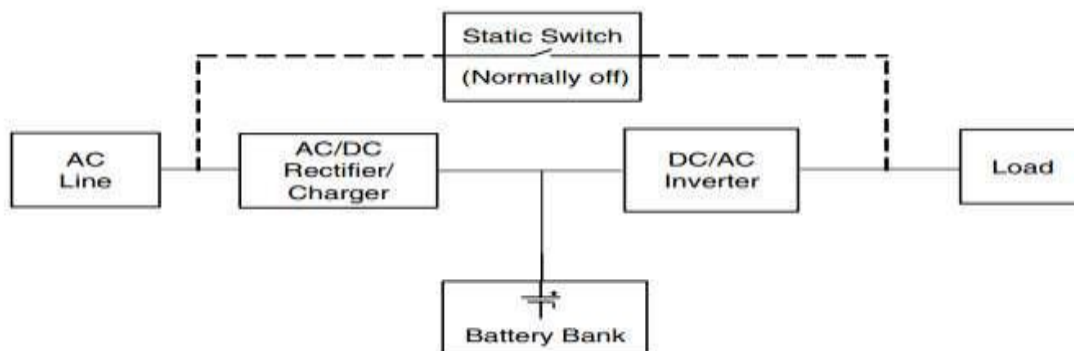
The advantage of Standby UPS

Simple design, low cost, and small size



Online UPS

Similar to standby UPS systems, on-line UPS systems also consist of a rectifier/charger, a battery set, an inverter, and a static switch. Other names for this configuration are "True UPS," "inverter preferred UPS," and "double-conversion UPS" shows the block diagram of a typical on-line UPS. The rectifier/charger continuously supplies power to the DC bus



Operation

During the normal mode of operation, the power to the load is continuously supplied via the rectifier/charger and inverter. In fact, a double conversion from AC to DC and then from DC to AC takes place. When the AC input voltage is outside the preset tolerance, the inverter and battery maintain continuity of power to the load. The duration of this mode is the duration of preset UPS backup time or till the AC line returns within the preset tolerance.

Advantages

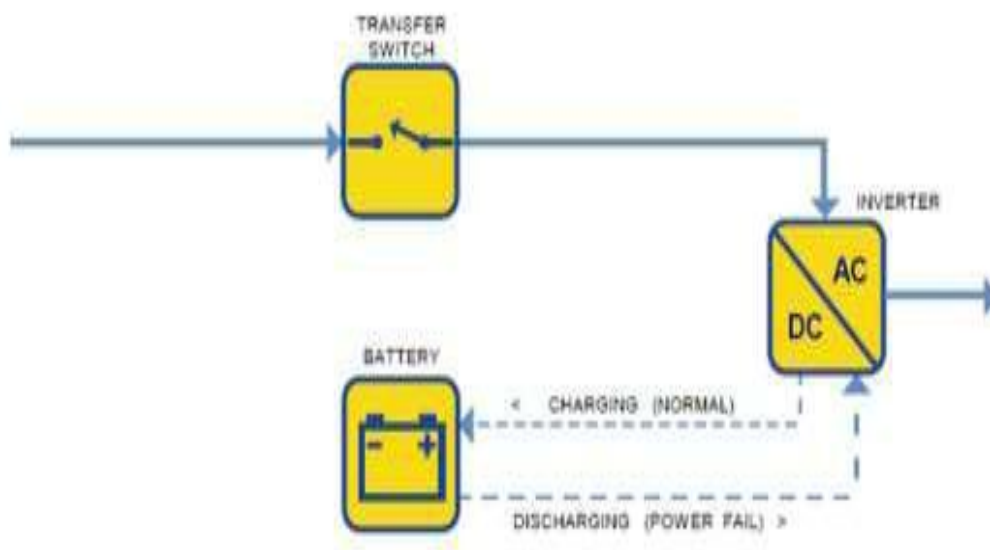
It is connected in series with the load; hence, there is no transfer time associated with the transition from normal mode to stored energy mode



Line-interactive UPS

Configuration

The inverter is connected in parallel acts to backup utility power. The battery to inverter is always connected to the output of the UPS.



Operating

At normal operation the load is supplied from the Ac-input power via a parallel connection of the UPS inverter with the Ac main. The inverter charges the battery (reversible operation)

When the input power fails, the transfer switch opens and the power flows from the battery to the UPS output. The inverter and battery continue to support load power until the end of the stored energy time or the Ac-input return to within UPS preset tolerances, at which point the UPS returns to normal mode

Advantages

➤ Low cost.



The selection of UPS is according to:

- 1) VA rating.
- 2) Efficiency.
- 3) Cost.
- 4) Operating time.

Chapter 10

Earthing systems



Introduction:

Types of Electrical Hazards:

1. **Direct hazards:** these are electrical hazards due to direct exposure to electricity, e.g. electrical shocks.
2. **Indirect hazards:** these are non-electrical hazards but due to electrical reasons, e.g. fires, burns, explosions.

Sources of Electrical Accidents

Electric shock: An Electrical shock is a sudden and accidental stimulation of the body's nervous system by an electric current. The degree of danger for the victim is a function of the magnitude of the current, the parts of the body through which the current passes, and the duration of current flow. When a current exceeding 30 mA passes through a part of a human body, the person will be in a serious danger if the current is not interrupted in a very short time.

Effect of Current on Human Body (For 3 Seconds)

Current (mA)	Effect
0.7-1.7	Beginning of sensation
6-10	Min. value of let-go current
15	Painful
20-30	Muscle paralysis
50-100	Difficult/failure in respiratory system
100-2000	Death

<i>Max current (mA)</i>	<i>Flow time</i>	<i>Reaction</i>
<i>0 - 0.5</i>	<i>constant</i>	<i>No Effect</i>
<i>0.5 - 5</i>	<i>constant</i>	<i>Let-go</i>
<i>5 - 30</i>	<i>Min</i>	<i>Difficult to let go</i>
<i>30 -50</i>	<i>Sec</i>	<i>Irregular heartbeat- Fainting</i>
<i>More than 100</i>	<i>More than 20 mS</i>	<i>Fainting - Death</i>

To calculate the max current safe

Direct and indirect shock:

Standards and regulations distinguish two kinds of dangerous contact

1- Direct shock:

Direct shock occurs when contact by persons or livestock is made directly with a live part.

An RCD (residual current devices) on the line side of the point of contact measures the current passing through the person and interrupts the current. When the risk is based on the installation environment or the persons and the risk is independent of the earthing system.

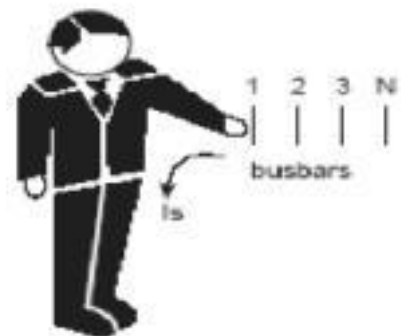


fig. G2: direct contact.

Is: touch current

Protection against Direct Shocks:

- 1. Insulation of live parts.*
- 2. Protection by barriers or enclosures.*
- 3. Protection by obstacles.*
- 4. Protection by placing out reach.*
- 5. Supplementary protection against direct contact by residual current device*

❖ *Sensitivity: $I_n = 30 \text{ mA}$*



- ❖ *Operation: 40 ms when 5 I_n (150 mA) is applied*
- ❖ *This measure cannot be used as a sole means of protection against direct shocks.*
- ❖ *An RCD is an electrical safety device designed specifically to switch the electricity immediately off when an electricity "leaking" (to earth) is detected at a level harmful to a person who is using electrical equipment.*

2- Indirect shock:

Indirect shock occurs when contact by persons or livestock is made with exposed conductive part during the period of earth fault.

An indirect contact refers to a person coming into contact with a conductive part which is not normally alive, but has become alive accidentally (due to insulation failure or some other cause). Indirect shock is mainly affected by arrangement of live conductors and type of earthing.

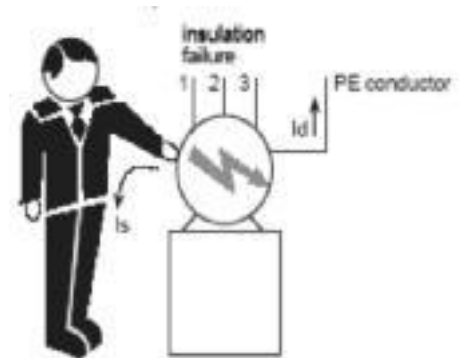


fig. G3: indirect contact.

I_d: insulation fault current

Protection against Indirect Shocks:

- 1. Making sure that when a fault occurs and makes the parts live, it results in the supply being cut off within a safe time. In practice, this involves limitation of earth fault loop impedance.*
- 2. Cutting off the supply before a fatal shock can be received using a residual current device.*
- 3. Applying local supplementary equipotential bonding which will ensure that the resistance between parts which can be touched simultaneously is so low that it is impossible for a dangerous potential difference to exist between them.*

Step potential and touch potential:

Step potential:

- *The flow of ground current between parts of the ground gives rise to a step potential.*

- *Step potential is defined as the difference in surface potential experienced by a person bridging a distance of 1 m with his feet without contacting any other grounded object.*
- *The value of the maximum safe step potential depends on the resistivity of the top layer of surface material, and on the duration of the current flow.*
- *For example, for a substation with a 10 cm layer of crushed rock and current flowing for 0.5 s, the maximum value of the step potential is approximately 3100 V. Touch potential: is the potential difference between a surface potential at a point a person is standing, and a grounded metallic structure at a normal maximum reach (1 m). For the same situation as above, the maximum safe touch potential is approximately 880 V.*

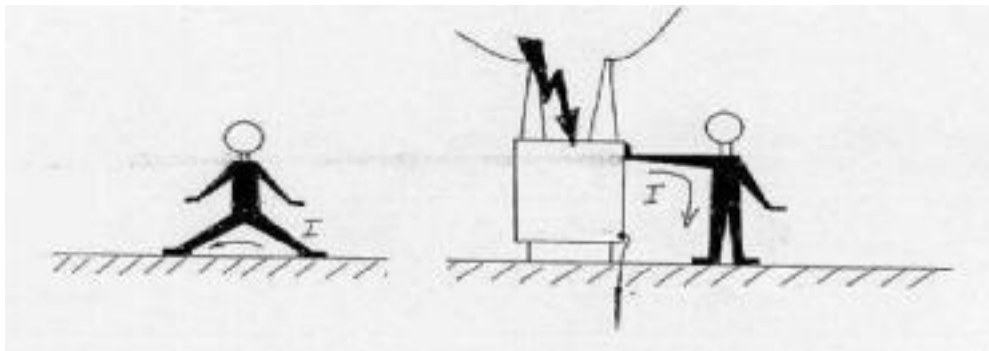


Illustration of a Step Potential and Touch Potential
Protection against electric shocks due to step potential and/or touch potential:

- ❖ *Increase the resistance of the ground surface area by using suitable insulating materials.*
- ❖ *Increase the human body resistance by wearing hand gloves, insulating wears.*
- ❖ *Decreasing the value of grounding resistance (5-25 ohms), to the least possible.*
- ❖ *Decrease the time duration of the electric shock by using fast and efficient protective devices.*

Grounding:

What is Grounding?

Grounding is the permanent and intentional connection of all metal parts of an electrical power system to the earth.

A Grounding Conductor must have a sufficient current rating to carry any possible fault current and sufficiently low impedance to limit the voltage rise above the ground potential.

Why Grounding?

1- Personal Safety: *Personal safety is provided by low impedance grounding and bonding between metallic equipment, chassis, piping, and other conductive objects so that currents, due to faults or lightning, do not result in voltages sufficient to cause a shock hazard.*

2- Equipment and Building Protection: *Equipment and building protection is provided by low impedance grounding and bonding between electrical services, protective devices, equipment and other conductive objects so that faults or lightning currents do not result in hazardous voltages within the building. Also, the proper operation of overcurrent protective devices is frequently dependent upon low impedance fault current paths.*

3- Electrical Noise Reduction:

Proper grounding aids in electrical noise reduction and ensures:

- The impedance between the signal ground points throughout the building is minimized.*
- The voltage potentials between interconnected equipment are minimized.*
- That the effects of electrical and magnetic field coupling are minimized.*

Types of Grounding

- System grounding: is the electrical connection of one of the current carrying conductors of the electrical system to the ground.*
- Equipment grounding: is the electrical connection of all the metal parts that do not carry current of all electrical equipment to the ground.*
- The equipment grounding circuit includes the metal enclosures, frames, and support structures of the circuit entrance equipment, transformers, switchboards, panels, switches, breakers, motors, generators, lighting equipment, cabinets, cable trays, racks etc.*

- These metal parts are interconnected by bare copper wires and metal sheaths and armors of cables.

- All parts of the grounding circuit must have adequate current carrying capacity for any current that is likely to flow through the circuit.

Earthing systems:

As far as earthing types are concerned, letter classifications are used.

The first letter indicates the type of supply earthing.

T- Indicates that one or more points of the Supply are directly earthed (For example, the earthed neutral at the transformer).

I- indicates either that the supply system is not earthed at all, or that the earthing includes deliberately-inserted impedance, the purpose of which is to limit fault current.

The second letter indicates the earthing arrangement in the installation.

T - All exposed conductive metalwork is connected directly to earth.

N - All exposed conductive metalwork is connected directly to an earthed supply conductor provided by the Electricity Supply Company.

The third and fourth letters indicate the arrangement of the earthed supply conductor system.

S- Neutral and earth conductor systems are quite separate. **C** - Neutral and earth are combined into a single.

1- **T-N system:** transformer neutral earthed, frame connected to neutral have three types

a) **TN-C system:** combined between neutral & protective conductor(PEN)

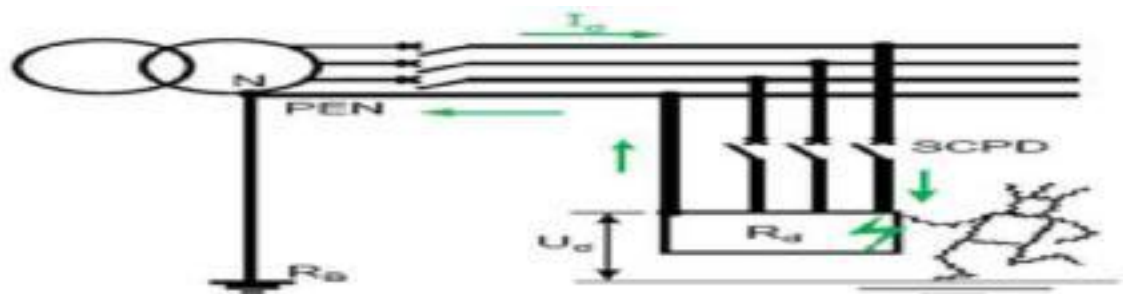


Fig. 1 a) : Insulation fault on a network operated in TN-C

b) ***TN-S system:*** separate the neutral & protective conductor

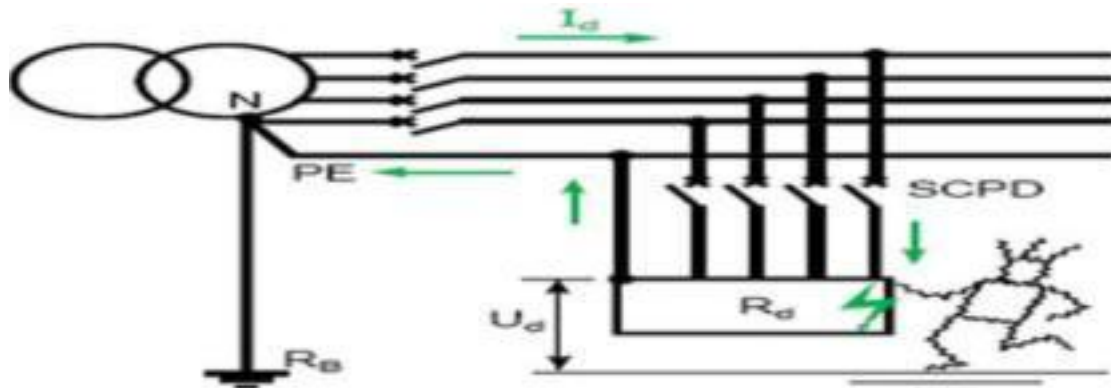


Fig. 1 b) : Insulation fault on a network operated in TN-S.

c) ***TN-C-S system:*** use of a TN-S downstream from a TN-C.

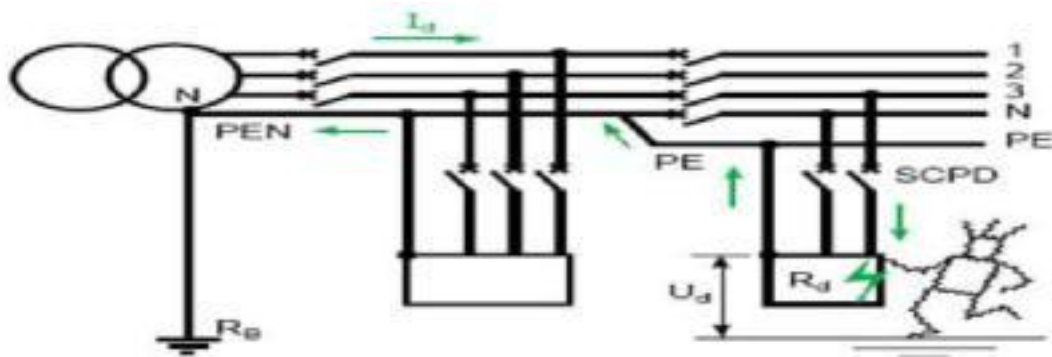


Fig. 1 c) : Insulation fault on a network operated in TN-C-S.

2- **T-T system:** transformer neutral earthed, frame earthed.

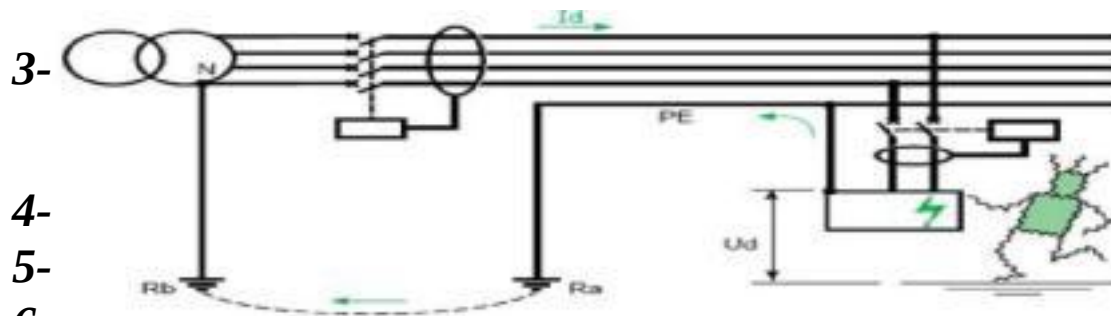


Fig. 2 : fault current and voltage in TT system.

3- I-T system: transformer neutral UN earthed, frame earthed.

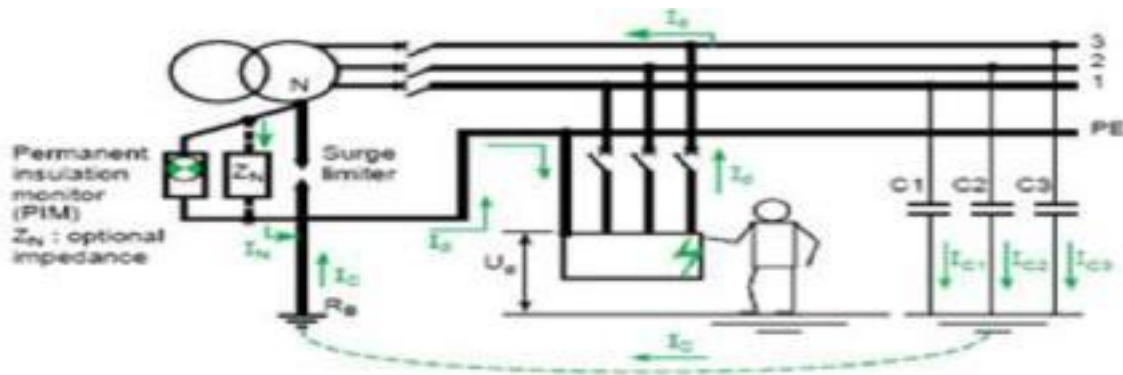


Fig. 3 a) : Single insulation fault on a network operated in IT.

Its application:

- 1- (T-T) system used for standby power supply "UPS".
- 2- (T-T) or (I-T) system used in load with high fault current "motor".
- 3- (T-T) or (I-T) system used in control or monitoring device.
- 4- (TN-S) system used in furnaces.
- 5- (TN-S) system used for very long network.

What is protective multiple earthing (PEN system):

►► If a continuous metallic earth conductor exists from the star point of the supply transformer to the earthing terminal of the installation, it will run throughout in parallel with the installation neutral, which will be at the same potential. It therefore seems logical that one of these conductors should be removed, with that remaining acting as a combined protective and neutral conductor (PEN). When this is done, we have a TN-C-S installation. The combined earth and neutral system will apply only to the supply, and not to the installation.

Substation Grounding:

Steps for Substation Grounding four steps

- Soil resistivity test
- Calculation of maximum ground fault current
- Design of grid
- Measurement of soil resistivity

1- Soil Resistivity Test:

Testing soil samples in the lab, the three pin method, and the four pin method are three procedures used to determine soil resistivity in the substation lot.

2- Maximum Ground Fault Current:

The value of maximum ground fault current must be calculated for use in the substation ground design.

3- Grid Design:

- a) Power transformers require two grounds, each capable of carrying full fault current.*
- b) Lightning arresters require one ground connected from the grid directly to the lightning arrester base. The ground must be capable of carrying the discharge.*
- c) Dead-end towers normally require one ground per each leg connected to the structure.*

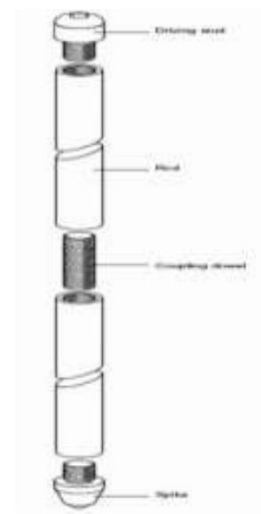
4- Measurement of Soil Resistivity:

The most common method used to measure the soil resistivity is the four pin method

Earthing components:

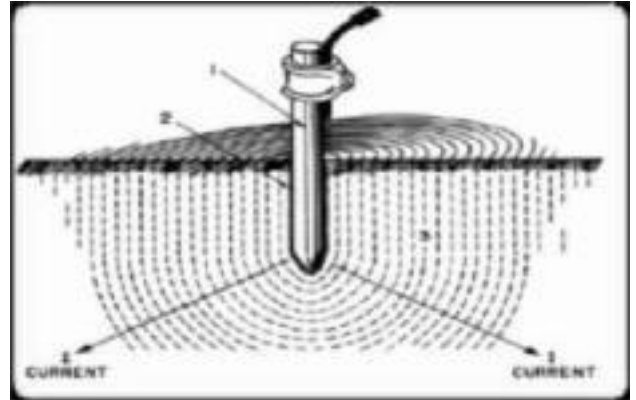
Ground clamps.

- 1) Ground electrodes & rod.*
- 2) Man hole.*
- 3) Sleeves.*
- 4) Coupling.*



THEORETICAL CALCULATIONS METHOD:

When current flows from a ground electrode into the surrounding soil, it is described as flowing through a series of concentric shells of increasing diameter.



From figure:

1 $\equiv$ Rod resistance (can be neglected).

2 $\equiv$ contact resistance of the electrode to the surround medium (can be neglected).

3 $\equiv$ Resistance of surrounding earth to current flow.

Theoretical, the ground resistance of a hemi-sphere derived from general formula

$$R_g = \frac{\rho}{2\pi L} \left(\ln \frac{4L}{d} \right)$$

Then the final form is

The most common formula for single cylindrical electrode,

$$R_g = \left(\frac{\rho}{2\pi L} \right) \left(\ln \frac{4L}{d} \right)$$

Where, ρ = Resistivity of the soil in Ω -m.

L = length of electrode, d = diameter of electrode.

r = radius of a hemi-sphere, R_g = ground resistance.

By making approximation calculation so, ground resistance for one vertical electrode

$$R_g = \frac{\rho}{2\pi L} \left(\ln \frac{4L}{d} \right)$$

Then, the total ground resistance for many vertical electrodes connected in parallel are

Where, L = length of vertical electrode.

N = number of electrodes connected in parallel.

P = Resistivity of the soil in Ω -m.

S = horizontal distance between two electrode.

η = screening coefficient "less than 1"

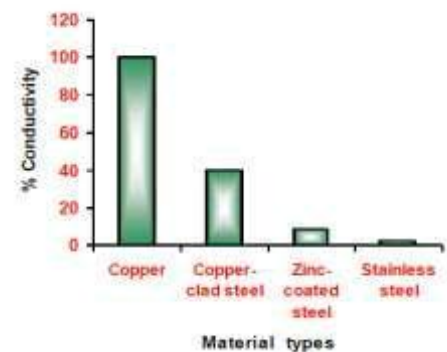
That depends on: 1) the ratio between (S/L)

2) The number of electrode (N)

η	N	/	η	N	/	η	N	/
0.95 - 0.97	2	3	0.93 - 0.95	2	2	0.8 - 0.87	2	1
0.92 - 0.95	3	3	0.9 - 0.92	3	2	0.76 - 0.8	3	1
0.89 - 0.91	5	3	0.86 - 0.89	5	2	0.67 - 0.72	5	1
0.82 - 0.88	10	3	0.8 - 0.85	10	2	0.56 - 0.62	10	1
0.79 - 0.81	20	3	0.76 - 0.79	20	2	0.5 - 0.47	20	1

Earth electrode materials: (IEC 60364-5-54-1980)

- Bare copper.
- Tin-coated copper.
- Zinc-coated copper.
- Hot-dip galvanized steel.
- Stainless steel. - Copper sheathed steel.
- Steel with electro-deposited copper coating.



Soil resistivity: it prefer to choose the soil with low resistivity to achieve the most economical grounding system and low ground resistance.

Factors affecting the soil resistivity:

- 1) Moisture content in soil.
- 2) Temperature of earth layer
- 3) Adding chemical salts.
- 4) Type of soil (moist-clay-bedrock-sand)

All this factors have inversely proportional with soil resistivity.

Typical resistivity ohm -m	Soil type
2-2.7	Marshy ground
4-150	Loam and clay
60-400	Chalk
90-8000	Sand
200 upwards	Peat
250-500	Sandy gravel
1000 upward	Rock

We assume its soil is Sandy gravel So, its soil resistivity = . .

Assume, we put the electrode on depth = 6 meter,

Where we make coupling between 7 rod connected in series each rod have length = 120 Cm =1.2 meter.

And we connected them by using horizontal tape (4*40mm²)

So, the length of one electrode = 7*1.2= 8.4meter, L = 8 m.

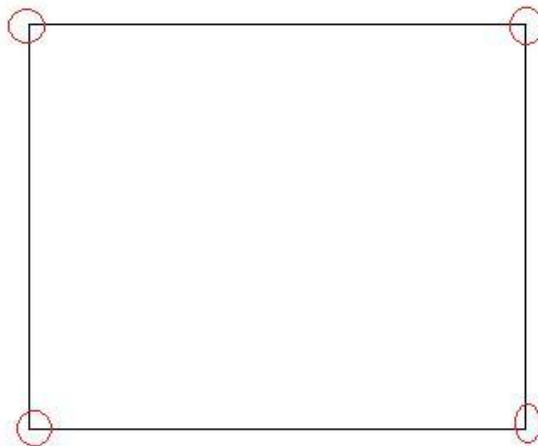
*According to (IEC 60364-5-54-1980) the ground resistance
for distribution substation not exceeds (1-5) Ω (suitable)*

	<i>R.Area</i>	<i>Area</i>	<i>N</i>		η				<i>Rg</i>	
	192	13*13	4	2	0.9		8.2	12		4.8Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω
	212	14*14	4	2	0.9		8.2	11		4.7Ω
	254	15*15	4	2	0.9		8.2	10.4		4.5Ω
	212	14*14	4	2	0.9		8.2	11		4.7Ω
	254	15*15	4	2	0.9		8.2	10.4		4.5Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	286	16*16	4	2	0.9		8.2	9.7		4.4Ω
	286	16*16	4	2	0.9		8.2	9.7		4.4Ω
	290	16*16	4	2	0.9		8.2	9.7		4.4Ω
	358	18*18	12	1	0.54		4.6	8.6		3Ω
	358	18*18	12	1	0.54		4.6	8.6		3Ω
	358	18*18	12	1	0.54		4.6	8.6		3Ω
	358	18*18	12	1	0.54		4.6	8.6		3Ω
	510	22*22	12	1	0.54		4.6	7.1		2.7Ω
	270	16*16	4	2	0.9		8.2	9.7		4.4Ω
	180	13*13	4	2	0.9		8.2	12		4.8Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	624	24*24	12	1	0.54		4.6	6.5		2.6Ω
	418	20*20	12	1	0.54		4.6	7.8		2.8Ω

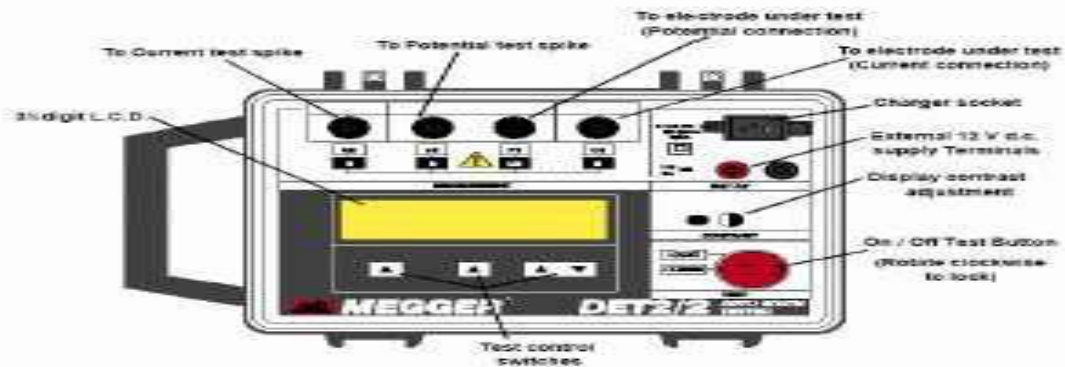
soil is chemically treated by using salts(magnesium sulfate salt, copper sulfate and sodium chloride salt or coal) to reduce the resistivity of the soil

Area	i_s A	t (break time)	Area(mm ²) $a=9\sqrt{t \cdot i_{sc}}$	stander(mm ²)	Length of cable (m)	resistance
13*13	1463.014711	0.1	4.163812863	6	5	0.02136
20*20	13103.53441	0.1	37.29331271	50	5	0.178
20*20	13950.18332	0.1	39.70291776	50	5	0.178
20*20	10760.64362	0.1	30.62532865	35	5	0.1246
14*14	9984.466865	0.1	28.41629086	35	5	0.1246
15*15	13446.27396	0.1	38.26876657	50	5	0.178
14*14	18038.86582	0.1	51.33951216	75	5	0.267
15*15	18038.86582	0.1	51.33951216	75	5	0.267
24*24	13985.77363	0.1	39.80420955	50	5	0.178
24*24	11648.86002	0.1	33.15323681	50	5	0.178

The Conductor Resistance From The D.B To The Earth Electrode



Measurement



Auto Earth Tester

Practical field METHOD:

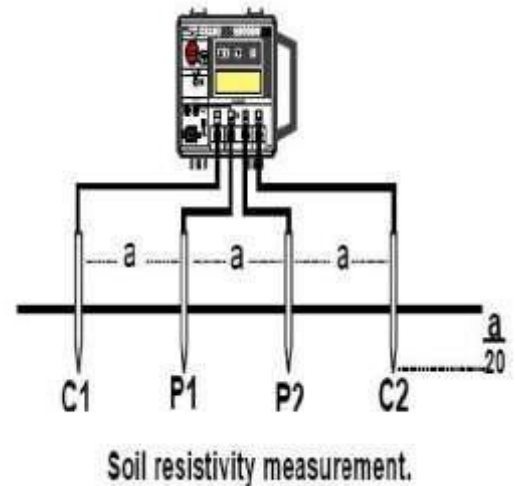
1) Soil resistivity measurement: (IEEE std. 81-1983)

Firstly, we measure the soil resistivity by "Four-point Method": Insert four-point with equally spaced into test area "Manage Building and factory", then pass a known current from a current Generator between outer electrodes then measure the Resistance between the inner electrodes by Megger device such that: =

Where:

a = distance between electrodes (Meters)

R = reading of Megger (Ω)



2)Ground resistance

Measurement: Secondly, we measure groundResistance by

" Three -pointMethod" that we use threeElectrodes, which are:

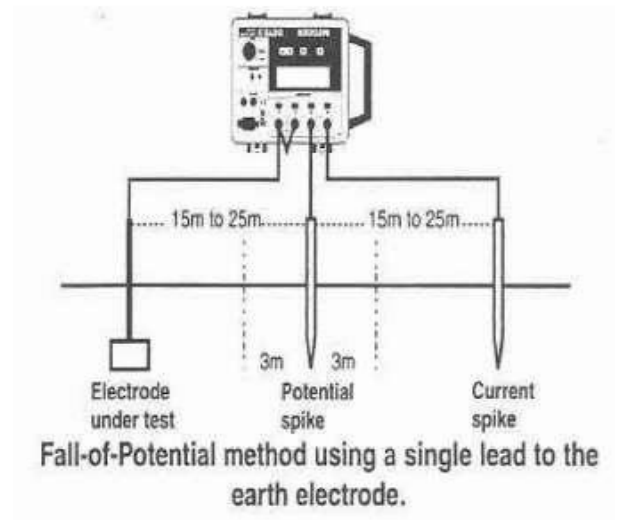
X = groundElectrode under test.

Y = auxiliary potential spike.

Z = auxiliary current spike.

That we apply current between (*X*, *Z*) electrode thenMeasure the current by ammeter (*I*),

And measure thePotential difference between (*X*, *Y*) electrode by voltmeter (*E*) From ohms law = * So, ground resistance = / (ohm)



Chapter 11

The protection devices



Introduction

All the distribution circuits are subjected to over loading and short circuits as a result of insulation failure or other reasons. devices that used to protect feeding circuits.

Circuit breaker specifications

The circuit breaker specification can be specified by two important values

- 1. Rated current of the circuit*
- 2. Short circuit capacity*

***The first value** → specifies rated current of the circuit specify the rated value that passes through the circuit breaker without making the breaker trips.*

And it has a standard values 6 , 10 , 15 ,20 ,25 ,32 , 40 ,50 ,63 ,80 , 100 , 125 and up to 6300A

***The second value** → it means the peak value of short circuit current that passes through the circuit breaker without making it burn.*

And it has a standard values 3,6,10,15,22,35,50,75,80,100.

Types of circuit breaker.

The circuit breaker used in low voltage are considered a relay and C.B.

- 1. Miniature C.B.*
- 2. Molded case Circuit Breaker.*
- 3. Ground fault Circuit Breaker.*

And there is another device is different in design and is similar to the fuse.

The main difference between the miniature C.B. and Molded Case Circuit breaker. The first type can carry short circuit current up to 10KA. Can carry short circuit currents up to 100KA.

Miniature C.B.

The figures shows different shapes of MCB the right figure are for single phase miniature C.B. and the left figure for three phase circuit breaker



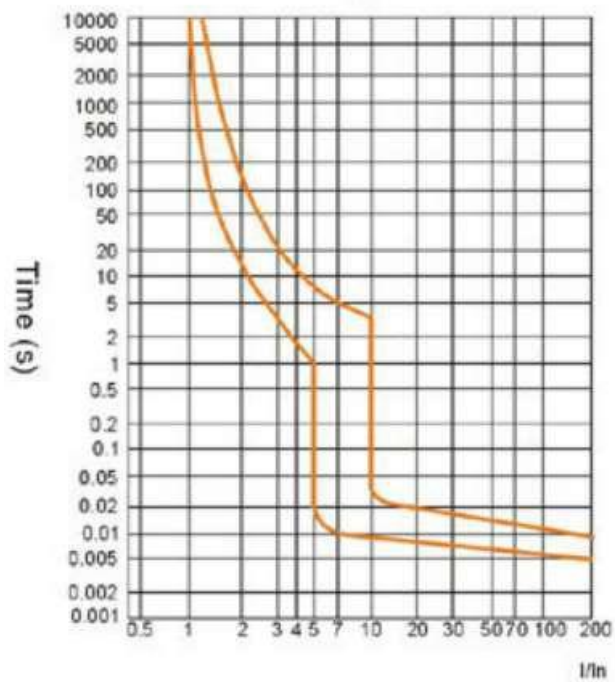
1-pole



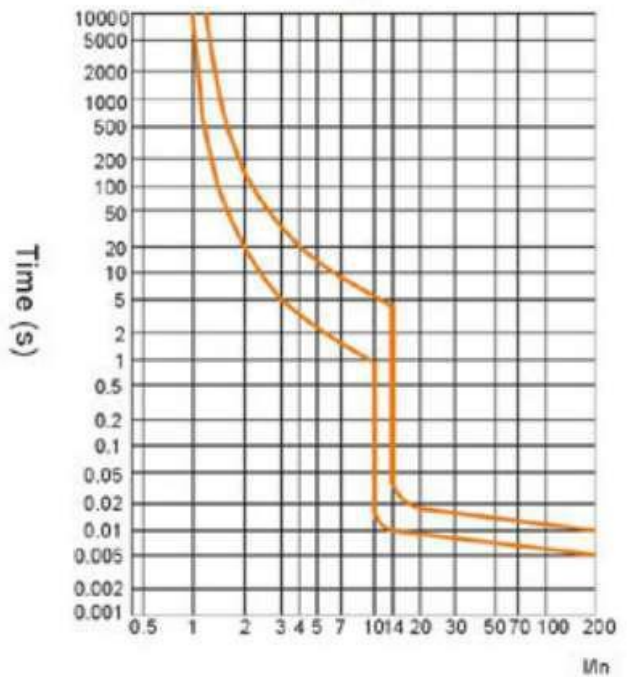
3-poles

In case of the MCB the relation between the fault current and the trip time And the figures below shows C-Type and D-Type

C type



D type



The first zone → and it includes the values below the rated breaker values and at these zone the breaker never trips.

The second zone → and this lies between the I_{RATED} and at this region the C.B. tripping current is inversely proportional with time in seconds the range of this zone for type-c is up to 7 times more than the rated value of current. But for type -d it is up to 10 times the rated value

The third zone → it is from the vertical region up to the highest short circuit at this region the C.B. trips instantaneously.

If short circuit current passes through the circuit breaker more than the short circuit breaker capacity .

Notes

- The type c and type D are used in the circuits of motors but type b used for the lighting and heating circuits
- At some applications because its low cost compared with C.B.
- The region before the C.B instantaneous, the fuse is faster than C.B.

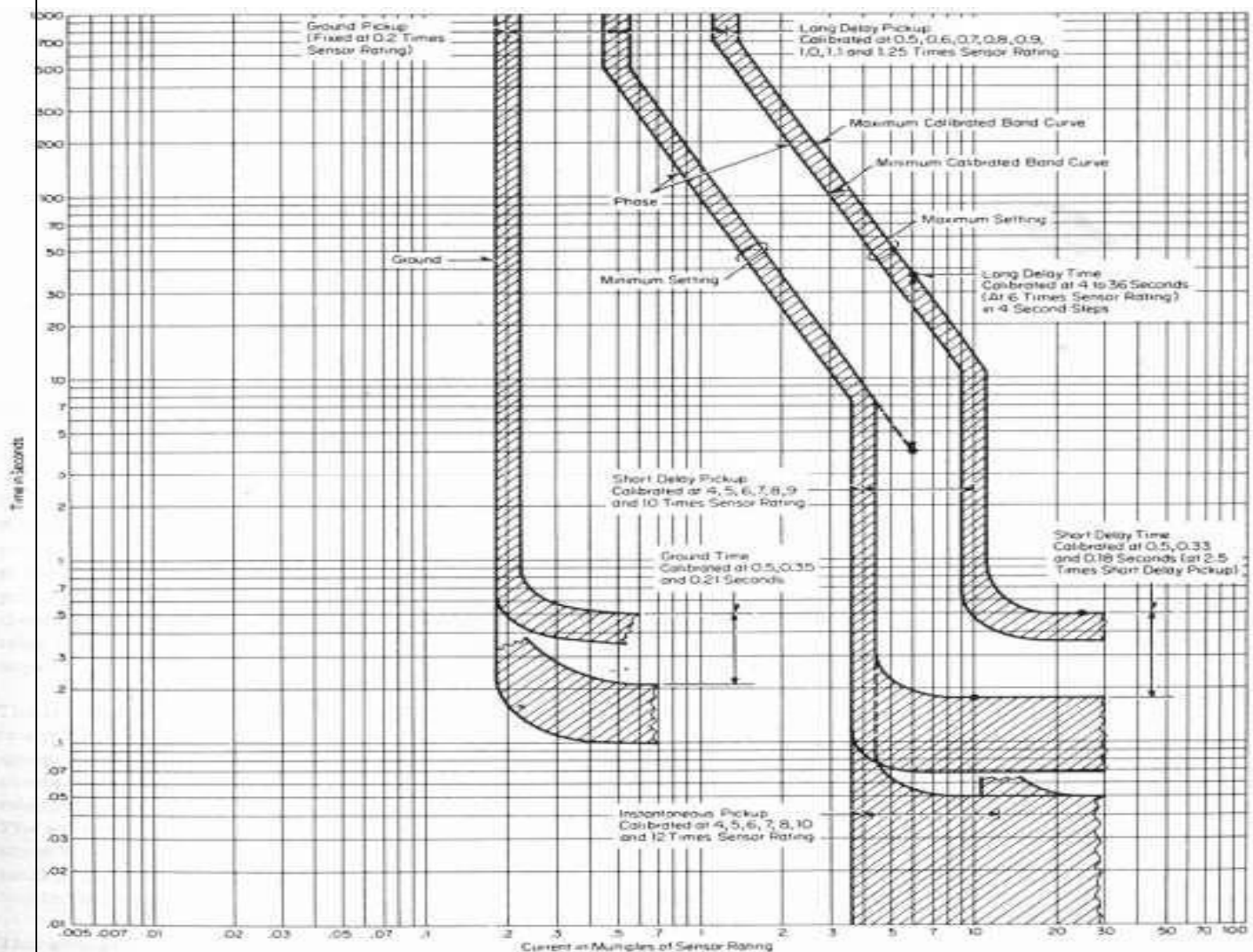
Molded case circuit breaker.

It's more complicated more than the miniature C.B. , It is used for protection of the main Feeders. It can carry S.C. currents more than miniature C.B. , this type is more flexible in the relation between the tripping time and the fault current.

We can adjust 3 parameters ($I_{thermal}$, T_{Mag} , I_{mag})



As the MCCB short circuit value increases as the more parameters are adjusted



For example the figure above the first and the second curve from right represent the range at which the C.B. can be adjusted (S.C. and tripping time)

But the third curve from right represents the relation between the tripping time and the ground faults.

The figure in the previous page is consisted of two parts

First part → long delay time it's concerned with the faults with the low values. And it starts with a vertical column passing through a sloping line and ends at a vertical column of the short delay part.

We can adjust the long delay time parameters

➤ *the pickup current can be adjusted at a value* of 0.5, 0.7, 0.9, 1, and 1.25 of the rated current of the C.B.

➤ *Second the tripping time can be adjusted*, the most common adjusted times are 4, 12, 20, 28 second. for example if we adjust the breaker current at a value say 100A, and adjust the tripping time for this current value say 4 seconds. But it's known that the fault current can exceed the rated current with a variable values so it's decided the breaker will trip at time of 4 seconds only when the current passes through it is 6 times the preset adjusted value (600A).

The sloping line is adjusted between the long delay time and the short delay time with a slope of -2 at the paper type (log - log).

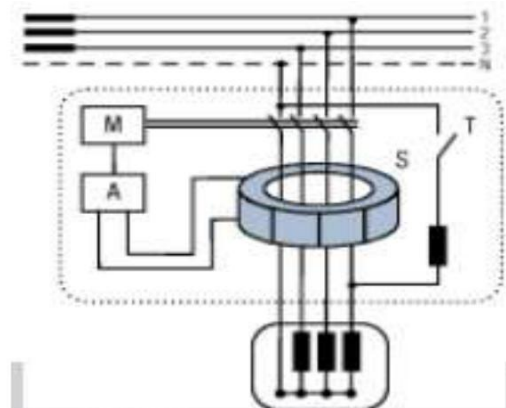
Second part → the short delay time this part concerned with the dangerous faults and the breaker will trip at a values of mille seconds. And starts from the second vertical column and this region can be adjusted at a value of time 0.18, 0.33, 0.5 second. And this is the actual tripping if value of current exceeds 2.5 times the pickup value of the adjusted current of short delay time. This means that if the adjusted value of time is 0.5 second for a current of value 6 times of rated current

600 ($6 \times 600 = 3600A$) this means that the breaker will trips at 0.5 second if the current passes through it is $2.5 \times 3600 = 9000A$

Ground fault circuit breaker

This type used for protection from ground leakage current , its idea is it compares between the phase value and the neutral in single phase or between the sum of the 3-phases current and the neutral in 3-phase applications. If the difference in currents is sufficient to make a magnetic force M can trip the breaker.

The figure below shows its idea.



The device is defined by its sensitivity to the leakage current from 5 mA to 30 mA

The 5 mA devices are very costly but it used for some sensitive applications.

But the 30 mA used for residential applications .and these devices are described by its rated currents and the most common used 32 , 40 ,63 ,80 ,100 A.

These devices can make the function of the MCB it can detect leakage currents and protect circuits from short circuit currents

Fuses

Fuse is the simplest current interrupting device for protection from excessive currents. It is used for over load or short circuit protection in medium voltage (up to 33K.V.) and low voltage (up to 400 v)

Principle of operations

It depends on the thermal effect of current that passes through it. As it melts at specific value of current and disconnects the current and the fuse element should be made from a good material and doesn' erode with time

Advantages of fuses

It is faster than C.B. in case of the very high currents. So it can be used without C.B.

Disadvantages of fuses

In case of three phase loads such as 3 phase induction motors the fuses of the three phases doesn't trip at the same time this makes the motor operates with two phases only

It has a lot of types

- 1- Cartridge fuses
- 2- High rupturing capacity fuse



Cartridge fuses

This type of fused is made to replace the old fuses that uses a thin wire and this wire melts and replaced with another may be isn't with the same cross sectional area of the melted fuse . but the cartridge type of fuse Cylindrical space of porcelaincontains a fusing element and thecylinderis full of silicon sand helps in quenching the spark.

H.R.C fuse.

Consists of a cylinder or a porcelain cubic and the fusing element is made from the pure silver and the cylinder is fill with silicon and this type can carry the high values of short circuit current and is provided with a fault indicators. The fuse element is single element or dual element . The single element is consists of one cross section . And the dual element is provided over the single element with another series element used to be affected with overloads.

Transformers Protection

1. Transformer Overcurrent Protection:

Fuses may adequately protect small transformers, but larger ones require overcurrent protection using a relay and circuit breaker, as fuses do not have the required fault breaking capacity.

1.1 Fuses :

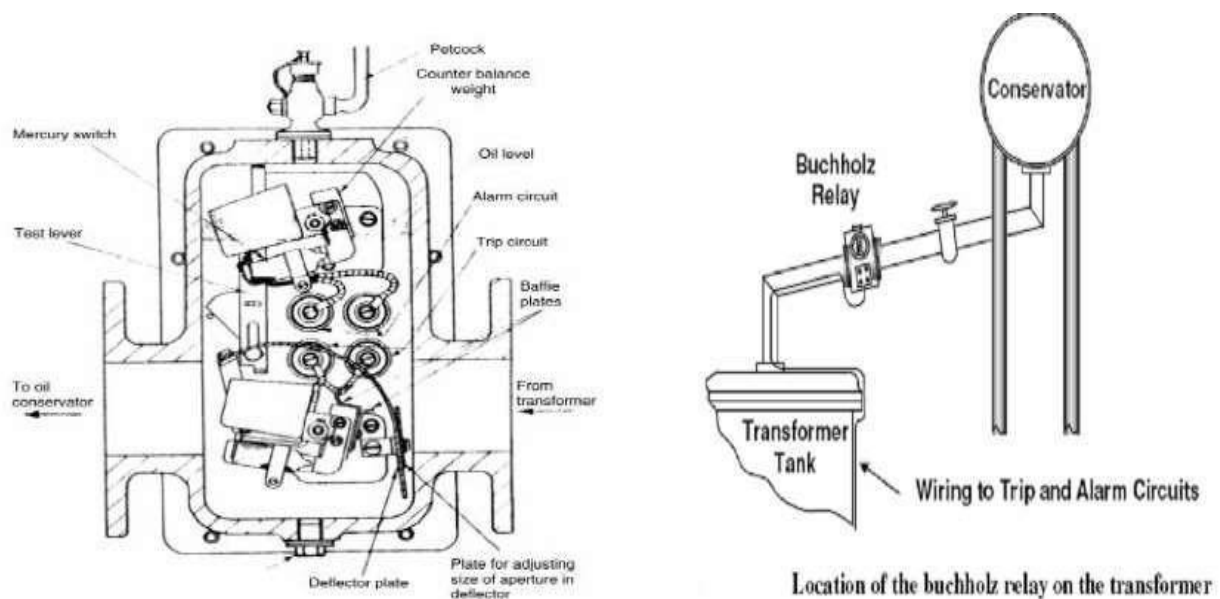
Fuses commonly protect small distribution transformers typically up to ratings of 1 MVA at distribution voltages.

Transformer rating		Fuse	
KVA	Full load current (A)	Rated current (A)	Operating time at 3xrating (s)
100	5.25	16	3.0
200	10.5	25	3.0
315	15.8	36	10.0
500	26.2	50	20.0
1000	52.5	90	30.0

1.2 Overcurrent Relays

With the advent of ring main units incorporating SF6 circuit breakers and isolators, protection of distribution transformers can now be provided by overcurrent trips .OC relay can be connected to transformer primary side or secondary side.

2.Buchholz relay



The following types of faults can be protected by buchholz relay and are indicated by an alarm:

- 1) Core insulation failure.*
- 2) Local overheating.*
- 3) Entrance of air into oil.*
- 4) Short circuited laminations (core) due to burns.*
- 5) Loss of oil due to leakage.*
- 6) Any bad electrical contacts will produce heat.*

➤ *The Buchholz relay has two oil-filled chambers with floats and relays arranged vertically one over the other. If high eddy currents, local overheating, or partial discharges occur within the tank, bubbles of resultant gas rise to the top of the tank.*

➤ *These rise through the pipe between the tank and the conservator. As gas bubbles migrate along the pipe, they enter the Buchholz relay and rise into the top chamber. As gas builds up inside the chamber, it displaces the oil, decreasing the level.*

➤ *The top float descends with oil level until it passes a magnetic switch, which activates an alarm.*

➤ *The float is located slightly below the top of the pipe so that once the top chamber is filled; additional gas goes into the pipe and on up to the conservator. Typically, inspection windows are provided so that the amount of gas and relay operation may be viewed during testing. If the oil level falls low enough (conservator empty), switch contacts in the bottom chamber are activated by the bottom float. These contacts are typically connected to cause the transformer to trip.*

Chapter 12

Fire Alarm



Fire alarm system

Aim of fire alarm system:

An automatic fire detection system is designed to detect a fire in its incipient stage and to automatically initiate programmed control function. All Fire Alarm Systems essentially operate on the same principle. If a detector detects smoke or heat or someone operates a break glass unit (manual break point), then alarm sounders operate to warn others in the building that there may be a fire and to evacuate and save life.

The choice of fire alarm system depends on the building structure, the purpose and use of the building.

Main components of fire alarm system:

1- Detectors

2- Manual call points

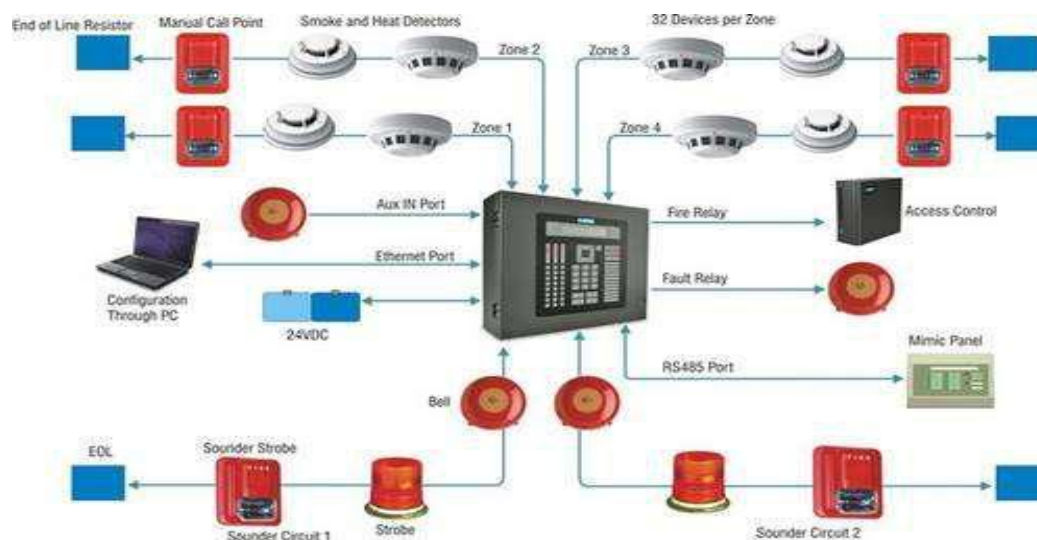
3-outputs (Sounders, Flashers and etc.)

4-control panel

5-power supply

6-cables

AS shown in the figure



Design of fire alarm system

1- Types of detectors:

A-smoke detector

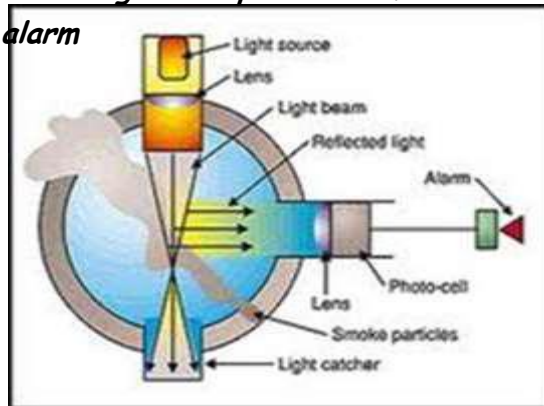
It mainly used to detect the visible and invisible combustion gases emanating from fires.

It has two types:

1/ Optical (photoelectric) smoke detector



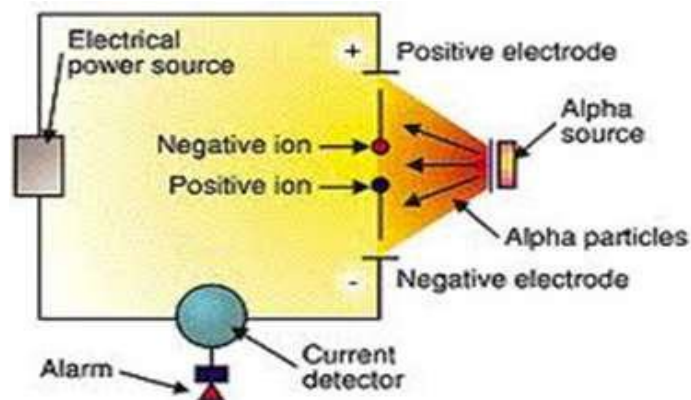
the best for detecting the smokes produced in home fires. It has light sensor that sense the scattered reversed light due to reach of any smoke then activate the siren of the danger. It consists of a photo cell and a light source and are normally applied where a fire is expected to generate large amount of visible particles of combustion. The principle of operation is that smoke entering the path of a light source will block the light and prevent it from striking a photo cell, thus generating an alarm



2/Ionization smoke detector



It's better for detecting the smoke generated by commercial installation. It capable of detecting the presence of both visible and invisible particles, but it more sensitive to invisible particles.



It consists of a source, detection circuit and ionization chamber containing a small radioactive alpha source. When particles of combustion enter the ionization chamber, they will attract themselves to the charged particles and reduce their flow. The detector circuitry will sense this flow reduction and generate an alarm signal. Generally smoke detector used in places where the smokes are not required. Such as (laboratories, corridors ,bed rooms.....ect)

B-Heat detector

It mainly used to detect the up normal rapid rise of temperature.

Heat detectors are normally used in environments where a smoke detector might generate false alarms for example kitchens or shower rooms.

It has two types:

1/Rate of rise heat detector

It's which differentiate the temperature with respect to time. It will be activated



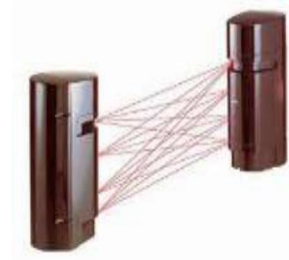
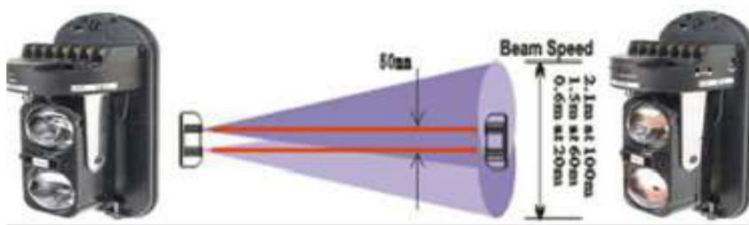
when undesired temperature rise occurs.

2/Fixed temperature heat detector



It will generate an alarm signal when the temperature of the internal detection element reaches its rated operating temperature. Used in places that normally has change in temperature.

C-Beam detector



Smoke detector includes one wired transmitter/receiver unit. When smoke enters the area between the unit and the reflector it causes a reduction in the signal. The alarm is activated when the smoke level reaches the predetermined threshold. Used in wide open areas.

D-Multi detector



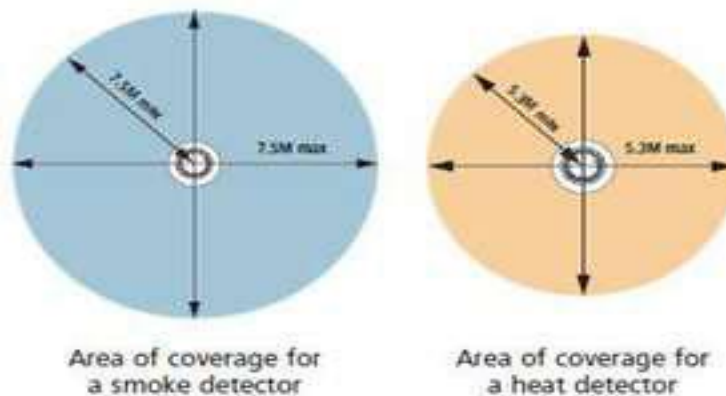
This detector combines optical smoke detector and heat detector. Used in electrical rooms.

E-Duct detector

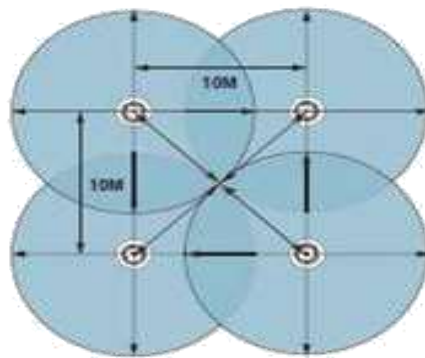
Used in air conditioning pipes. designed for prevention of smoke recirculation in areas by the air handling systems. Fans, blowers, and complete systems may be shut down in the event of smoke detection.



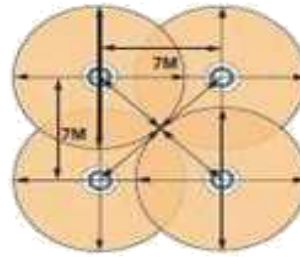
Spacing of detectors:



All smoke detectors have similar spacing distance, all heat detectors also have similar spacing distance. For general areas the spacing between any point in a protected area and the detector nearest to that point should not exceed 7.5m for a smoke detector and 5.3m for a heat detector. To ensure complete coverage, spacing between detectors should be reduced to 10.0m between smoke detectors and 7.0m between heat detectors.

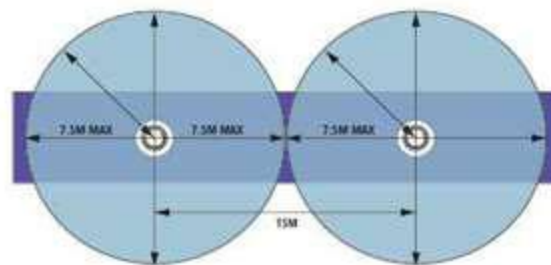


Spacings between smoke detectors



Spacings between heat detectors

For corridors less than 2m wide only the center line need be considered as it is not necessary to reduce detector spacing in order to provide complete coverage. Then for smoke detectors spacing becomes 7.5m from a wall and 15.0m between detectors. For heat detectors the spacing becomes 5.3m to a wall and 10.6m between detectors.



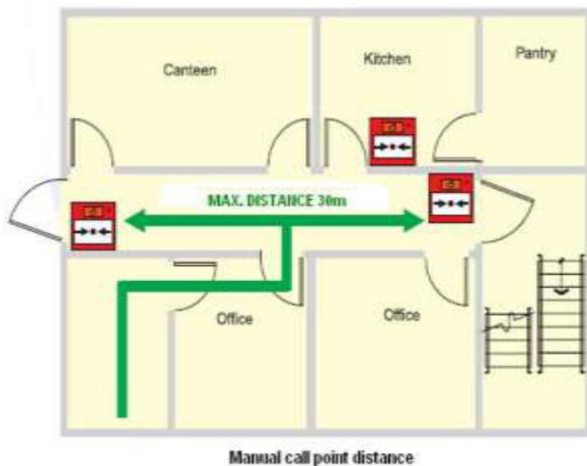
Corridor spacing for smoke detectors

If the ceiling height more than 3m from the floor, the spacing between detectors should be reduced so that it is equal to a percentage of the distance interfaces in the table below from EC.

Heat detector spacing

Ceiling height (feet)	Percent (%) of listed spacing
0-10	100
10-12	91
12-14	48
14-16	77
16-18	71
18-20	48
20-22	84
22-24	85
24-26	84
26-28	85

2- Manual call points:



Manual call point is used to initiate an alarm signal. It can be manual alarm system or part of automatic alarm system. Under normal conditions push button will be in the depressed condition. In the case of fire when the glass cover is broken, the push button will be released by the spring action and will actuate an alarm at the control

panel through its switching contacts. In addition to this, there will be an LED indicator on the monitor module for visual indication to locate the call point easily. Manual call points should be mounted on all escapes routes, and at all exit points from the floors of a building.

A fire alarm call point should be installed at a height of 1.4m above floor level at easily accessible positions. This includes on exit routes, at the entry floor landing of staircases and at all exits to the open air.

A fire alarm call point should also be spaced so that one may always be found within a maximum distance of 30m apart and the maximum distance between two call points 45m.

3- Fire alarm sounder with flasher:

The notification appliances types are

Audible → Horns - Bells -Sounders -Sirens - speakers

Visual → Strobes.

Physicals → Bed Shakers.

Olfactory→ Smell.

Audible Devices

1- Bells



Only used if they are only for fire, or have a distinctive sound from other bell signaling devices. Often used as an external gong to indicate the flow of water in the sprinkler system.

2- Horns



Loud and distinctive output. Often used in high-noise environments, such as manufacturing plants.

3- Sounders



Electronic or mechanical audible devices, which are capable of producing a variety of tones. Often, the tone is selectable during installation of the device.

4- Sirens



Extremely loud devices generally limited in use to outdoor or heavy industrial areas.

5- Speakers



Audible devices used in conjunction with voice evacuation messages. Life-Safety speakers are not generally associated with Muzak systems.

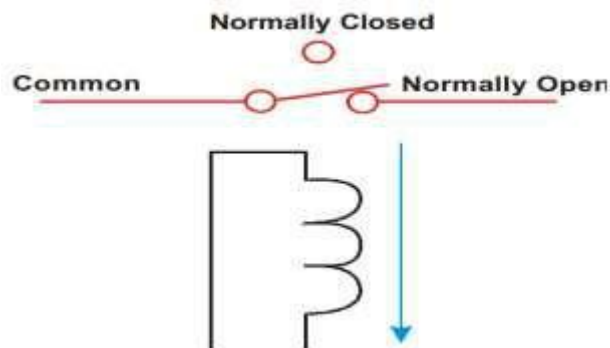
RELAYS AND CONTACTS

*Form A-refers to a relay with contacts for common
And normally open ONLY.*

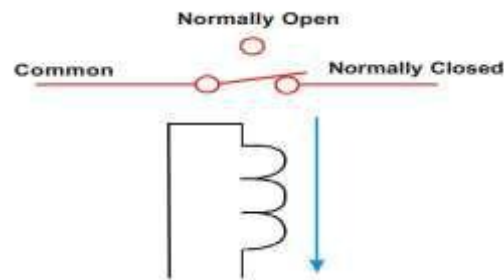
*Form B-refers to a relay with contacts for
common And normally close ONLY.*

*Form C-refers to a relay with contacts for common
, Normally open, and normally closed.*

*Dry Contacts-By magnetically controlling the state
Of the contacts, the control panel is electrically isolated (and thus
protected) from power connected to the contacts..*



When the relay is activated, current passing through the relay coil magnetically influences the common "wiper", moving it to the opposite position.

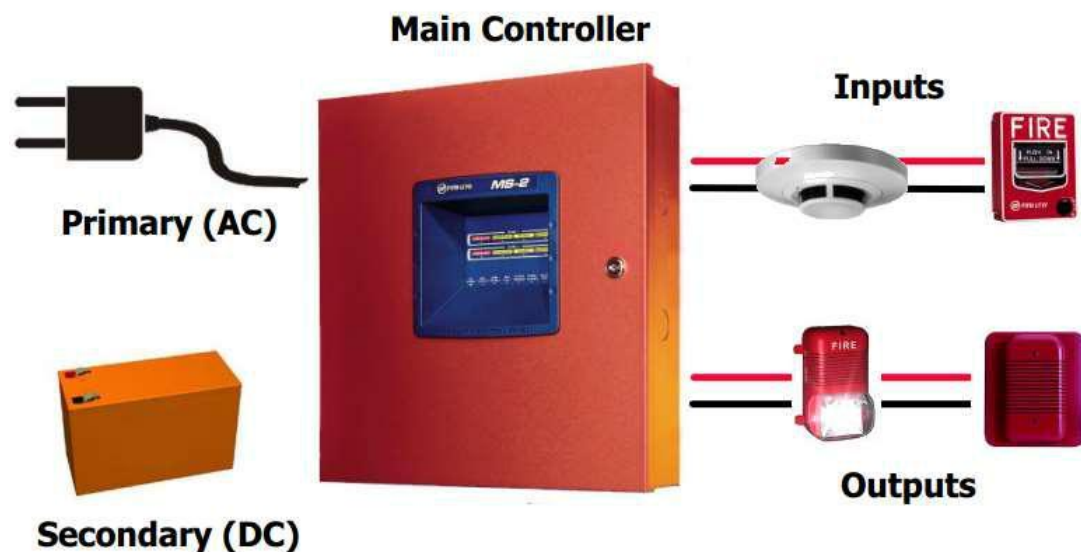


"Fail-Safe" relays are energized during "normal" conditions. The relay is activated by deenergizing the coil, guaranteeing activation of the desired signal during loss of all power to the system.

4- control panel

Main Controller

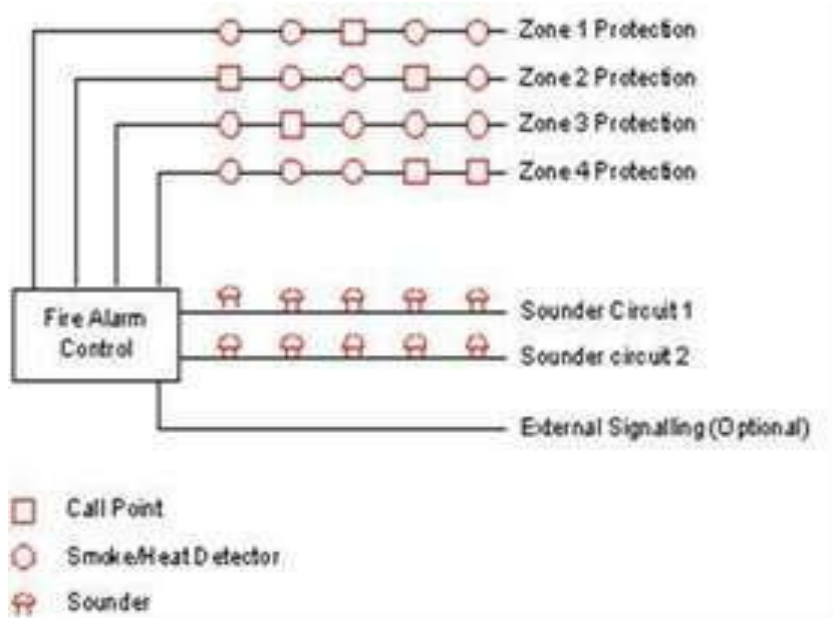
The brains of the system, It Provides power to the system, monitors inputs and controls outputs through various circuits. It performs other functions



Types of fire alarm control panel

A/conventional system:

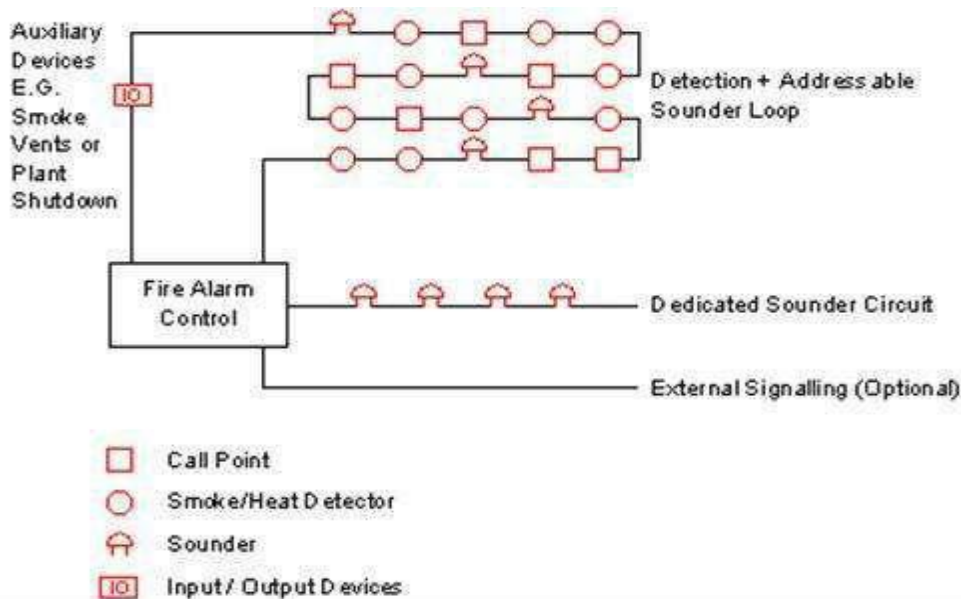
Simplest type of control unit. Limited special functions and capabilities. In a Conventional Fire Alarm System, a number of call points and detectors are wired to the Fire Alarm Control Panel in Zones. A Zone is a circuit and typically one would wire a circuit per floor or fire compartment. The Fire Alarm Control Panel has a number of Zone Lamps. A group of initiating devices is used to protect a certain area, if any of those devices senses a fire, an alarm is sent without detection of the zone of fire or which device sends an alarm.



The Control Panel is wired to a minimum of two sounder circuits which could contain bells, electronic sounders or other audible devices. Each circuit has an end of line device which is used for monitoring purposes.

B/Addressable system:

It is similar to a Conventional System except that the Control Panel can determine exactly which detector or call point has initiated the alarm. The detection circuit is



wired as a loop and up to 022 devices may be connected to each loop.

The address in each detector is set by switches and the Control Panel is programmed to display the information required when that particular detector is operated, so it help to determine quickly the correct location of fire.Used in large buildings or systems require more complications.

[There are other types of this system but more intelligent.](#)

1- Wireless Fire Alarm System

Wireless fire alarm systems are an effective alternative to traditional wired fire alarm systems for all applications. It is a simple concept, which provides many unique benefits and is a full analogue addressable fire detection system without the need for cable.

2- Analogue Addressable Fire Alarm Systems

It is known as Intelligent Fire Alarm Systems. With a true Intelligent Analogue System each detector incorporates its own computer which evaluates the environment around it, and communicates to the Control Panel whether there is a fire, fault or the detector head needs cleaning.

5- Power supply:

1- Normal supply

The normal supply for the system should be derived from the public supply system

2- Standby supplies

Secondary batteries; the most commonly used type of standby supply is a secondary battery with an automatic charger. For small fire alarm systems; mains and battery are normally built into the fire alarm control panel. On the medium and larger sized Fire Alarm systems, the standby batteries will often not fit within the control panel.

6- Cables:

Must be copper wire and fit its purpose. Must provide protection and be inserted inside tubes if exposed to mechanical factors or shocks. Wire cross section areas of 1 mm² and must be chosen so that it does not cause low voltage and effect on the efficiency of devices. The outer shell must be resistant to moisture and non-flammable. Electrical connection must be installed for the detection and alarm system at a distance not less than 50mm from any special electrical connections lighting parallel to the track.

Cable must withstand temperature of 1050 deg. C.

Fire resistance cables usually used have a 0cores of 1.5 mm^2 .

For distances less than 1000 M the 1.5 mm^2 is used for higher distance $2 * 2.5 \text{ mm}^2$ is used



Other components used in this system:



Repeaters

The repeater panels are connected to the main panel, but the user can't reset any Alarms. The main control is from the main panel any message can appear in any repeater panel also the user of the main panels can send a message to the other repeater



Control Module

The control modules are usually used for controlling something for example in case of fire there is control module to open the doors of the safetyregions ,to turn off the chiller and to turn on the shutters and so on.



Isolator

In case of faults or short circuits in the detector so the detector is provided with isolators so If a fault occur between in the loop the detectors that the fault between them opens and isolate the fault

The interface between the other systems

We should make interface with the following systems

1- Elevator

To make the elevators in case of fire go to the ground floor automatically

2- Audio System

To send a warning message in case of fire to make people pay attention and be ready to escape.

3- CCTV

To make the cameras make zoom in the area where the detector made alarm to check is there a fire or a false alarm.

4- Fire Pump.

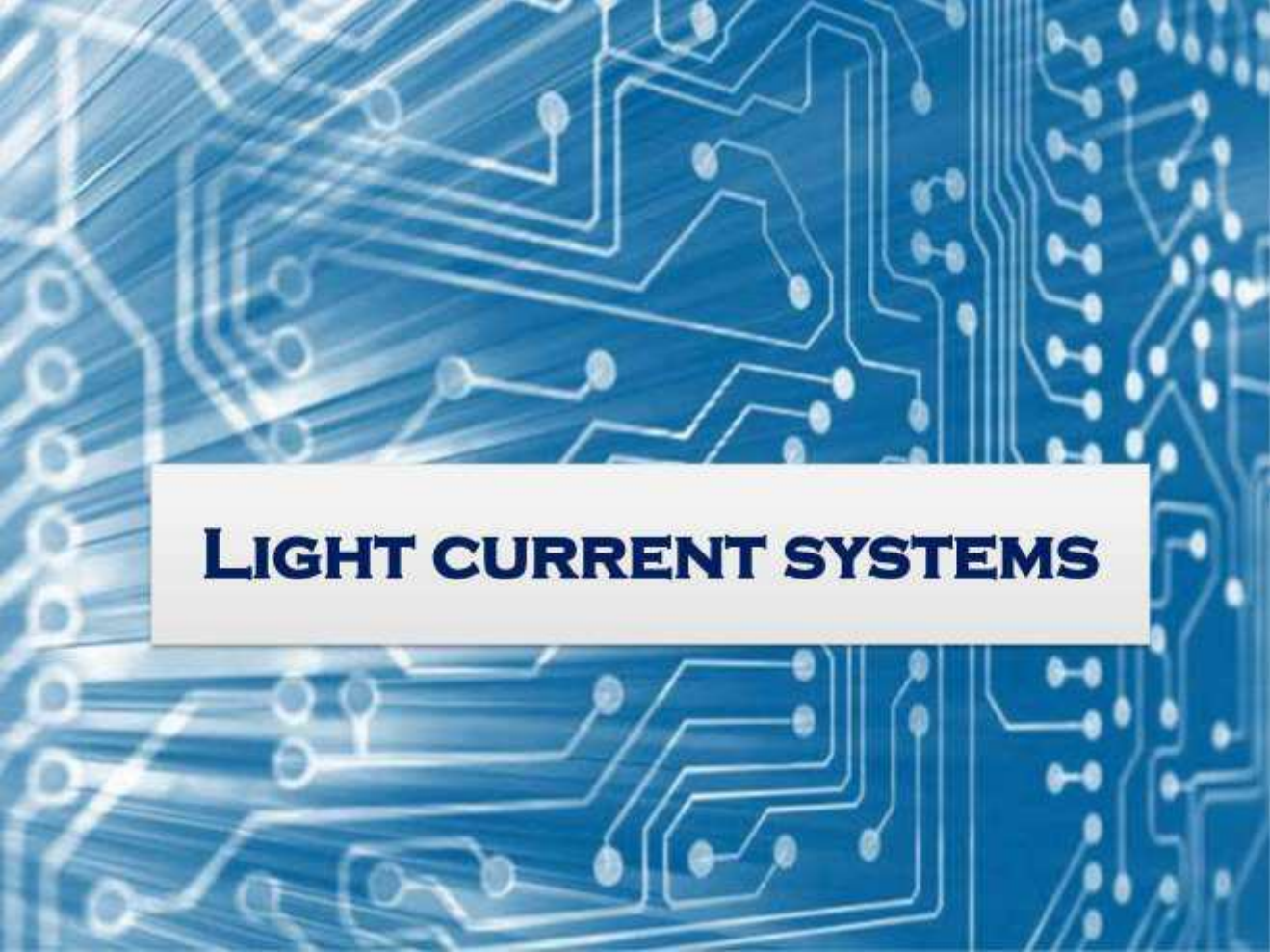
To operate the fire pump in case of fire to pump water to fight the fire.

5- Main Distribution board

To turn off the electric power from the building.

Chapter 13

CCTV and IP telephone system



LIGHT CURRENT SYSTEMS

Light current systems



systems

Fire alarm

CCTV

Sound
system

Telephone

Data

TV

Nurse call
system

C.C.T.V



Closed circuit television system

Closed circuit television system



Main element of CCTV

- camera
- video matrix
- digital video recorder [DVR]
- monitor
- control keypad
- wiring system

Conventional or IP systems – IP to contact with network

CCTV cameras to rating

- Regular cameras and the cameras DOME
- Cameras with auto iris lenses that adapt to change the intensity of lighting the place
- Day and Night cameras that work in low light
- Cameras Moving Pan / Tilt / Zoom

Camera types

pan tilt zoom [PTZ]



Dome shaped camera

fixed camera



An ordinary camera and be on a stable stand and be either color or black and white

motorizing camera



The camera on the trolley and can be color or black and white and used outdoor / indoor

day and night camera



is a security camera that can see the picture during the day hours, and during the night in total darkness

Closed circuit television system



- Assume camera cover area with radius $R = 7\text{ m}$
- Video matrix many cameras as input one screen display output
- Multiplexer example: 16 camera input and 8 screen display output
- Digital video recorder [DVR]: using hard disk to record cameras output
- monitor: color or black and white
- Control keypad: used to run the cameras and displayed on the screen and are determined by any camera operates and O do not work
- cameras feed from power source 220v & 50 HZ AC (from UPS)
- Using Co-axial, RG6, 4pairs UTP cat6 cable
- Implementation is often by the partners, and they determine the supply of the product demands, whether the cables or pipes

- **Specification of power box**

the PWR-24AC-16-8UL is a UL listed CCTV power center engineered to manage remote camera and accessory power requirements. The PWR-24AC-16-8UL distribution module is designed with 16 individually fused outputs for powering up to 16 CCTV cameras and accessories. It will provide 24Volts AC to sixteen fused outputs with a total of 8.0 Amps continuous supply current via two 4.0 Amp supplies. The PWR-24AC-16-8UL power center is housed in a premium heavy gauge metal enclosure



Closed circuit television system



What about our project :

-specification of reception camera" 360 DEG "& "IR"

Image Sensor	1/3" Sony CCD Image Sensor
Effective Pixels	480 TV Lines Color
Resolution	720 (H) x 540(V)
Minimum Illumination	0.1 Lux
Lens	3.7mm Fixed Lens
S/N Ratio	More than 50 dB
Signal System	NT SC
Power	12V DC or 24 v AC
Operations Temperature	-10°C to 50°C (14°F to 122°F)

-specification of corridor camera" 180 DEG "& "IR"

Image Sensor	1/3" Sony CCD Image Sensor
Effective Pixels	480 TV Lines Color
Resolution	720 (H) x 540(V)
Minimum Illumination	0.1 Lux
Lens	2.8mm Fixed Lens
S/N Ratio	More than 50 dB
Signal System	NT SC
Power	12V DC or 24 v AC
Operations Temperature	-10°C to 50°C (14°F to 122°F)

* NOTE : Operating Distance is 15 m

specification of outdoor camera" 180 DEG "& " IR"

Image Sensor	1/3" Sony CCD Image Sensor
Effective Pixels	680 TV Lines Color
type	PTZ
Minimum Illumination	0.1 Lux
Lens	2.8mm Fixed Lens
S/N Ratio	More than 65 dB
Signal System	NTSC
Power	12V DC or 24 v AC
Operations Temperature	-10°C to 50°C (14°F to 122°F)

Closed circuit television system



* NOTE : Operating Distance is 50 – 60m

How we choose camera:

When buy camera we should consider the following:

- 1- type of camera >>>> NTSC or PAL
- 2- Resolution of camera >>>> measure by "TVL"
- 3- Volume of lense>>>> measure by" mm"
- 4- IR or not >>>> measure by "lux"



Telephone system

Telephone system



The old system:

main component of telephone system

- *EPABX - Electronic Private Automatic Branch Exchange*
- *main distribution frame [MDF]*
- *Intermediate distribution frame (IDF)*
- *telephone outlets*
- *cable of telephone ((cat6 , thermal cu telephone cable))*

PBX:

- *PBX (Private Branch eXchange) is a privately owned telephone switching system for handling multiple telephone lines without having to pay the phone company to lease each line separately.*
- *Normally a telephone line is connected to the phone company's local Central Office through "a trunk." The Central Office is responsible for routing incoming and outgoing calls. It also provides other services like voice mail, call forwarding, caller ID and so on. For this service the phone company receives a monthly fee.*

Main distribution frame (MDF):

- *Main Distribution Frame (MDF) is a distribution frame on one part of which the external trunk cables entering a facility terminate, and on another part of which the internal user.*
- *Subscriber lines and trunk cabling to any intermediate distribution frames terminate.*



Intermediate distribution frame (IDF):

An intermediate distribution frame (IDF) is a free-standing or wall-mounted rack for managing and interconnecting the telecommunications cable between end user devices and a main distribution frame (MDF). For example, an IDF might be located on each floor of a multi-floor building routing the cabling down the walls to an MDF on the first floor. The MDF would contain cabling that would interconnect to the phone company or to other buildings.

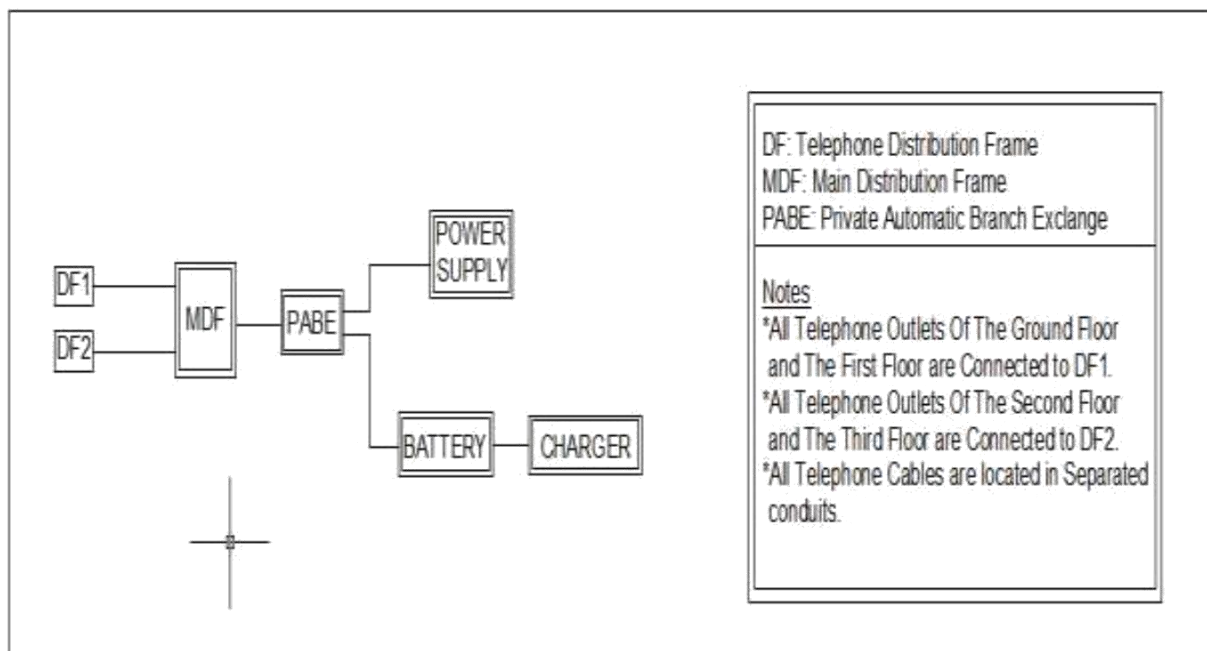


Telephone plug :

A telephone plug is a type of male connector used to connect a telephone to the telephone wiring in a home or business, and in turn to a local telephone network. It is inserted into its female counterpart, a telephone "jack", commonly fixed to a wall or baseboard. The standard for telephone plugs varies from country to country, though the RJ11 "modular connector" has become by far the most common.



TELEPHONE SYSTEM SINGLE LINE DIAGRAM



Cable types:

1. *Unshielded Twisted-Pair Cable.*
2. *Shielded Twisted-Pair Cable.*
3. *Coaxial Cable.*
4. *Fiber-Optic Cable.*
5. *Wireless LAN.*

In our project we use fiber optic cable we classified by:

A-Category:

- 1- cat3
- 2- cat4
- 3- cat5
- 4- cat6
- 5- cat 6e
- 6- cat 5e



b- Pair of cables:

- 1- 2 pair
- 2- 4 pair
- 3- 8 pair



When the design and development of telephone lines connected to the building must take into account the following:

Socket Phone:

- *socket phone is kind of immersion compound wall and determine their locations depending on the graphics and the direction of opening the doors and installed at a height of 35 cm from the surface of the floor also be of connectors distinct from other outlet and the separation of the Chinese Full generally polished aluminum oxidized and supplied complete Balvic for her ride This outlet on Plastic Bmsmaran tray bolts and be the outlet of the same made-luminescent keys.*

General technical specifications for the installation of telephone lines:

- *Phone wire diameter 0.6 mm 2 plastic insulation according to these specifications.*
- *take into account when installing telephone lines to be isolated from the rest of the electrical connections and allocate have separate pipes do not share with them the power cords.*
- *parties must be numbered telephone wires either at distribution or cans outlet.*
- *per telephone socket installed wires and one plastic casing also and can not be the work of the welds of the wire pipes bystanders.*
- *Capacity pipes for telephone cables shall be according to general specifications.*
- *The resistance of the Department, telephone, one of the public distribution box to its outlet at most 25 ohms.*
- *The telephone cable consisting of insulated wire with a diameter of 0.6 mm 2*



Ip telephony system



What is IP Telephony?

IP Telephony principally satisfies a simple need: Enabling the use of open standard LAN and distributed data networks for the transport of voice, replacing the use of traditional (and proprietary) telephone networks and equipment for that purpose. With IP Telephony, voice conversations are converted into packets of data and then transmitted over data networks much in the same manner as e-mails or instant messages. Using standard Internet Protocol (IP) data networks and methods to transport all Forms of information media - voice, data and video is more efficient and opens the door for the creation of new value-added cross-media features and applications.

Routers:

is a device that forwards data packets between computer networks, creating an overlay internetwork. A router is connected to two or more data lines from different networks. When a data packet comes in one of the lines



Switch:

That it connects devices to allow them to act as a single segment.



Patch panel:

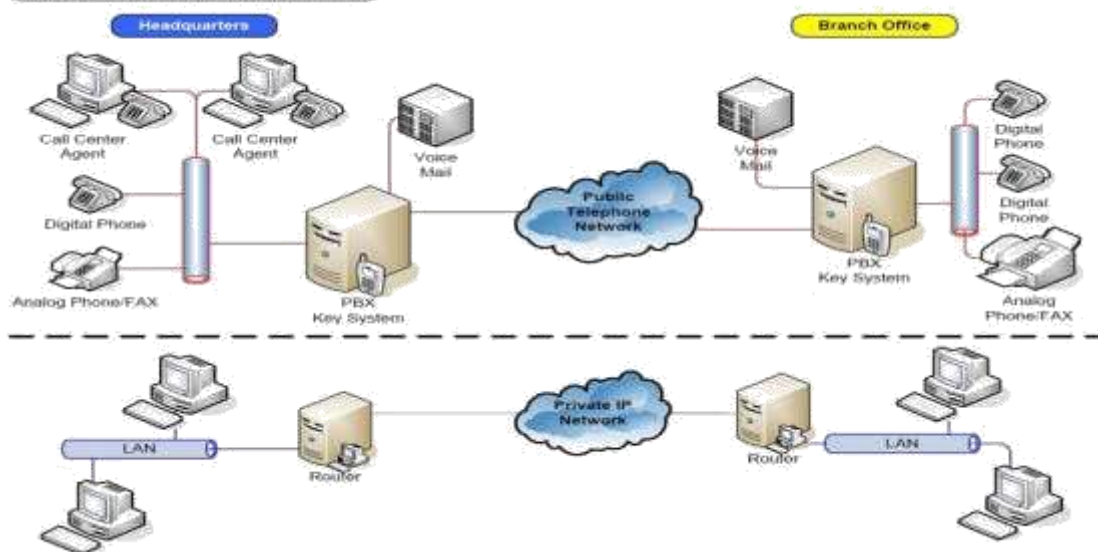
arrange and organize the cables coming from various points of the network assume there is network of 10 points lying in the building, there will be a network cable for each point the first end of the cable at the point and the second party is in the room side exchanger Switch.

In general if not used directly Patch panel then Connect these cables will have a network and operate without any problems

But in the event of a hardware problem For example, the device shows you that the device is not connected to the network with a network cable plugged into it You must make sure that the second end of the link cable with a switch Here you will find the problem in finding this cable between the 10 cables connected _ of course, in this example, ten cables and in some cases you may have to have more than 100 cable _ and you have to guess which cable is that you want Similarly, if you want to physically separate network for one of the points you should experience all the points to do so Here comes the role Patch panel where you numbering of the points that you have



Figure 1 – Separate Voice & Data Networks

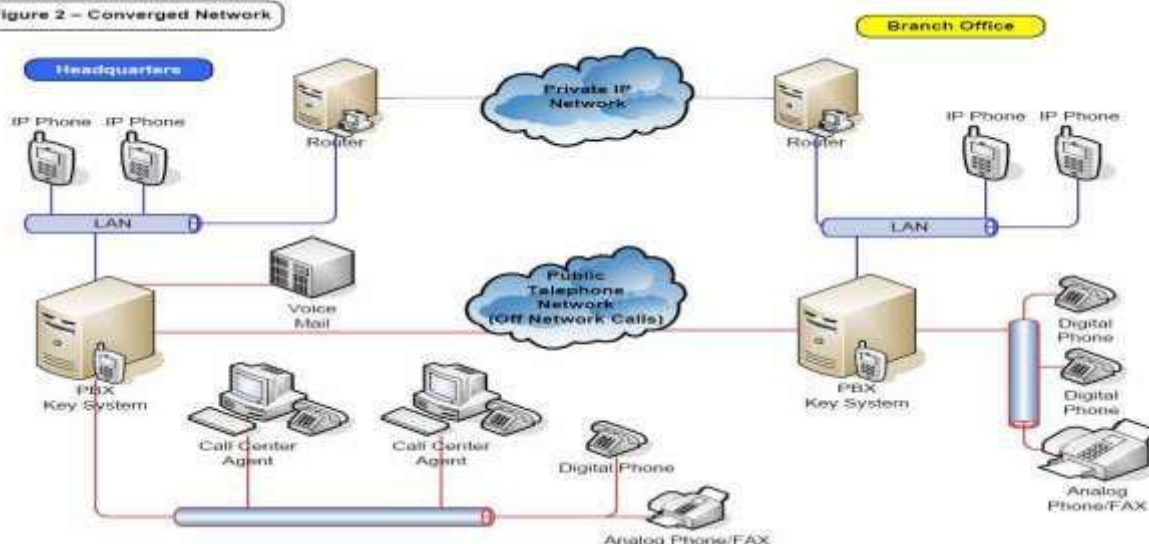


The above diagram reflects a typical separated network topology deployed to provide data and voice connectivity between headquarters or regional business offices and remote branches and employees. Data connection needs were enabled by point-to-point circuits, frame relay and in more recent years by Internet connections. Private voice tie-lines were used to interconnect stand-alone PBXs and key systems. This design, although state of the art for many years resulted in a number of inefficiencies:

- Inefficient use of communications network bandwidth
- Higher equipment costs as each site requires dedicated voice systems
- Higher equipment management costs since each site must be managed separately

Converged Networks, Devices and Applications with IP telephony The wide adoption of IP networks and the emergence of IP Telephony have led to the convergence of what were once disparate applications (data, voice and video) onto a single network. This has eliminated many of the inefficiencies of legacy networking models.

Figure 2 – Converged Network



Features and Benefits of IP Telephony :

Much easier to install &configure than a proprietary phone system:

An IP PBX runs as software on a computer and can leverage the advanced processing power of the computer and user interface as well as Windows' features. Anyone proficient in networking and computers can install and maintain an IP PBX. By contrast a proprietary phone system often requires an installer trained on that particular proprietary system

Lower Telecom Costs:

Initial IP Telephony migrations were easily justified by the cost savings generated by avoiding expensive long distance phone services that linked geographically dispersed business locations. Inresponse, traditional telecommunications carriers over time have continued to lower prices of such services, reducing this specific cost advantage of network convergence. This does not impact, however, the cost advantage produced by better utilization of available network bandwidth enabled by convergence. Since dedicated bandwidth is no longer needed for voice conversations, businesses an transmit more voice, data and video per unit of available bandwidth with IP Telephony. The combination of lower transport costs and less bandwidth requirements can produce telecom cost savings from 20 - 40%.

Simplified Communications Management:

The day-to-day provisioning, operations and management expenses for traditional PBX systems can add up quickly. The simple need of activating a phone or moving an extension can cost anywhere from \$100 to \$250 per event when involving service technicians from the PBX manufacturer. With IP Telephony this is no longer an issue. Common adds,moves and changes, can be administered with simple web based tools. In addition, since each IPTelephone contains a unique identifier, all a user needs to do to relocate a phone is to simply unplugit from the data network, place it at its new location (within the original office or any facility with access to the corporate data network) and plug it back in. Service to the user is instantly restored with all features and profiles previously assigned in tact - with no involvement required from the system administrator

Significant cost savings using VOIP providers:

With an IP PBX you can easily use a VOIP service provider for long distance and international calls. The monthly savings are significant. If you have branch offices, you can easily connect phone systems between branches and make free phone calls

Eliminate phone wiring:

An IP Telephone system allows you to connect hardware phones directly to a standard computer network port (which it can share with the adjacent computer). Software phones can be installed directly onto the PC. You can now eliminate the phone wiring and make adding or moving of extensions much easier. In new offices you can completely eliminate the extra ports to be used by the office phone system!

Access to Centralized Applications :

For distributed enterprises IP Telephony can provide remote and mobile access to advanced communications applications, including call management, messaging and call center. These and their applications can now be transparently distributed over IP networks, eliminating the need for standalone PBX's or key systems for small remote office locations, and can extend significant benefits to remote and mobile teleworkers.

Scalable:

Proprietary systems are easy to outgrow: Adding more phone lines or extensions often requires expensive hardware modules. In some cases you need an entirely new phone system. Not so with an IP PBX: a standard computer can easily handle a large number of phone lines and extensions - just add more phones to your network to expand!

Better customer service & productivity:

With an IP PBX you can deliver better customer service and better productivity: Since the IP telephone system is now computer-based you can integrate phone functions with business applications. For example: Bring up the customer record of the caller automatically when you receive his/her call, dramatically improving customer service and cutting cost by reducing time spent on each caller. Outbound calls can be placed directly from Outlook, removing the need for the user to type in the phone number.

Allow hot desking& roaming

Hot desking - the process of being able to easily move offices/desks based on the task at hand, has become very popular. Unfortunately traditional PBXs require extensions to be re-patched to the new location. With an IP PBX the user simply takes his phone to his new desk - No patching required!

Users can roam too - if an employee has to work from home, he/she can simply fire up their SIP software phone and are able to answer calls to their extension, just as they would in the office. Calls can be diverted anywhere in the world because of the SIP protocol characteristics!

the adoption of IP Telephony is gaining momentum, with primary business drivers being:

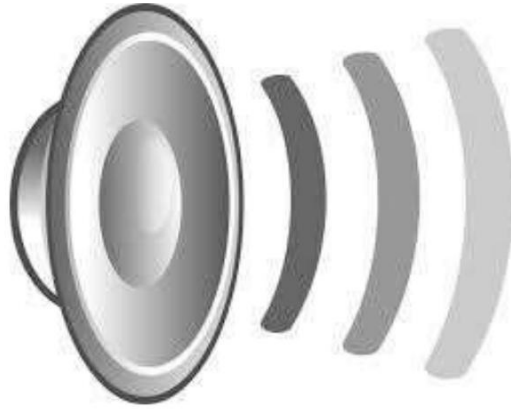
- *Lower voice networking costs*
- *Lower equipment administration costs*
- *Centralized network control and management*
- *Increased communications capabilities and productivity for remote and mobile employees*
- *Increased customer satisfaction through the use of distributed call center applications*

CHAPTER 14

SOUND SYSTEM



ABRIEF OVERVIEW ABOUT SOUND SYSTEM



MAIN IDEA OF SOUND



FOR OUR PROJECT

We use two types of sound system:

- *NORMAL SOUND SYSTEM*
- *VOICE ALARM SOUND SYSTEM (AVAC)*

FIRST NORMAL SOUND SYSTEM



COMPONENTS OF SYSTEM:

- *INPUT DEVICES*
- *AUDIO MIXER*
- *AMPLIFIER*
- *OUTPUT DEVICES*

➤ **INPUT DEVICES:**

- *Its function is to convert sound signal to electrical signal*
- *SUCH AS :*
- *MICROPHONE (types : dynamic - condenser - carbon)*



CD PLAYER



RADIO FM



➤ **AUDIO MIXER :**

Its function is connecting the different input devices and for selecting any of them for broadcasting.

The mixer provides the ability to connect effects processors to add reverb, echo, delay, etc. to the sound

Mixers are divided into two parts:

1- The master section

Controls the output of the mixer to all of the various other components

2- The channels

A mixer usually handles 8, 16, 24, 32, 40, or 48 channels or inputs. Individual channel equalization and balancing on the Mixer.



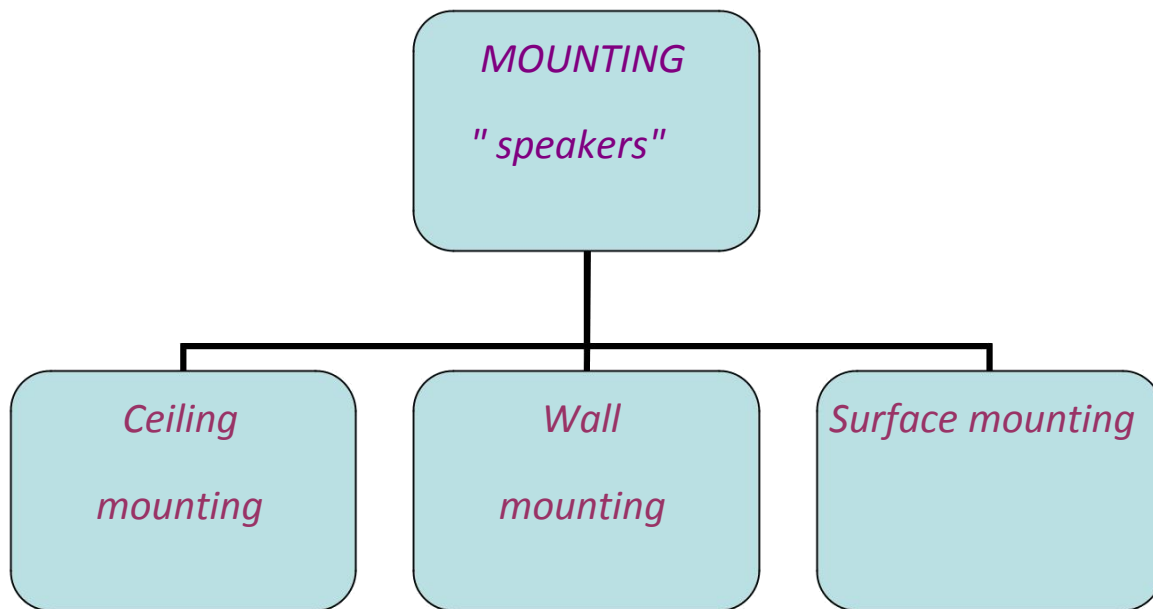
AMPLIFIER :

Its function is to amplify the electrical signal



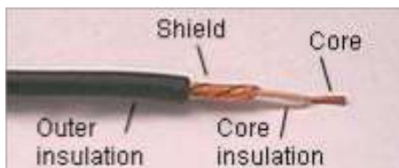
➤ **Output DEVICES :**

Its function is to convert electrical signal to sound signal

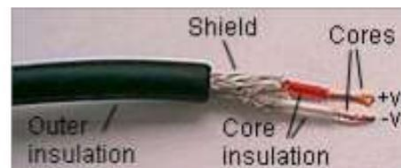


AUDIO CABLES OF SYSTEM :

- *There are two main types of audio cable we will look at: Single core / shielded (unbalanced) and One pair / shielded (balanced)*
- *In a single core / shielded cable, the single core is used for the +ve, and the shield is used for the -ve*
- *A one pair / shielded cable has one core as the +ve, and the other core is ve. The shield is earthed*



SINGLE



DOUBLE

DESIGN OF SYSTEM :

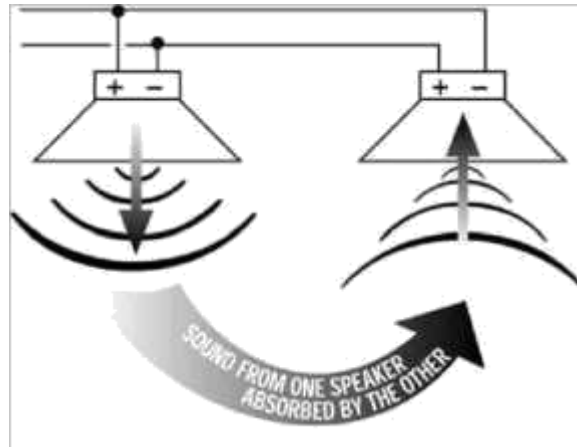
- *We select speaker by determine the noise level and application We find no of speakers by two methods :*
 - A. *from the speaker data sheet determine the speaker coverage and then find no by dividing area of the place on speaker coverage*
 - B. *by knowing length , width and height of the place we can determine no from standard tables*

" TABLE OF DESIGN"

- *We find watt of Amp by producting no of connecting speakers on Amp and speaker watt , then we select the suitable AMP*

NOTES :

- *It is prefer to connect speakers parallel to avoid fault all speakers*
- *We must connect speakers with correct polarity to avoid out*
- *phasing that effect on sound quality*



Out phasing

SECONDLY. AVAC SOUND SYSTEM

COMPONENTS OF SYSTEM :

- *Digital message store*
- *Mixer priority*
- *Amplifiers*
- *Battery*
- *Conventional trigger inputs*
- *POWER SUPPLY*
- *INPUT DEVICES*

AS BELOW

AVAC is suitable for category V1 and V2 systems complying with BS5839-8 (2008).

In one compact wall-mountable enclosure, the AVAC comprises:-

- *A Context Plus XP95 specific interface for direct connection to a Context Plus fire alarm loop*
- *A straightforward fire alarm interface that can be connected to any analogue addressable fire alarm system via a series of input/output units on the loop*
- *A high-quality digital message store containing programmable Evacuate, Alert and Test messages*
- *A prioritised mixer*
- *Two x 60-Watt continuous average power Class D amplifiers (plus an optional standby amplifier)*
- *An EN54-4 compliant switch mode power supply and battery charger*
- *Three balanced line level inputs for the (optional) connection of an emergency microphone, paging/public address microphone(s) and a background music source*

OPERATION OF SYSTEM :

The principal function of AVAC is to generate clear voice messages to alert people to the presence of fire, normally under the control of a fire detection system. When AVAC receives a message trigger from the fire detection system, it responds by playing an appropriate message (Evacuate, Alert or Test) from its digital message store. This message is amplified and broadcast around the site via the system's loudspeakers

AVAC's

prioritized mixer ensures that only the most urgent audio signal is broadcast as indicated below:-

Priority	Description
1	Emergency MIC
2	evacuated , alert or test message
3	Paging MIC
4	Music

FINALY. FOR OUR PROJECT

PLACE	SOUND TYPE	SPEAKER TYPE	MODEL OF SPEAKER	WATT OF SPEAKER	MODEL OF AMPLIFIER	WATT OF AMPLIFIER
RESIDENTIAL & RECEPTIONS	AVAC	RECESSED CIELING	CS80R	6W	V60	60W
OFFICE & RESTURANTS	AVAC	RECESSED CIELING	CS80R	6W	V60	60W
DOUBLE HEIGHT ROOM	Normal	WALL & RECESSED CIELING	6RSHV250 & CS80R	6W& 6W	WV100	100W
OUTDOOR AREA	Normal	WALL	TR920	30W	M300	300W

HINT : " TABLE OF DESIGN"

Look Up LONGER Dimension Of Area On This Side																				
	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	170	180	190	200	
20	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9	9	10	10	11
30	1	1	2	2	2	3	3	3	4	4	4	5	5	5	6	6	7	7	7	8
	2	3	4	5	5	6	6	7	7	8	8	9	9	10	10	11	11	12	12	13
40	3	4	5	6	6	7	7	8	8	9	9	10	10	11	11	12	12	13	13	14
	4	5	6	7	7	8	8	9	9	10	10	11	11	12	12	13	13	14	14	15
50	5	6	7	8	8	9	9	10	10	11	11	12	12	13	13	14	14	15	15	16
	6	7	8	9	9	10	10	11	11	12	12	13	13	14	14	15	15	16	16	17
60	7	8	9	10	10	11	11	12	12	13	13	14	14	15	15	16	16	17	17	18
	8	9	10	11	11	12	12	13	13	14	14	15	15	16	16	17	17	18	18	19
70	9	10	11	12	12	13	13	14	14	15	15	16	16	17	17	18	18	19	19	20
	10	11	12	13	13	14	14	15	15	16	16	17	17	18	18	19	19	20	20	21
80	11	12	13	14	14	15	15	16	16	17	17	18	18	19	19	20	20	21	21	22
	12	13	14	15	15	16	16	17	17	18	18	19	19	20	20	21	21	22	22	23
90	13	14	15	16	16	17	17	18	18	19	19	20	20	21	21	22	22	23	23	24
	14	15	16	17	17	18	18	19	19	20	20	21	21	22	22	23	23	24	24	25
100	15	16	17	18	18	19	19	20	20	21	21	22	22	23	23	24	24	25	25	26
	16	17	18	19	19	20	20	21	21	22	22	23	23	24	24	25	25	26	26	27
110	17	18	19	20	20	21	21	22	22	23	23	24	24	25	25	26	26	27	27	28
	18	19	20	21	21	22	22	23	23	24	24	25	25	26	26	27	27	28	28	29
120	19	20	21	22	22	23	23	24	24	25	25	26	26	27	27	28	28	29	29	30
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150	25	26	27	28	28	29	29	30	30	31	31	32	32	33	33	34	34	35	35	36
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180	31	32	33	34	34	35	35	36	36	37	37	38	38	39	39	40	40	41	41	42
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190	33	34	35	36	36	37	37	38	38	39	39	40	40	41	41	42	42	43	43	44
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200	35	36	37	38	38	39	39	40	40	41	41	42	42	43	43	44	44	45	45	46
	36	37	38	39	39	40	40	41	41	42	42	43	43	44	44	45	45	46	46	47

Chapter 15

Smart Homes

Making Your Life Easier



ABRIEF DISCUSSION ABOUT :



SMART HOME

Introduction

Objectives of smart home :

- *Saving energy*
- *Security*
- *Comfort*
- *Automation*
- *Convenience*

What about comfort system :

- * *Comfort talks to you in a clear human (not synthetic) voice, operated through the Keypads, any phone in the premises or any outside phone through a comprehensive Voice menu system.*
- * *Comfort is an advanced tapeless Answering Machine with up to 8 personal mailboxes which even can call your pager or mobile phone when a new message for your mailbox is recorded.*
- * *Comfort is a practical Home Automation System which allows you to operate your lighting, HVAC or home or office appliances via telephone, Time program, or when triggered by Events*
- * *Comfort's Keypads with one-touch arming keys and short-cut keys make it easy to operate the system, and double as intercoms for communications within the premises.*

- * *Comfort' Door Station (DP01) * intercom rings your home phones when you are in the premises, and calls you on your mobile phone when you are out, and even takes a message from visitors.*
- * *Comfort's Reminder Messages can ring the phone or call your mobile phone to remind you of regular events or even act as an alarm clock.*
- * *Comfort's Time Programs can automatically switch off lights and appliances or arm the security system or perform many other functions at programmed times.*
- * *Comfort can be controlled from the RC01 handheld remote control or any universal learning remote control (eg Philips Pronto)*



Modules of the system :

UCM " Universal communication module :

- The Master control panel that give orders to all modul



Ports

8 Inputs & 8 outputs

Types

UCM/USB - UCM/ ETHERNET - UCM/232

Interface

can be interface to other systems like : UCM/KNX - UCM/C-bus - UCM/Z-wave

Power

transformer using to convert 220 Volt to 12 or 14 or 16 Volt

Check of connecting correctly

from this table

LED	Behavior	State
D1	Cont green	ON
D2 & D3		OFF
D4		OFF
D9	Flash slow	ON
D10	Flash fast	ON
D11 & D12		OFF

KEYPAD :

Construction

- *16 programmable keys*
- *LEDs for status*
- *Baby monitor feature*
- *Up to 8 keypads in the system, all with IR Receivers*

EX

KP03: no LCD

KP04: LCD

KP05: LCD horizontal

KP06: LCD H/V

KT03: Touch screen



Functions

- *Acts as intercom with (Phone - door station - other keypads)*
- *offer trouble condition on lcd*
- *arming system in different modes as shown :*

Button	Function
--------	----------

Arm to AWAY Mode

Arm to NIGHT Mode

Arm to Day Mode



The F key selects a Function by pressing any of the other keys



LEM " Local Expansion Module " :



Ports

8 Inputs & 8 outputs

NO

1 LEM only can connected to Main Comfort and each SEM



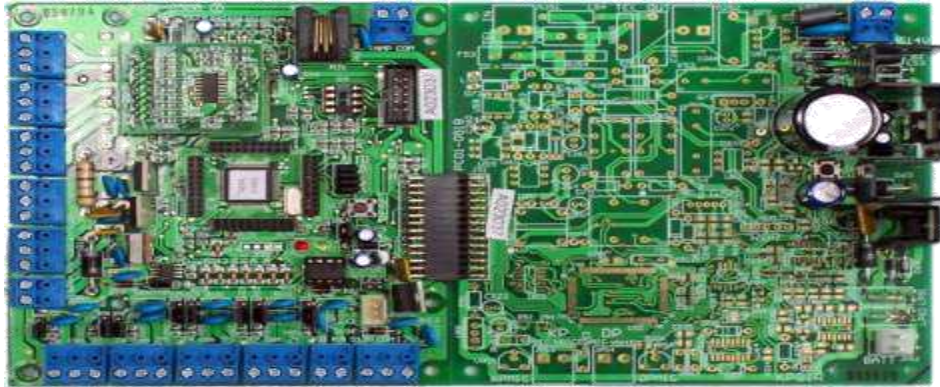
SEM " Slave Expansion Module " :

Ports

8 Inputs & 8 outputs

NO

Maximum of 5 SEMs for 1 Main Panel, hence max 96 Inputs and Outputs per system for Comfort and SEMs



RIO " Remote input/output " :

Ports

8 Inputs & 8 outputs

NO

Max 15 RIO per Ultra II system giving 120 non-alarm inputs and open-collector outputs



Door Station:

Construction

Speaker/microphone in-built

NO

Up to 3 supported by system

Function

- *Pressing Doorbell rings phones*
- *Talk to visitor at door station using the internal phones*
- *Dials to user in away mode allowing user to talk to the visitor*
- *User able to control a gate or door to let visitor in*



Remote control:

Function

Sends IR codes to

- *IRR IR receiver (with RIO)*
- *IRD Dome (with RIO)*
- *SCS04/IR*

KT01/KP03/KP04

Room buttons to control lights/appliances in rooms



Scene Switch:

Function

- Programmed to make alot of Responses



Wiring of the system :

Bus cable : used to connect between UCMS

Net cable : used to connect between UCM & Modules



net cable

bus cable

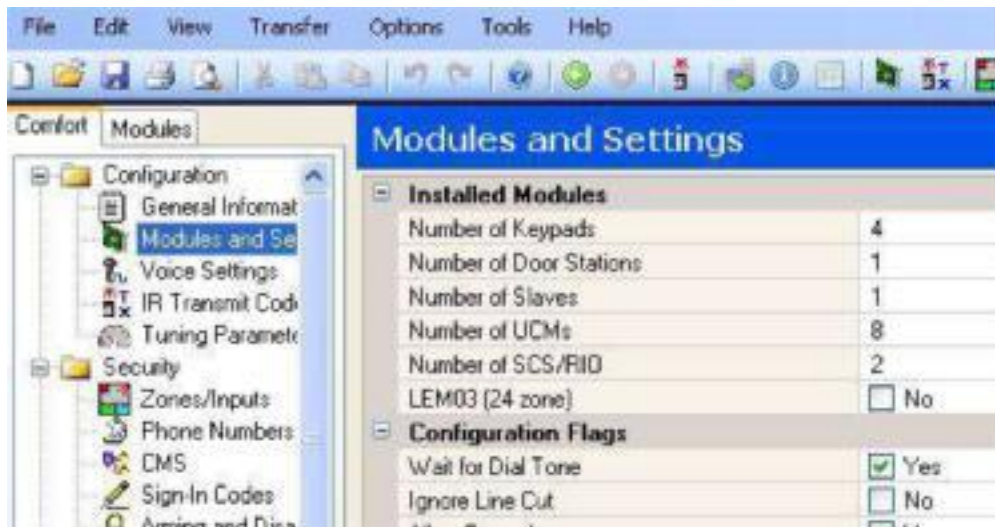
Software of the system :

Programm

Configurator Programming Software

Hint about program

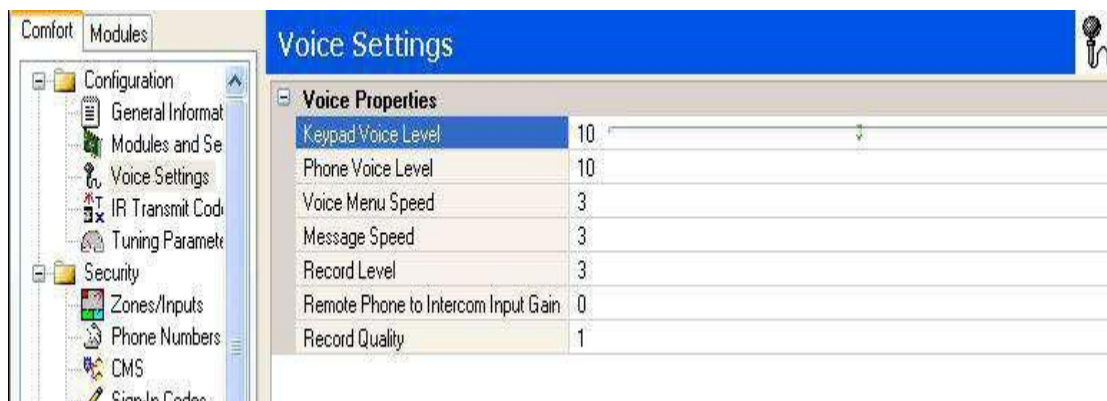
Modules & settings In Configurator, go to Configuration > Modules and Settings



In this screen, the Number of Keypads, Door Stations, Slaves (SEM01 or SEM02), UCMs, and SCS/RIO are shown

Voice settings

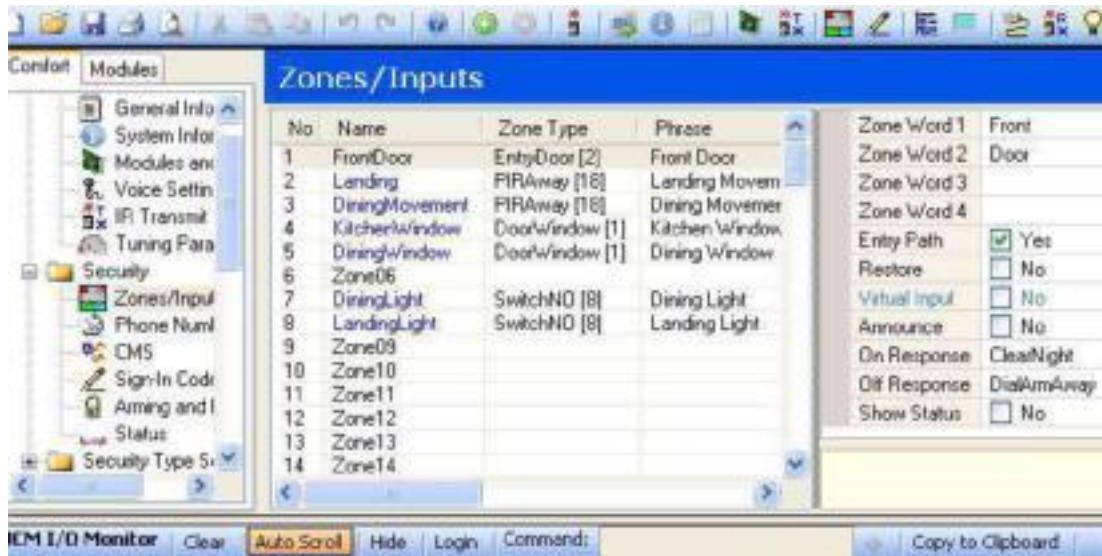
Configuration > Voice Settings gives the Voice Settings



This screen is used to specify settings associated with voice output or record

ZONES/INPUT

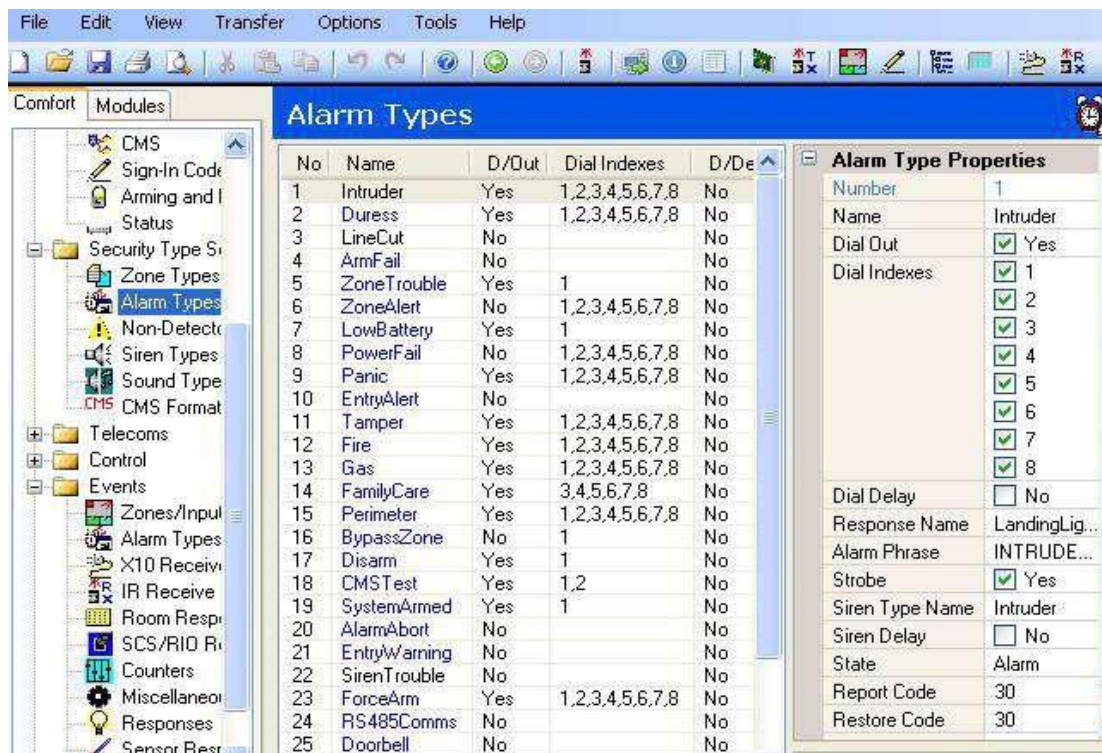
In Configurator, select Security > Zones/ Inputs



This screen is used to show security modules in system (96 zone)

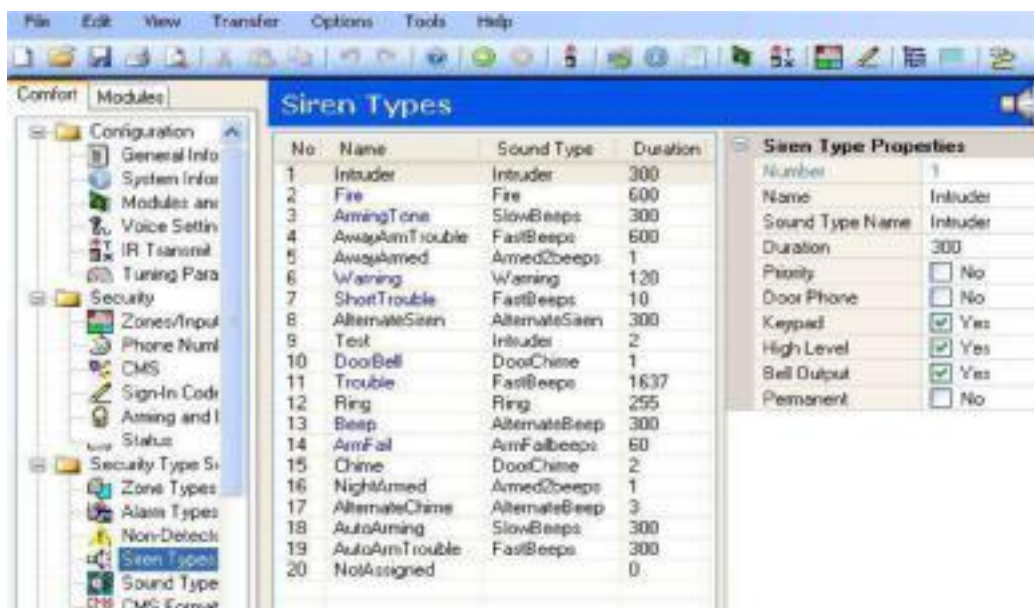
Alarm Types

In Configurator, go to Security Types > Alarm Types-



- 31 programmable alarm types determine the behaviour of each Alarm
- Intruder, fire, smoke, gas, etc
- Dial-out to combination of 8 phones
- 20 siren types
- Dedicated strobe output
- Alarm response to trigger lights or announce message

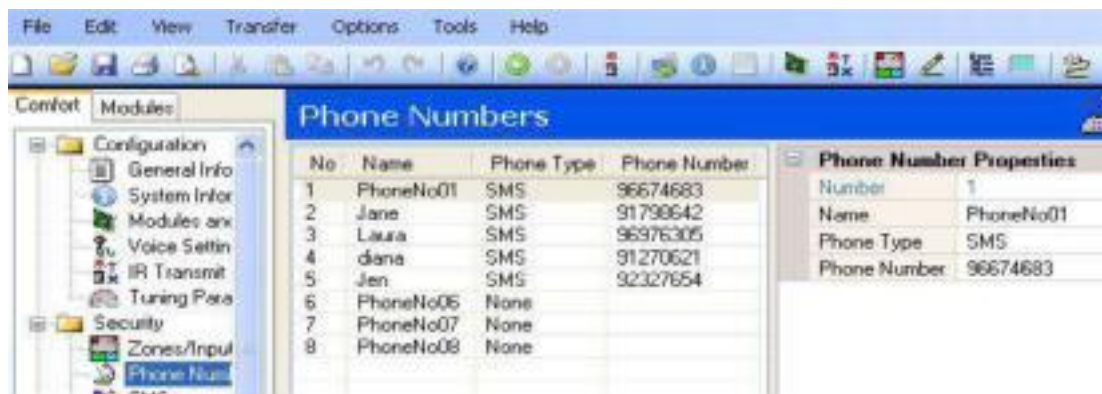
Siren Types → In Configurator, go to Security Types > siren Types



This screen allows the Siren Types to be defined. This allows for example, Intruder and Fire alarms to have different siren sounds to distinguish the audible alarms from each other

Phone Numbers

In Configurator, go to Security > phone number



Up to 8 telephone numbers can be programmed

References

- **IEC for electrical distribution in building**
- **NEC for electrical distribution in building**
- **El-Sweedy cables catalogue**
- **Schneider catalogue**
- **Philips & dialux for indoor and outdoor lighting design catalogue**
- **THE LECTURES**
- **Internet Searches**
- **ABB catalogue**
- **Electrical Installation Practice**
- **Electrical Installation. Dr/M.Jelany**
- **Electrical transformers. Dr/M.Jelany**
- **Cooper catalogue**
- **Egyptian code for electrical distribution in building**
- **Electrical Protection. Dr/ M.Jelany**